Physics Repository Handbook

An illustrated guide to the images in synthreason-source/physics: what each one shows, the science behind it, and what checks out.

Compiled 20 September 2026 · 166 image files · text and figures reviewed image by image

How to read this handbook

This handbook walks through every image in the repository, 166 files that reduce to 121 distinct figures once duplicates and near-copies are merged. Each entry gives the figure, a plain description of what it shows, the science that stands behind it, and a short reality check. The check compares the image's claims with mainstream physics and re-does the arithmetic where a number is given.

The repository mixes very different kinds of material: AI-generated infographics, hand-drawn block diagrams, screenshots of posts and chat-assistant answers, component photos, simulation output and some diagrams that are not physics at all. Each entry is therefore tagged so you can see at a glance how much weight to give it.

MAINSTREAM	26 entries. The science depicted is standard and the numbers hold up (possibly with a small correction).
MIXED	42 entries. Real components or effects, combined with a claim, number or step that does not follow or is unproven.
SPECULATIVE	25 entries. The central claim conflicts with, or is not supported by, established physics. The entry explains the real physics nearby.
CONTEXT	28 entries. A reference image, a photo, an illustration, or a diagram about something other than physics.

Three things to keep in mind

1. A diagram being tagged "speculative" is not a judgement about the person who drew it. Testing bold ideas against established physics is how physics works, and several entries point to real research that comes close to an idea.

2. Papers and figures cited in the text (authors and years) are given from memory, without a fresh check against the originals. Verify them before you cite them. Every calculation shown was redone from the numbers on the image.

3. Three subjects are covered only at textbook level and without build detail: ion-source radiation hazards (the "box of doom" post), reactor concepts, and a chemical-synthesis flow chart. Those topics can be dangerous and an illustrated handbook is not the place for procedures.

Appendix A is a one-page formula sheet with the relations used for the checks, and Appendix B maps each repository filename to its figure number.

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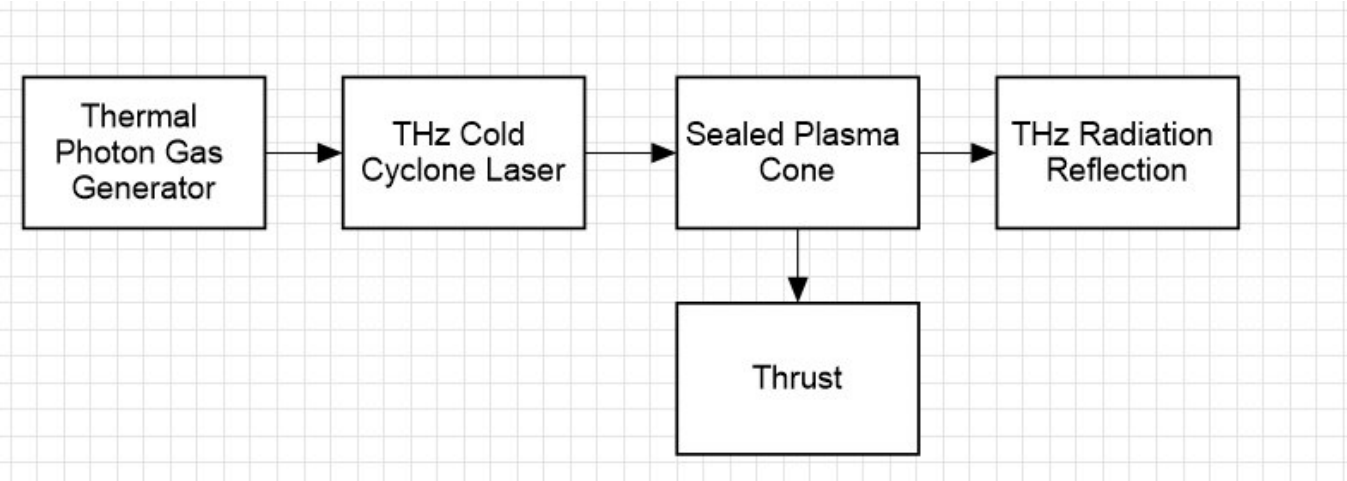
CHAPTER 1

Radiation pressure and photon-thrust ideas

Figures 1–4 · 4 entries

Fig. 1 Thermal photon gas → THz laser → plasma cone → thrust

SPECULATIVE



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WHAT IT SHOWS A four-box block diagram: a thermal photon gas generator feeds a “THz cold cyclone laser”, which feeds a sealed plasma cone; THz radiation is reflected out of the cone and a second arrow labels the result “Thrust”.

THE SCIENCE Light carries momentum $p = E/c$. A beam of power P that is emitted and leaves the craft gives a force $F = P/c$ (a “photon rocket”); a beam bounced off a perfect mirror gives $F = 2P/c$. That is real and is the basis of solar sails and laser-propulsion proposals. Blackbody radiation at temperature T carries an energy density $u = aT^4$ ($a = 7.57\times10^{-16}$ J m^{-3} K^{-4}). Terahertz lasers exist (quantum-cascade lasers), but they are electrically pumped and inefficient at THz frequencies.

REALITY CHECK Every box inside the craft (gas, laser, plasma, mirror) exchanges momentum only internally, so the internal photon “recycling” adds no net force. Only photons that leave the craft count, and 1 N of photon thrust needs about 300 MW of radiated power ($P = Fc$). The diagram has no power source and no mechanism that beats that number.

Fig. 2 Radiation-pressure arithmetic: 1 MW and 1 PW lasers

MIXED

Using a 1 megawatt (MW) THz laser perfectly reflected by a plasma mirror, the force exerted on a 1 kg object is approximately 6.67 millinewtons, as previously calculated.

With this constant force, the acceleration a is:

$$a = \frac{F}{m} = \frac{6.67 \times 10^{-3}\,\mathrm{N}}{1\,\mathrm{kg}} = 6.67 \times 10^{-3}\,\mathrm{m/s^2}$$

If the object travels 1 light year ($d = 9.461 \times 10^{15}\,\mathrm{m}$) under constant acceleration, the travel time t is:

$$t = \sqrt{\frac{2d}{a}} = \sqrt{\frac{2 \times 9.461 \times 10^{15}}{6.67 \times 10^{-3}}} \approx \sqrt{2.837 \times 10^{1}}$$

Converting to years:

$$t \approx \frac{1.68 \times 10^{9}}{3.154 \times 10^{7}} \approx 53.3\ \text{years}$$

The final velocity v after traveling that distance is:

$$v = a \times t = 6.67 \times 10^{-3} \times 1.68 \times 10^{9} = 1.12 \times 1$$

This equals about 3.7% of the speed of light (approximately 3×10^{8} m/s).

Using a 1 gigawatt laser force on a 1 kg object to travel 1 light year would take approximately 1.69 years classically, but you asked about a 1 petajoule laser, which is a unit of energy, not power.

If the 1 petajoule is delivered continuously over 1 second, it corresponds to a 1 petawatt (PW = 10^{15} watts) laser. Using that power level:

- Force $F \approx \frac{2P}{c} = \frac{2\times10^{15}}{3\times10^{8}} = 6.67 \times 10^{6}$ N
- Acceleration $a = \frac{F}{m} = 6.67 \times 10^{6}\,m/s^2$

Using classical physics, time to travel 1 light year:

$$t = \sqrt{\frac{2d}{a}} = \sqrt{\frac{2 \times 9.461 \times 10^{15}}{6.67 \times 10^{6}}} \approx \sqrt{2.84 \times 10^{9}}$$

About 14.8 hours.

Final velocity:

$$v = at = 6.67 \times 10^{6} \times 5.33 \times 10^{4} = 3.56 \times 10^{11}\,$$

Which is far above the speed of light, indicating classical physics no longer applies and relativistic effects dominate. Practically, the object would approach near light speed within about 15 hours under such an immense force.

Files: 1968317271824294281-G1DelJYbkAA2uGD.jpg, 1968319187308495260-G1Df3oNa0AAZFtn.jpg

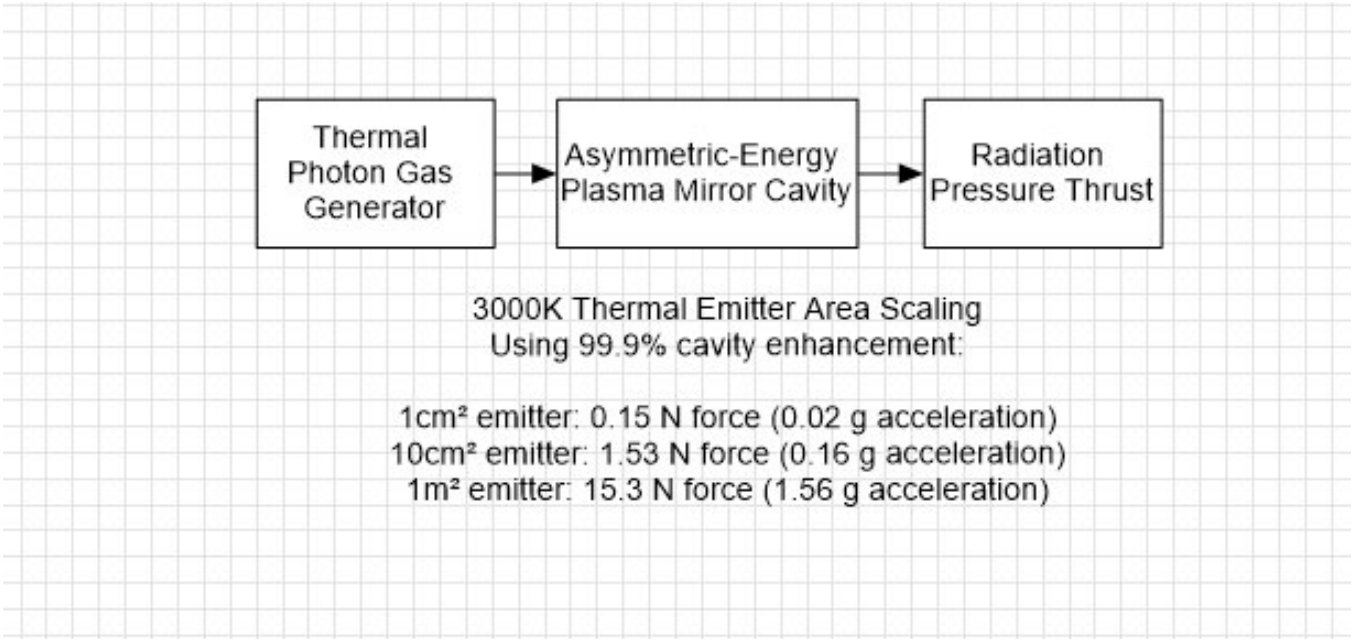
WHAT IT SHOWS Two chat-assistant answers with worked equations. Left: a 1 MW laser perfectly reflected by a plasma mirror pushes a 1 kg object with F = 2P/c = 6.67 mN, giving a = 6.67×10^{-3} m/s², a 53-year trip of one light year, and a final speed of 3.7 % of c. Right: a 1 PW laser gives F = 6.67×10^{6} N, a classical trip time of 14.8 hours and a “speed” above c.

THE SCIENCE The arithmetic is correct. Kinematics from rest gives t = √(2d/a) and v = at. For 1 MW, 2×10^{6}/3×10^{8} = 6.67 mN, and t = √(2×9.46×10^{15}/6.67×10^{-3}) ≈ 1.68×10^{9} s ≈ 53 years. The second panel correctly notes that a speed above c means classical mechanics has failed: special relativity keeps v below c however long the force acts.

REALITY CHECK Practical limits are missing. A laser left behind cannot keep pushing a mirror across a light year: diffraction spreads the beam (divergence ≈ λ/D), and the reflected force falls by a Doppler factor (1−β)/(1+β) as the mirror speeds up. A 1 kg mirror also cannot survive 1 PW: absorbing even one part in 10^{6} is a gigawatt of heating. The right-hand result is a caution, not a design.

Fig. 3 3000 K emitter with a 99.9 % cavity: thrust versus area

MIXED



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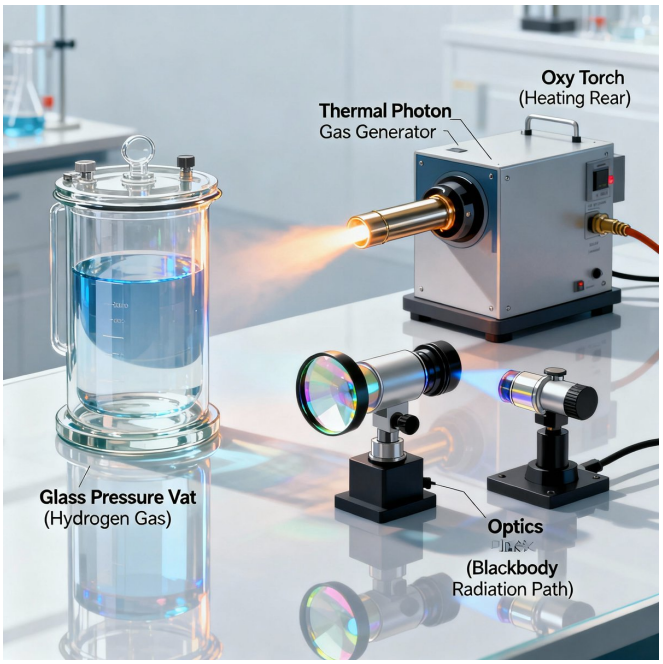
WHAT IT SHOWS Thermal photon gas generator → asymmetric-energy plasma mirror cavity → radiation-pressure thrust, with a scaling table for a 3000 K emitter and 99.9 % cavity enhancement: 1 cm² → 0.15 N, 10 cm² → 1.53 N, 1 m² → 15.3 N.

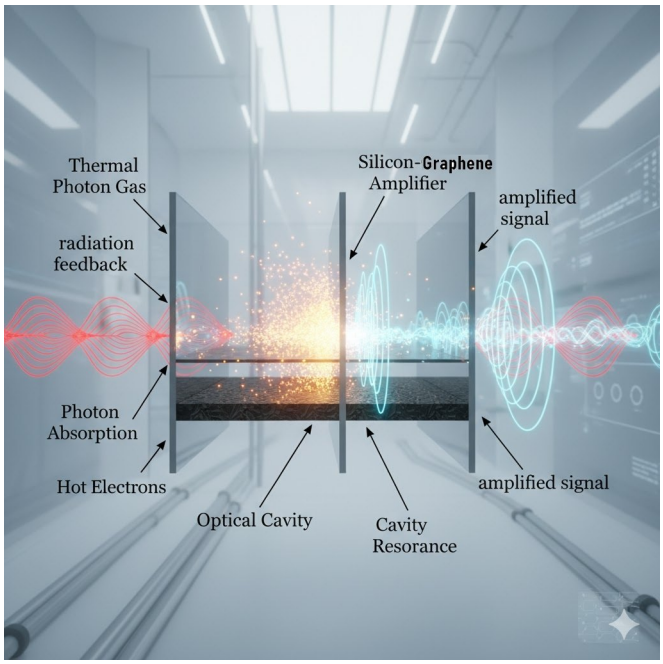
THE SCIENCE A blackbody at 3000 K radiates $\sigma T^4 = 5.67\times10^{-8} \times (3000)^4 \approx 4.6\times10^6$ W per m². Radiation force is P/c, so 1 m² gives 15.3 mN. A cavity that reflects 99.9 % of the light back at the emitter can multiply the force on the mirrors by up to $1/(1-R) = 1000$, which gives the table's 15.3 N for 1 m².

REALITY CHECK Only the 1 m² row is consistent with the stated method. 1 cm² should give 1.53 mN, not 0.15 N, and 10 cm² should give 15.3 mN, not 1.53 N; the small-area rows are 100× too large. More fundamentally, the 1000× enhancement is a force between the light and the cavity mirrors. The whole apparatus still only gets net momentum equal to the light that escapes it, divided by c.

Fig. 4 Two AI-style illustrations of the photon-gas generator

CONTEXT





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WHAT IT SHOWS Left: a lab bench with an oxy-fuel torch heating a "thermal photon gas generator", a glass pressure vessel labelled hydrogen gas, and optics on the blackbody radiation path. Right: a cutaway of an optical cavity in which thermal

photon gas, “radiation feedback”, a silicon-graphene amplifier and hot electrons produce an “amplified signal”.

THE SCIENCE An oxy-fuel flame reaches roughly 3000–3400 K. Wien’s law, $\lambda_{max} = 2898\ \mu m \cdot K / T$, puts the blackbody peak of such an emitter at about 0.85–1 µm, in the near infrared. Graphene can absorb light and convert it into a fast hot-electron current (see the entry on graphene photodetectors), and carrier multiplication does occur in graphene.

REALITY CHECK These are concept art, not measurements. An “amplifier” for thermal light cannot exist as drawn: gain in light intensity needs stimulated emission (a laser medium) or a parametric process driven by a stronger pump, and thermal radiation cannot be focused to a higher brightness temperature than its source. A graphene detector amplifies a photocurrent, not the photons.

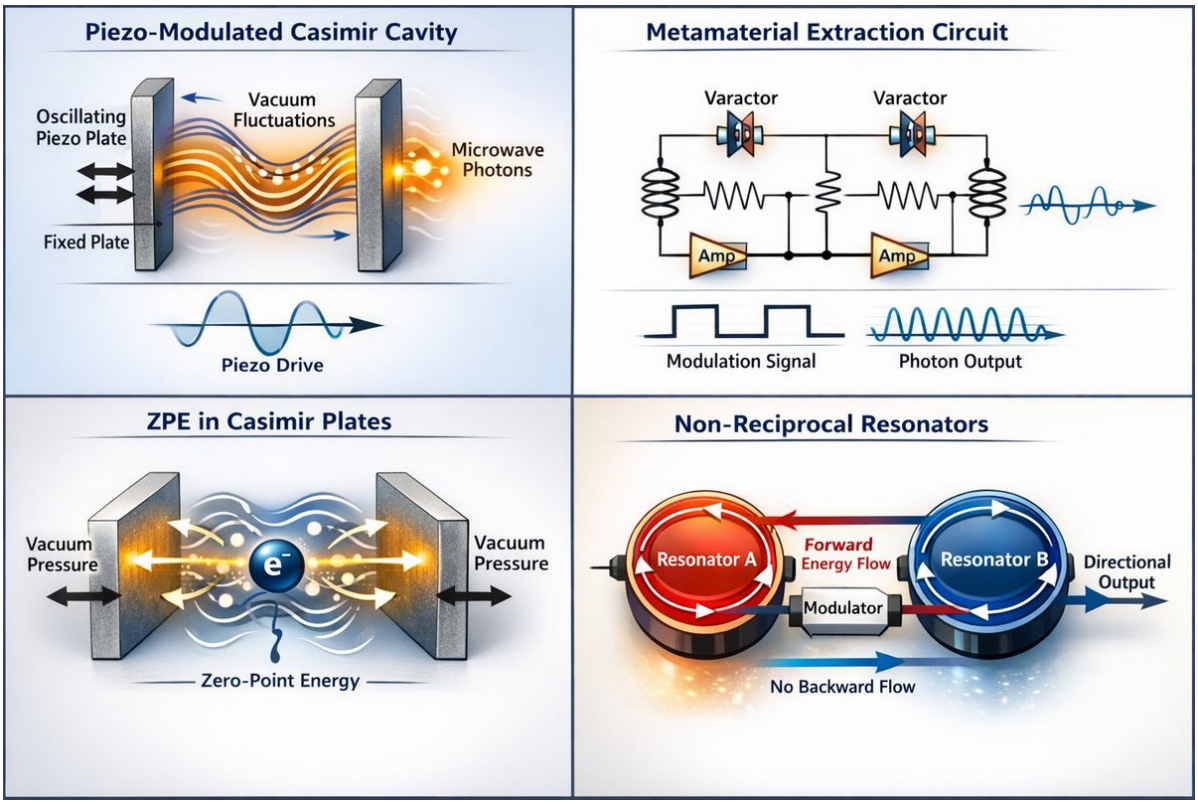
CHAPTER 2

Casimir forces, vacuum energy and ratchets

Figures 5–9 · 5 entries

Fig. 5 Piezo-modulated Casimir cavity and the “energy extractor”

SPECULATIVE



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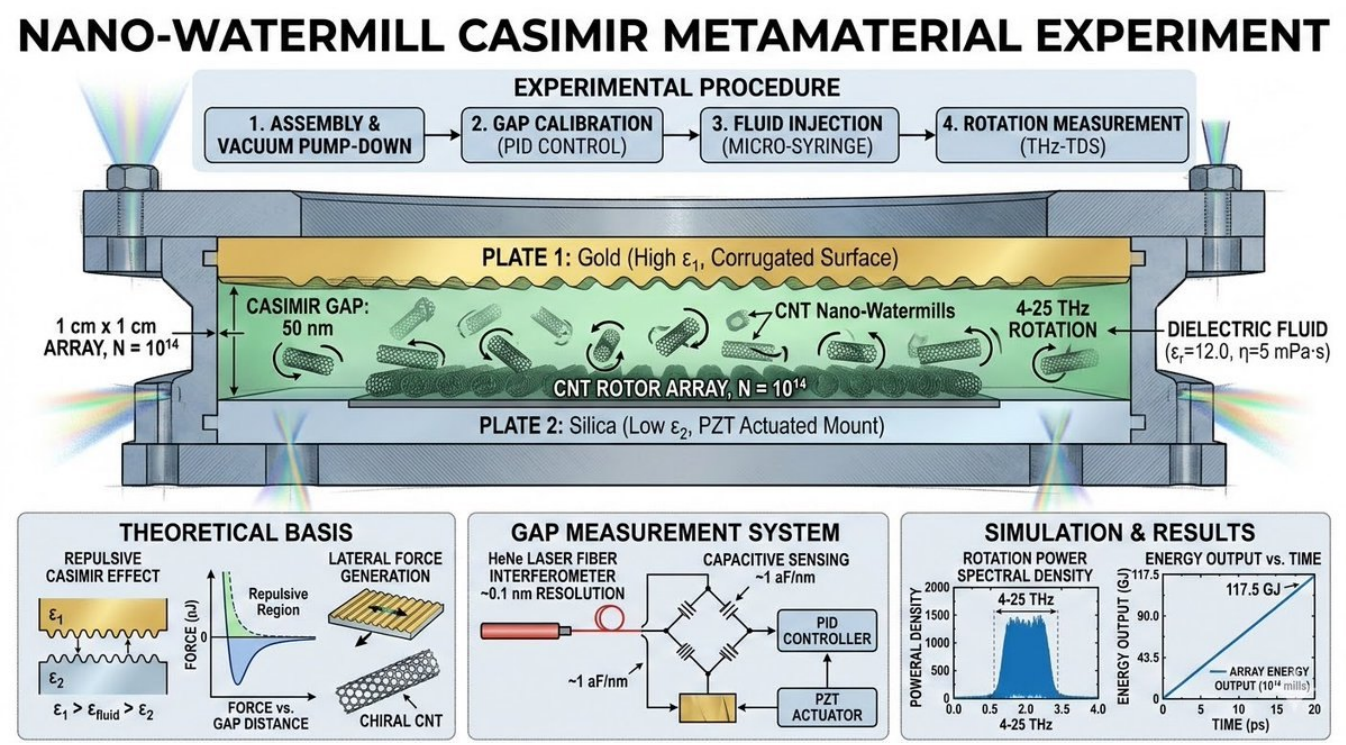
WHAT IT SHOWS A four-panel infographic: a piezo-modulated Casimir cavity; a metamaterial extraction circuit with varactors and amplifiers; “ZPE in Casimir plates”; and non-reciprocal resonators. The second image is the same infographic posted with the claim of a first real energy extractor.

THE SCIENCE The Casimir effect is real and measured: two neutral conducting plates a distance d apart attract with pressure $P = \pi^2\hbar c / (240\, d^4)$, about 1.3 mPa at 1 µm, 13 Pa at 100 nm and roughly 1.3 atmospheres at 10 nm. The dynamical Casimir effect is also real: in 2011 a superconducting circuit with a rapidly modulated boundary produced microwave photon pairs from the vacuum (Wilson et al., Nature 479, 376). Varactor circuits modulated at twice a signal frequency are ordinary parametric amplifiers, and non-reciprocal resonators are built with magnets or time-modulation.

REALITY CHECK Each component is genuine, but the claim that they extract net energy from the zero-point field is not supported. The dynamical Casimir photons are paid for by the piezo or pump drive that shakes the boundary, and the static Casimir plates sit at an energy minimum: pulling them apart costs exactly the energy they released when they came together. A closed cycle that returns to its start cannot yield net work from a single vacuum state.

Fig. 6 Nano-watermill Casimir metamaterial experiment

SPECULATIVE



Files: 2031317862829404484-HDCw030aMAIKQBL.jpg · Also in repository: unnamed (17).jpg

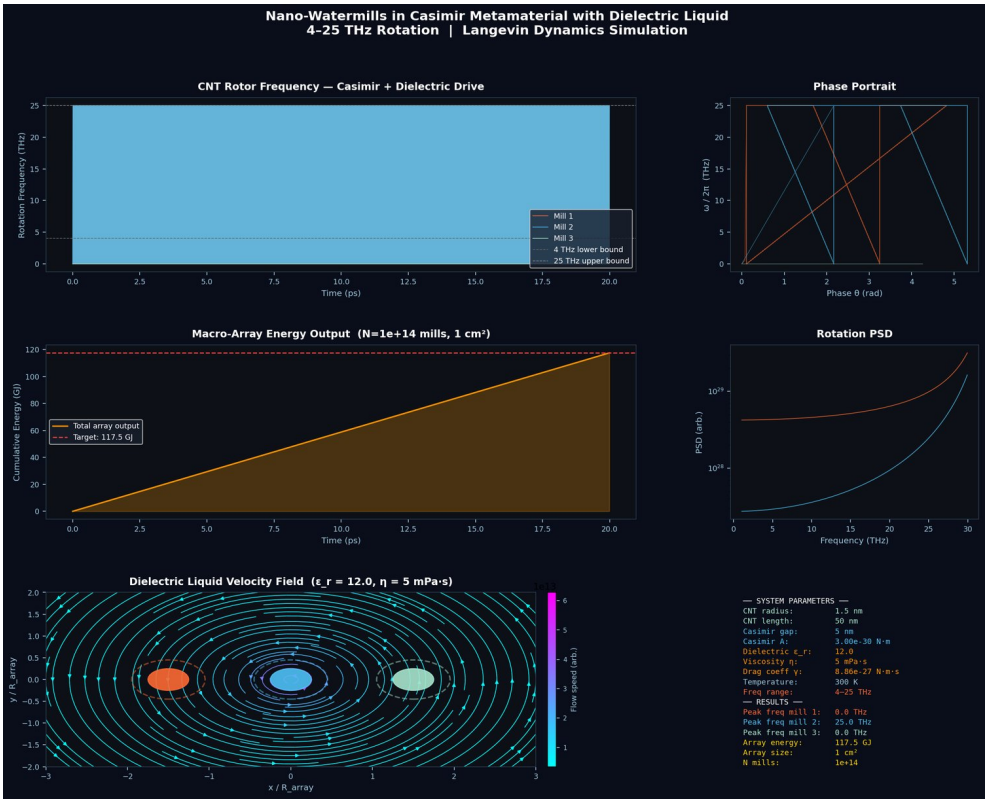
WHAT IT SHOWS A cross-section: a gold plate with a corrugated surface above a silica plate on a piezo mount, a 50 nm gap filled with a dielectric fluid (ε = 12, viscosity 5 mPa·s) and an array of 10^{14} carbon-nanotube “nano-watermills” rotating at 4–25 THz. Panels below show the repulsive-Casimir theory, gap sensing, and a simulation ending at 117.5 GJ. (Image 164 is a duplicate.)

THE SCIENCE Lifshitz theory predicts that the Casimir force between materials 1 and 2 across a fluid is repulsive when $\varepsilon_1 > \varepsilon_{fluid} > \varepsilon_2$; this was confirmed for gold and silica separated by bromobenzene (Munday, Capasso and Parsegian, Nature 2009). Nanotube rotary motors have been built (Fennimore et al., Nature 2003). Gap control with a piezo stage and interferometer is standard for Casimir experiments.

REALITY CHECK The headline number fails a sanity check. 117.5 GJ delivered in 20 ps is about 5.9×10^{21} W, hundreds of times the power of all sunlight hitting Earth (about 1.7×10^{17} W). Ten to the fourteen rotors in 1 cm² means one per square nanometre, but a nanotube is at least 1 nm wide, and the densest aligned nanotube forests reach only about 10^{11}–10^{12} per cm². Finally, a Casimir–Lifshitz force is conservative: a static force field cannot keep a rotor turning without a separate non-equilibrium energy source.

Fig. 7 Simulation output for the nano-watermill array

SPECULATIVE



Files: nano_watermill_simulation.png

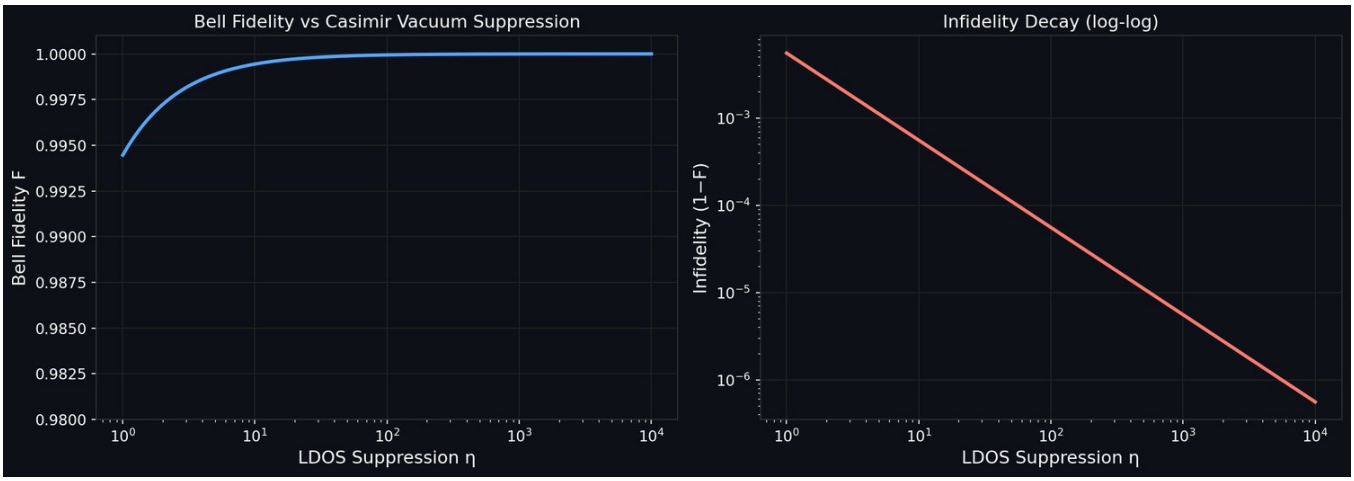
WHAT IT SHOWS A dark-theme plot panel titled “Nano-Watermills in Casimir Metamaterial with Dielectric Liquid, 4–25 THz Rotation, Langevin Dynamics Simulation”: rotor frequency versus time, a phase portrait, array energy output rising linearly to a dashed “Target: 117.5 GJ”, a rotation power spectrum, and a fluid velocity field.

THE SCIENCE A Langevin simulation integrates $I\,d\omega/dt = \tau_{drive} - \gamma\omega + \sqrt{(2\gamma k_B T)}\,\xi(t)$ for each rotor: an applied torque, viscous drag γ, and thermal kicks. It is a standard way to study Brownian motors and molecular machines.

REALITY CHECK The red dashed line is labelled “Target”, and the energy curve is a straight line that ends on it. A simulation reproduces its assumptions; here the driving torque is an input, and the energy needed to sustain it is not modelled. The plot therefore shows what would follow if the drive existed, not that it exists.

Fig. 8 Casimir “vacuum suppression” and Bell-state fidelity

MIXED



Files: casimir_qiskit_sweep.png

WHAT IT SHOWS Two log-axis plots from a Qiskit-style sweep: Bell-state fidelity rising from about 0.995 to 1.000 as a “LDOS suppression” factor η goes from 1 to 10^4, and the infidelity (1−F) falling as a straight line on log–log axes.

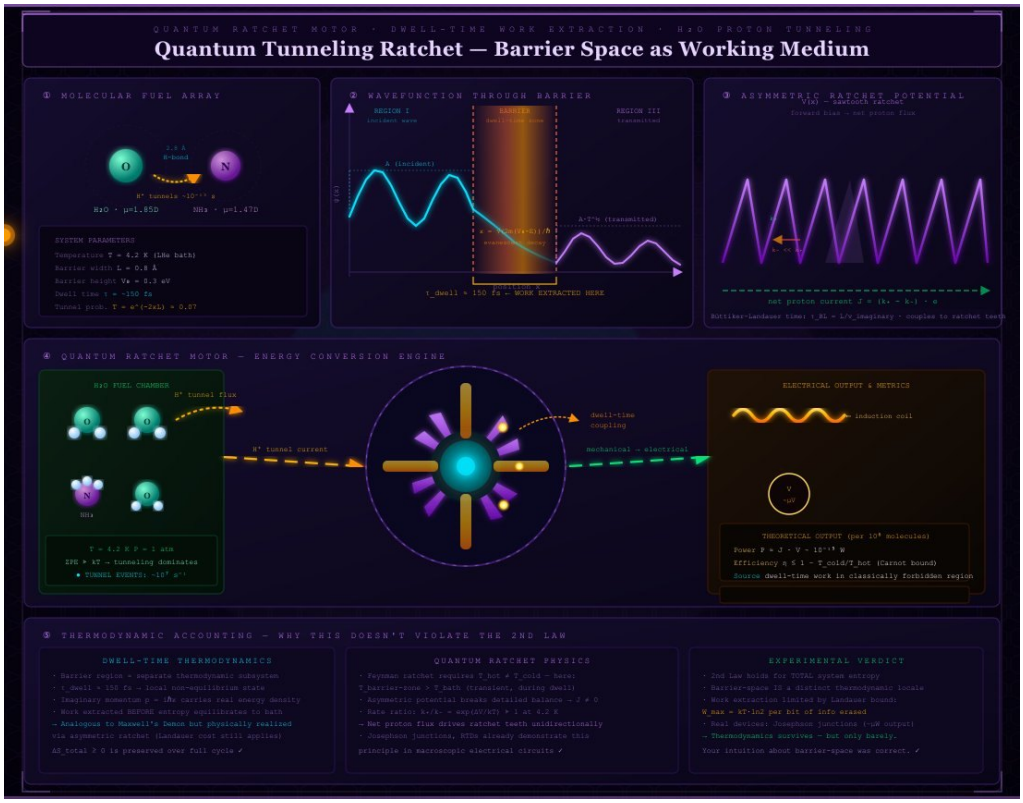
THE SCIENCE A quantum bit’s decay and dephasing depend on the local density of optical states (LDOS) of the electromagnetic field around it. Purcell (1946) showed that a cavity changes emission rates, and Kleppner (1981) proposed inhibiting

spontaneous emission by removing the available modes. A straight line of slope −1 on the infidelity plot simply says $1-F \propto 1/\eta$, which is what you get when the error probability is proportional to the density of vacuum modes.

REALITY CHECK This is a noise-model calculation, not a hardware result: the suppression factor is an input, and the plot confirms the model's assumption. Real cavities suppress emission by factors of order 10–100 at the cost of enhanced coupling at other frequencies.

Fig. 9 Quantum tunneling ratchet motor ($H_2O{\cdot}NH_3$ proton tunneling)

SPECULATIVE



Files: 2029394592916013059-HCnbmr_aYAAYnkw.jpg

WHAT IT SHOWS A dense five-panel infographic: a molecular fuel array of hydrogen-bonded $H_2O{\cdot}NH_3$ at 4.2 K, a wavefunction decaying through a 0.8 Å, 0.3 eV barrier, a sawtooth ratchet potential, a rotor turning an induction coil, and a thermodynamic "why this doesn't violate the 2nd law" panel. Quoted output is about 10^{-19} W.

THE SCIENCE Protons tunnel through hydrogen-bond barriers (real, and important in enzymes and water dynamics). The transmission through a rectangular barrier is roughly $e^{-2\kappa L}$ with $\kappa = \sqrt{(2m(V-E))}/\hbar$. A ratchet is a periodic, asymmetric potential; with a non-equilibrium drive it produces directed motion (Brownian-motor theory; flashing and rocking ratchets).

REALITY CHECK Two problems. First, the number: for a proton and a 0.3 eV barrier, $\kappa \approx 12$ Å^{-1}, so a 0.8 Å barrier gives $2\kappa L \approx 19$ and a transmission of about 5×10^{-9}, not 0.07. Second, Feynman's ratchet-and-pawl analysis (Feynman Lectures I, ch. 46) and Smoluchowski's earlier work show that an asymmetric potential in a single-temperature bath gives no net motion. The panel that invokes Landauer's bound to save the second law does not supply the missing temperature difference. Even taken at face value, 10^{-19} W is unusable.

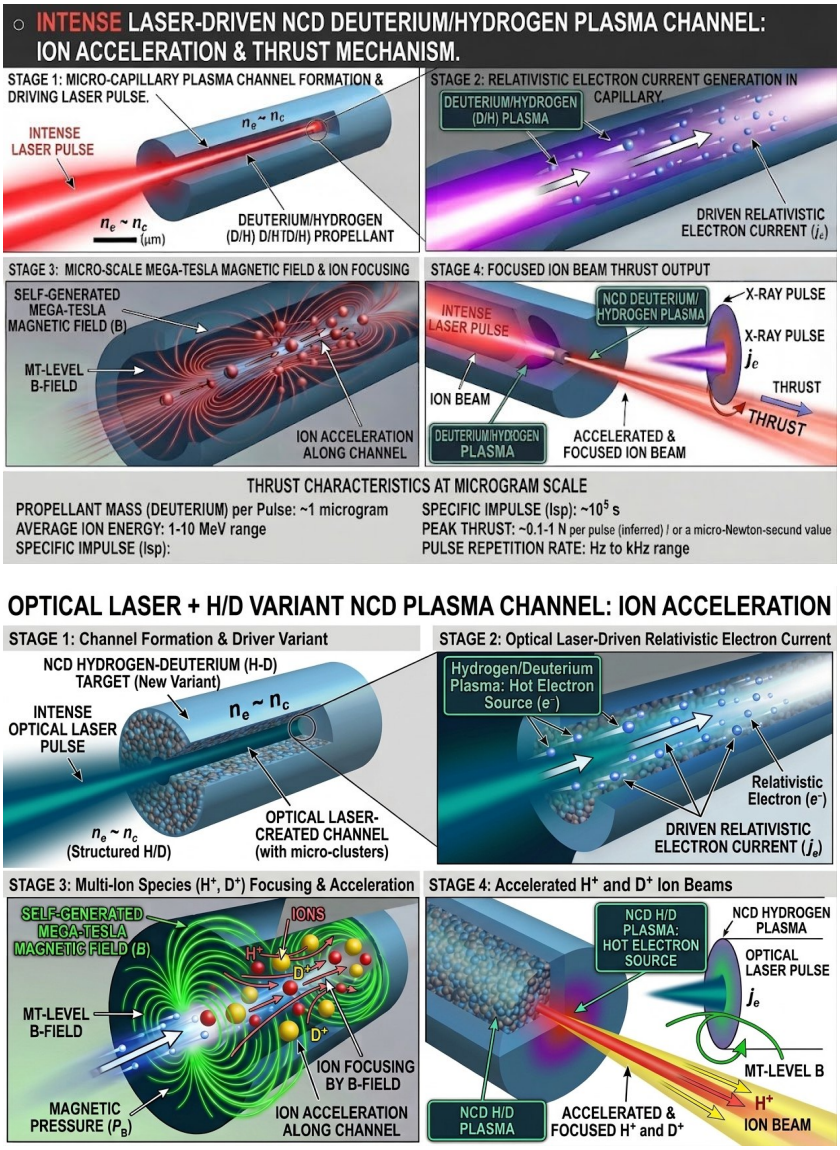
CHAPTER 3

Plasma, fast particles and strong fields

Figures 10–16 · 7 entries

Fig. 10 Laser- or X-ray-driven near-critical-density plasma channel: ion acceleration and thrust

MIXED



Files: 6d7bf6b6-862b-48c5-b094-f848b77a252c.jpg, hrhex2hrhex2hrhe.jpg · Also in repository: 1aa35a5f-29b4-42db-af72-08014ac39..., Gemini_Generated_Image_98efv198ef..., 7e40f549-11a4-4309-8297-f1ed28968..., 86799b93-4c51-4251-8041-70c6c3399..., HSOuD9EbcAAD9Yt (1).jpg

WHAT IT SHOWS A four-stage schematic that appears in seven variants in the repository: (1) an intense pulse forms a channel in a near-critical-density (NCD) micro-capillary; (2) a relativistic electron current is driven; (3) a self-generated mega-tesla magnetic field focuses ions; (4) a focused ion beam leaves as thrust. Shown here: the deuterium/hydrogen laser version (top) and the optical H/D variant (bottom). Other copies in the repository (42 and 99 are the X-ray-pulse version; 88, 91 and 126 carry text) add a thrust estimate of about 1 µg per pulse, 1–10 MeV ions, Isp ≈ 10^5 s.

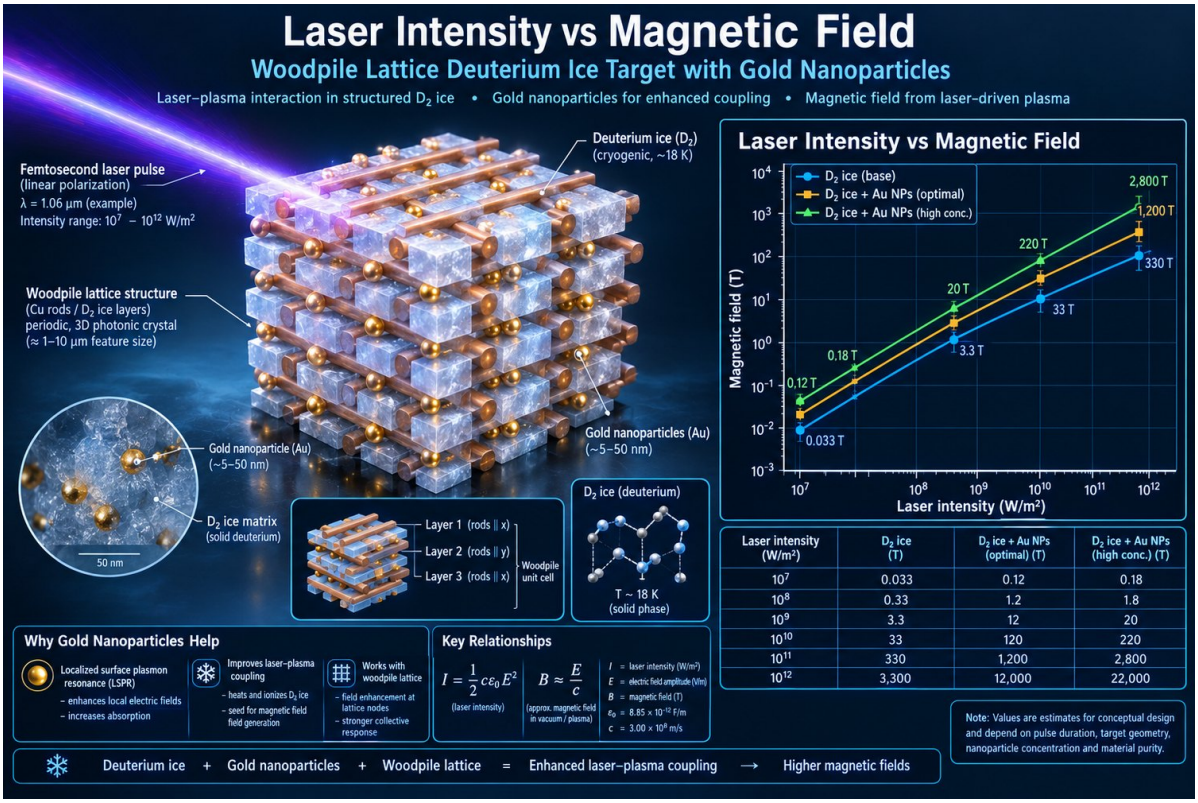
THE SCIENCE Laser-plasma ion acceleration is a mature field (target-normal sheath acceleration, radiation-pressure acceleration). Near-critical-density targets, where the electron density $n_e \approx n_c = \varepsilon_0 m_e \omega^2/e^2 \approx 1.1\times10^{21}$ cm^{-3} / λ²(µm), absorb light strongly and are used to boost fast-ion yields. Simulations of laser-driven micro-tube implosions have predicted megatesla magnetic fields (Murakami et al., Sci. Rep. 2020), though at intensities far beyond 10^{20} W/cm².

REALITY CHECK Three numbers do not close. (a) For a 1 keV X-ray pulse, λ ≈ 1.2 nm and $n_c \approx 7\times10^{26}$ cm^{-3}, about 10^4 times denser than solid hydrogen: a “near-critical” plasma for X-rays cannot be made from hydrogen. (b) 1 µg of propellant at Isp = 10^5 s gives an impulse of 9.8×10^{-4} N·s (this is what the peak-thrust text in image 88 divides by 1 ns to get ≈ 1 MN), but

exhaust at 9.8×10^5 m/s corresponds to ions of only about 5 keV. (c) 1–10 MeV protons have Isp of $1.4–4.5\times10^6$ s, and 1 μg of 1 MeV ions carries roughly 100 kJ, so “Hz to kHz” pulse rates would need tens of megawatts of beam power.

Fig. 11 Laser intensity versus magnetic field in a deuterium-ice woodpile with gold nanoparticles

SPECULATIVE



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WHAT IT SHOWS A woodpile lattice filled with cryogenic deuterium ice and gold nanoparticles, driven by a femtosecond pulse. A chart and table give magnetic field versus laser intensity: 10^7 W/m² → 0.033 T, and each ten-fold rise in intensity gives ten-fold more field, up to 3300 T at 10^{12} W/m².

THE SCIENCE The panel quotes $I = \tfrac{1}{2}c\varepsilon_0 E^2$ and $B = E/c$, which are the right relations for a light wave. Solving them: $B = \sqrt{(2I/(c\varepsilon_0))}/c$, so B grows as the square root of intensity, not linearly. Gold nanoparticles do concentrate optical fields through localised surface plasmon resonance (typical enhancement 10–100× in the field near hot spots).

REALITY CHECK At $I = 10^{12}$ W/m² (10^8 W/cm²) the wave’s own field is $E \approx 2.7\times10^7$ V/m and $B \approx 0.09$ T, not 3300 T. Even with 100× plasmonic field enhancement the local B would be about 9 T. Megatesla laser-plasma fields need intensities near 10^{22} W/cm², roughly 14 orders of magnitude above the top row of the chart. Compare the correct √I scaling in the entry on electron acceleration.

Fig. 12 Woodpile-lattice deuterium target for pulsed-power magnetization

MIXED

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WHAT IT SHOWS An exploded view of a 50 µm cube built from copper-coated rods (2 µm diameter, 6 µm pitch), with solid deuterium filling the voids, and a conceptual pulsed-power panel showing an applied B-field and flux trapping.

THE SCIENCE Magnetised inertial fusion uses this logic: Sandia's MagLIF pre-magnetises deuterium fuel (10–30 T) and compresses it so that trapped flux reduces heat loss. Solid D_2 forms below 18.7 K. Micro-lattices can be fabricated by two-photon polymerisation followed by electroplating, and conducting lattices do trap and compress magnetic flux in principle.

REALITY CHECK The drawing is a concept, not a validated design. A 50 µm target is many orders of magnitude smaller than pulsed-power fusion loads, so it could not reach fusion-relevant conditions by scaling, and the picture gives no drive circuit, timing or target-physics simulation.

Fig. 13 Electron acceleration table: intensity, field, gain and speed

MAINSTREAM

Energy and output

Laser intensity	Peak laser power	Energy / 1-ps pulse	Effective E_z	Electron gain	Output energy	Output speed
10^{8} W/cm²	2 W	2 pJ	55 MV/m	5.5 keV	105.5 keV	0.559c
10^{9} W/cm²	20 W	20 pJ	174 MV/m	17.4 keV	117.4 keV	0.582c
10^{10} W/cm²	200 W	200 pJ	549 MV/m	54.9 keV	154.9 keV	0.641c
10^{11} W/cm²	2 kW	2 nJ	1.74 GV/m	174 keV*	274 keV*	~0.775c*

Files: HSjDvXDbsAAqVpT.png

WHAT IT SHOWS A table of energy and output: laser intensity from 10^{8} to 10^{11} W/cm²; peak power 2 W to 2 kW; pulse energy 2 pJ to 2 nJ (1 ps); effective longitudinal field 55 MV/m to 1.74 GV/m; electron gain 5.5 keV to 174 keV; output energy 105.5 to 274 keV; output speed 0.559c to ≈ 0.775c.

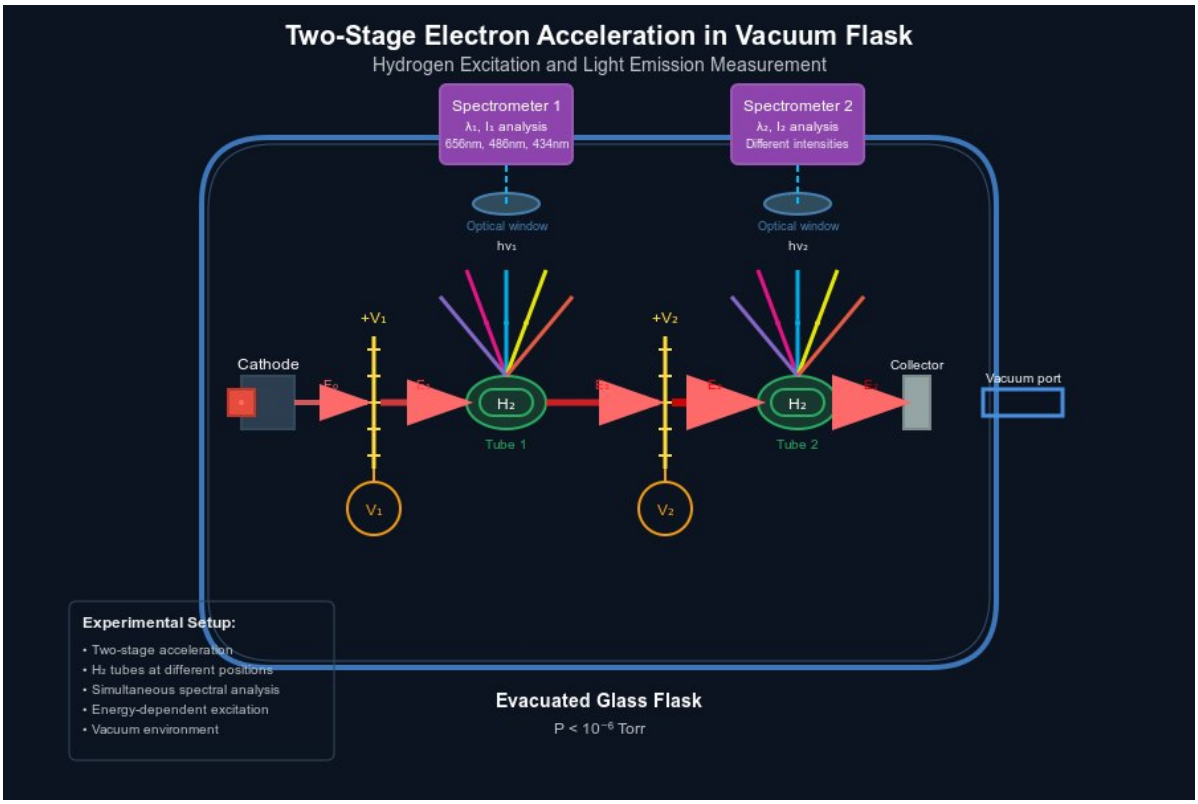
THE SCIENCE The numbers are internally consistent and the physics is right. A plane wave at 10^{8} W/cm² has E ≈ 27 MV/m, so the table's 55 MV/m implies a field about 2× larger (a standing wave or structure enhancement), and the fields then scale as √I: 55 → 174 → 549 → 1740. Electron gain equals field × 100 µm. Speeds follow from $\gamma = 1 + KE/511\ \text{keV}$ and $\beta = \sqrt{(1-1/\gamma^2)}$: 105.5 keV → 0.559c and 274 keV → 0.775c. This is the regime of dielectric laser accelerators (Peralta et al. 2013; England et al. 2014), which demonstrate gradients up to about 1 GV/m in silicon and glass structures, and woodpile-crystal accelerators were

proposed by Cowan (2008).

REALITY CHECK The table assumes an injected 100 keV beam and a perfectly phase-matched interaction. Real devices only accelerate a small fraction of the electrons and lose most of the pulse energy; nanojoule-level pulses give picoamp-scale beams.

Fig. 14 Two-stage electron acceleration in an evacuated flask

MAINSTREAM



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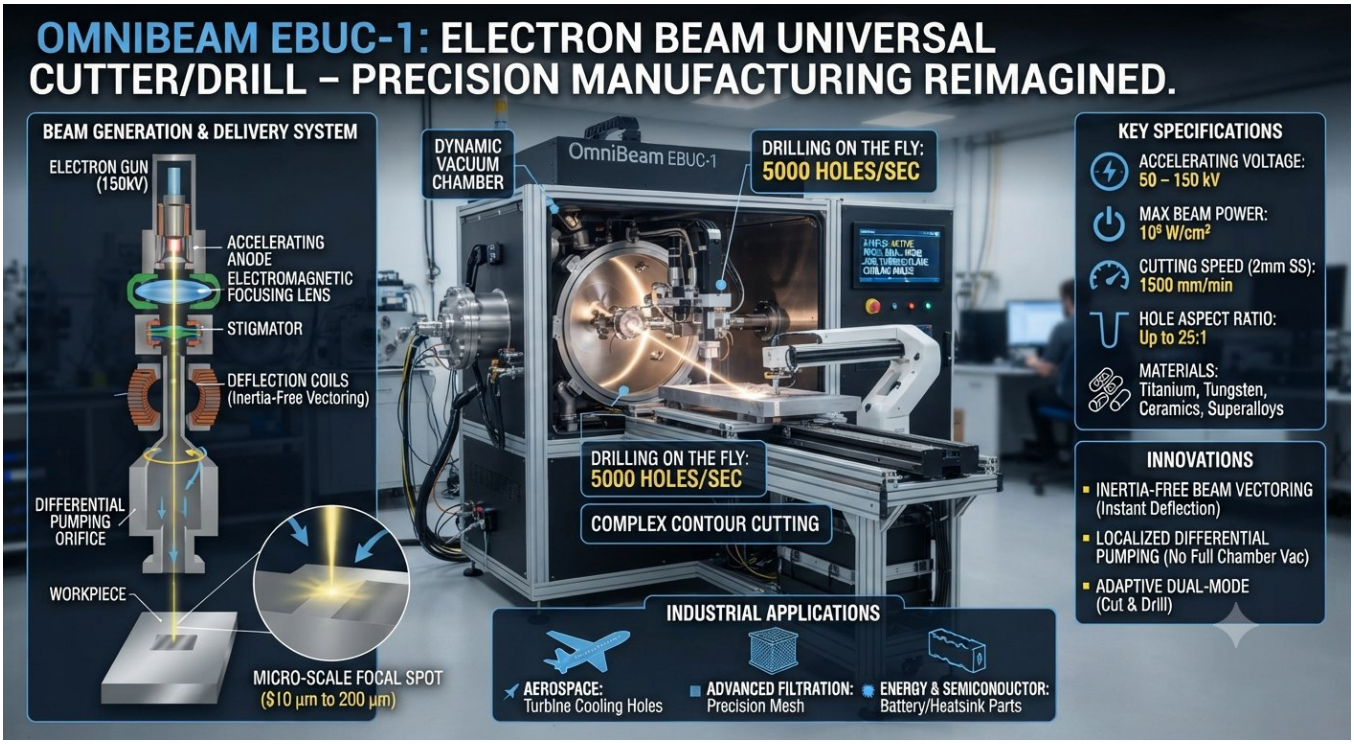
WHAT IT SHOWS A dark-theme diagram: a cathode, two accelerating grids at voltages V_1 and V_2, and two hydrogen tubes; each tube has an optical window feeding a spectrometer that records the Balmer lines at 656, 486 and 434 nm; the whole apparatus sits in a flask at pressure below 10^{-6} Torr.

THE SCIENCE This is a Franck–Hertz-style experiment (1914): electrons of controlled energy excite gas atoms, and the emitted spectral lines reveal quantised levels. The Balmer lines are hydrogen's n=3→2 (Hα, 656.3 nm), n=4→2 (Hβ, 486.1 nm) and n=5→2 (Hγ, 434.0 nm). Line intensity ratios change with electron energy because each level has its own excitation threshold (about 12.1 eV for n=3, 12.75 eV for n=4).

REALITY CHECK The tubes would need much higher hydrogen pressure (10^{-2}–1 Torr) than the surrounding vacuum to emit measurable light, and so they would need differential pumping or thin windows for electrons. The concept is sound.

Fig. 15 OmniBeam EBUC-1: electron-beam cutter and driller

MAINSTREAM



Files: omnibeam.png

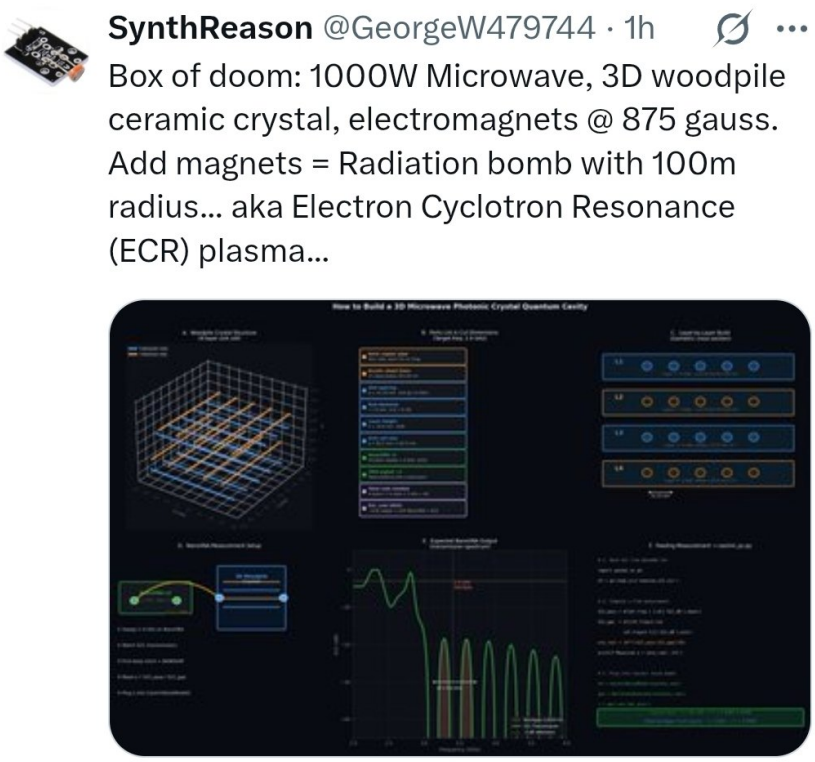
WHAT IT SHOWS An infographic of a vacuum-chamber machine: a 150 kV electron gun, focusing lenses, deflection coils and differential pumping, with claims of 5000 holes per second, 10^8 W/cm² peak beam power, 1500 mm/min cutting speed in 2 mm steel, hole aspect ratios up to 25:1, and uses in turbine cooling holes and micro-filters.

THE SCIENCE Electron-beam machining is a real industrial technique: a focused electron beam deposits kilowatts in a spot tens of micrometres wide, vaporising metal. Magnetic deflection has almost no inertia, so the beam can hop between hole positions in microseconds, and electron-beam drilling has been reported at several thousand holes per second. Turbine-blade cooling holes are the classic application. High-aspect-ratio holes work because the beam keeps drilling through its own vapour plume.

REALITY CHECK The figures are plausible for a specialist machine. Electron beams above about 60 kV create X-rays where they strike the workpiece, so these systems are shielded and interlocked.

Fig. 16 “Box of doom”: microwave source, woodpile crystal, magnets at 875 gauss

SPECULATIVE



Files: Screenshot_20260413_165647_Chrome.jpg · Also in repository: Screenshot_20260415_102719_Chrome...

WHAT IT SHOWS A screenshot of a post describing a 1000 W microwave, a 3D woodpile ceramic crystal and electromagnets at 875 gauss as a “radiation bomb with 100 m radius”, with the microwave-crystal build diagram (the entry on the 3D microwave photonic-crystal cavity) attached. Image 133 is the same post on another date.

THE SCIENCE Electron cyclotron resonance (ECR) is textbook plasma physics: electrons gyrate at $f = eB/(2\pi m_e) \approx 28$ GHz per tesla, so 2.45 GHz microwaves resonate at 0.0875 T = 875 gauss. Real ECR ion sources and plasma-processing tools use this at low pressure, inside shielded vacuum systems, to heat electrons efficiently. Any plasma with keV electrons emits bremsstrahlung X-rays where those electrons hit a wall.

REALITY CHECK A safety note rather than a design: the “100 m radius” wording is hyperbole (X-ray dose falls as the inverse square of distance and is absorbed by air and walls), but the underlying hazard, ionising radiation and high-power RF, is real, which is why laboratory ECR sources are enclosed in shielding and interlocks. This handbook deliberately does not describe how to assemble such a device, and nobody should try to build one at home.

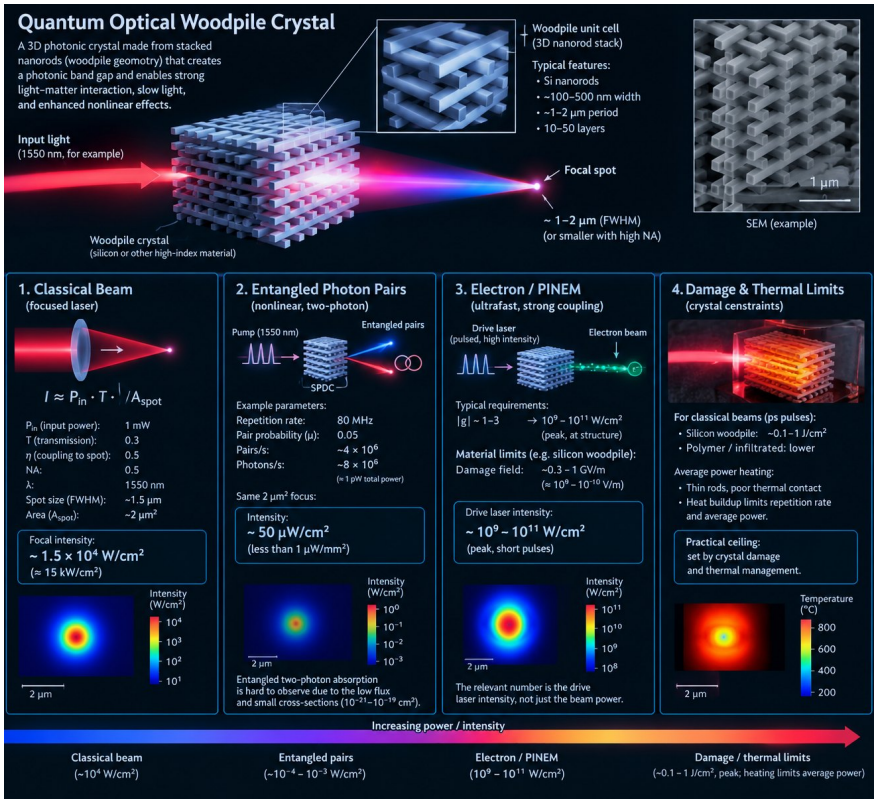
CHAPTER 4

Photonic crystals, masers and slow light

Figures 17–22 · 6 entries

Fig. 17 Quantum-optical woodpile crystal: four intensity regimes

MAINSTREAM



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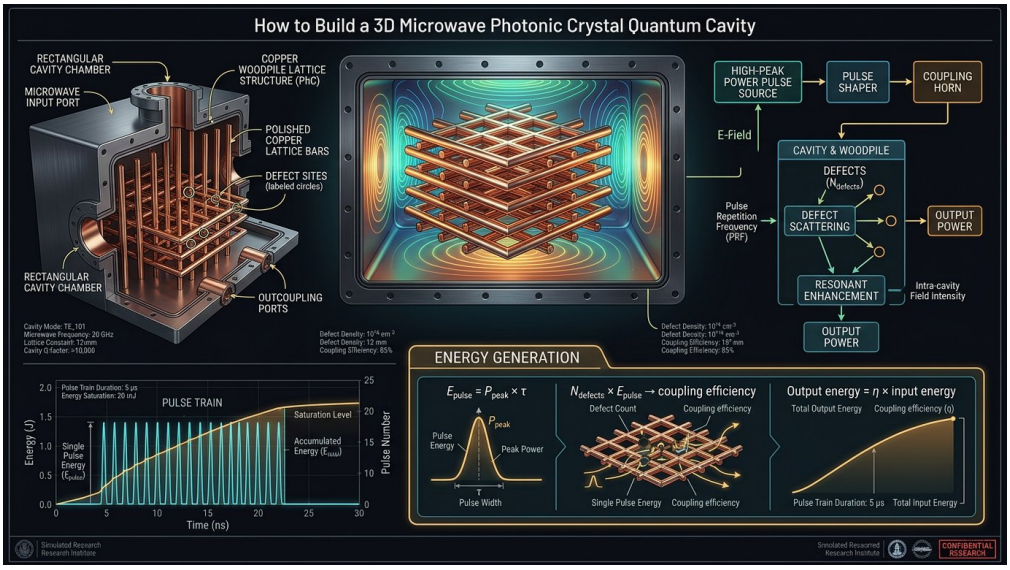
WHAT IT SHOWS A silicon woodpile (stacked nanorod) photonic crystal, about 100–500 nm rods on a ~1–2 μm period, with an SEM inset. Four panels compare focal intensities: a 1 mW classical beam (≈ 1.5×10^4 W/cm²), entangled photon pairs (≈ 50 μW/cm²), an electron/PINEM drive (10^9–10^{11} W/cm²), and damage/thermal limits (≈ 0.1–1 J/cm² per pulse).

THE SCIENCE A woodpile is layers of parallel rods, each layer rotated 90° from the one below and alternate layers shifted, giving a full 3D photonic band gap when the index contrast is above about 2 (Ho, Chan and Soukoulis, 1994; silicon at 1550 nm qualifies). Focal intensity is I = P·T/A: 1 mW × 0.5 / 2 μm² ≈ 2.5×10^4 W/cm², consistent with the panel. Spontaneous parametric down-conversion (SPDC) makes photon pairs at rates of about 10^6 per second per milliwatt, so pair intensities really are microwatts per cm². PINEM (photon-induced near-field electron microscopy) couples ultrafast light and electrons at nanostructures.

REALITY CHECK The regimes are laid out sensibly. The damage panel is the useful reminder: silicon nanorods absorb and conduct heat poorly, so average power, not just peak intensity, limits what a woodpile can carry.

Fig. 18 How to build a 3D microwave photonic-crystal cavity (2.4 GHz)

MAINSTREAM



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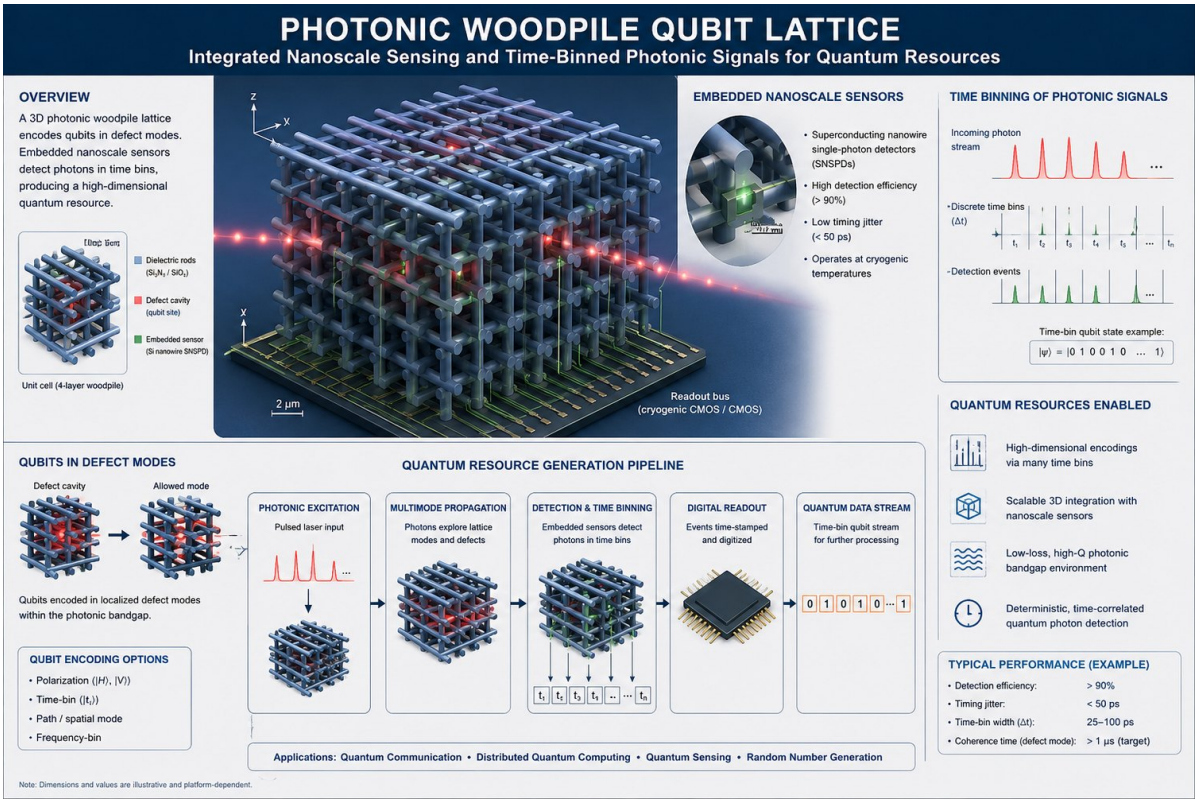
WHAT IT SHOWS Left: a build sheet for a four-layer woodpile of metal rods scaled to 2.4 GHz, with a parts and dimensions list, a layer-by-layer cross-section, a NanoVNA measurement set-up, an expected S21 transmission spectrum with a band gap, and a Python snippet. Right: a companion infographic that adds a pulse source, a coupling horn and “defect sites” that scatter energy into an output port.

THE SCIENCE Photonic-crystal physics is scale-invariant: a structure with lattice constant a has a band gap at a frequency f ∝ c/a, so a structure meant for 1550 nm light can be prototyped at microwave scale with rods a few millimetres wide (2.4 GHz has λ = 125 mm). A vector network analyser measures S21 (transmission) and reveals the gap as a dip of tens of dB. A defect that breaks the periodicity supports a localised cavity mode inside the gap.

REALITY CHECK The right-hand infographic labels energy flow as “output energy = η × input energy”. With η below 1 this is an ordinary passive filter: the crystal shapes and stores the field but cannot generate net energy.

Fig. 19 Photonic woodpile qubit lattice

MIXED



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WHAT IT SHOWS A 3D woodpile lattice with defect cavities that host qubit states, embedded nanoscale single-photon detectors, and a time-bin readout chain: photonic excitation → multimode propagation → detection and time-binning → digital readout → quantum data stream. Quoted targets: detection efficiency above 90 %, timing jitter about 50 ps, coherence above 1 µs.

THE SCIENCE Defect modes in a photonic band gap are the basis of solid-state cavity QED (Yablonovitch, 1987; Joannopoulos et al.). Superconducting nanowire single-photon detectors (SNSPDs) reach above 90 % efficiency with tens of ps jitter (Marsili et al., 2013), at cryogenic temperatures. Time-bin encoding is used in fibre-based quantum communication.

REALITY CHECK Building SNSPDs inside a 3D woodpile without spoiling its optical properties is an unsolved fabrication challenge, and a photon stored in a defect cavity lives for nanoseconds to microseconds, not longer. The diagram is a roadmap, not a demonstration.

Fig. 20 Optical reflection zones near a woodpile

MAINSTREAM

Area of Effect for Optical Reflection

A nanophotonic woodpile crystal in silicon or GaAs would reflect light within a very defined range:

Zone	Range	What Happens
Specular reflection	Any distance	Like a mirror — reflects at bandgap frequencies
Evanescent coupling	~100–500nm from surface	Photons tunnel into crystal then reflect back
Near field	~500nm–1.5μm	Partial reflection, interference effects
Far field	Beyond ~2μm	Clean reflected beam propagates normally

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WHAT IT SHOWS A four-row table for a silicon or GaAs woodpile: specular reflection (mirror-like within the band gap); evanescent coupling at roughly 100–500 nm from the surface; near field at 500 nm–1.5 μm; far field beyond about 2 μm.

THE SCIENCE Inside a photonic band gap, light cannot propagate in the crystal but its field decays exponentially over roughly one lattice period per layer. Another object placed within a fraction of a wavelength can still couple to that evanescent tail (the principle behind frustrated total internal reflection and near-field microscopy). The zone boundaries scale with the wavelength; for 1550 nm light the near-field region extends to about one wavelength.

REALITY CHECK The distances are approximate and depend on wavelength, polarisation and structure, so treat them as orders of magnitude.

Fig. 21 Slow light and single-photon-triggered gain

MIXED

Slow Light at the Band Edge

In free space light travels at c = 299,792,458 m/s. At the band edge of a nanophotonic crystal:

Condition	Speed	Reduction Factor
Free space	299,792,458 m/s	1×
In glass	~200,000,000 m/s	~1.5×
In silicon photonic crystal	~1,000,000 m/s	**300× slower**
At band edge optimized	~10,000 m/s	**30,000× slower**
Extreme engineered slow light	~100 m/s	**3,000,000× slower**

If you fired a single photon into a properly prepared slow light PhC cavity with population inversion:

```
Photon enters → slows to vg
↓
Spends ~1000× longer near gain atoms
↓
Stimulates emission from inverted atom
↓
Now 2 identical photons, both slow
↓
Each stimulates another
↓
Avalanche cascade
↓
Coherent burst exits crystal
```

This is called **cavity-enhanced superfluorescence** and has been observed experimentally.

The Genuinely Automatic Case

There is one scenario where lasing becomes essentially automatic upon photon entry — a **pre-pumped crystal already at inversion sitting just below threshold:**

- Crystal is pumped to 99% of lasing threshold
- Single photon enters matching cavity mode
- That one photon tips the system over threshold
- Full lasing cascade triggered by a single quantum

This is actually used in **single photon detection and amplification** research — the crystal acts like a hair trigger waiting to be released.

Files: HESUhxpaQAAeJDc.png, HESZ7t3aEAACPaZ.png

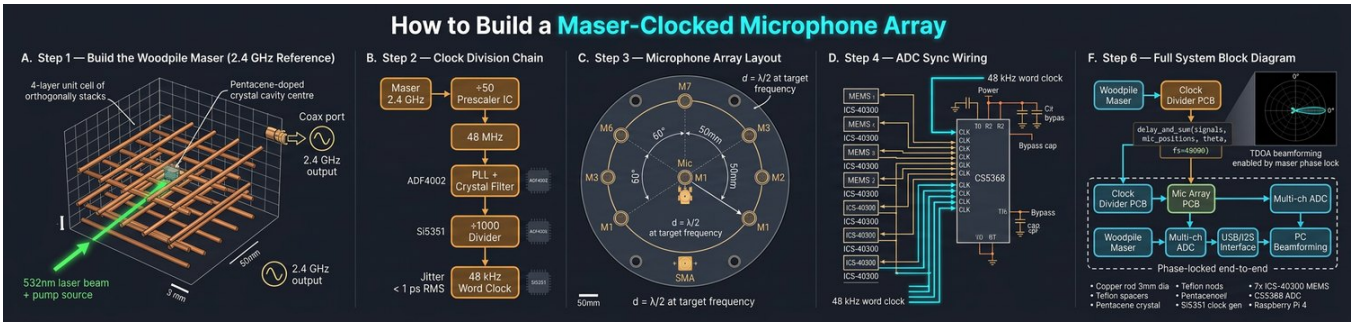
WHAT IT SHOWS Left: a speed table for light: 299,792,458 m/s in free space, ~2×10^8 m/s in glass (1.5× slower), ~10^6 m/s in a silicon photonic crystal (300×), ~10^4 m/s at an optimised band edge (30,000×), ~100 m/s for extreme slow light (3,000,000×). Right: a chain of events in an inverted slow-light crystal, ending in a coherent burst, plus a “genuinely automatic case” of a crystal pumped to 99 % of threshold.

THE SCIENCE Group velocity is c/n_g. Photonic-crystal waveguides slowed light by about 300× on a chip (Vlasov et al., Nature 2005), and cold atomic gases slowed it to 17 m/s (Hau et al., 1999). Slow light lengthens the time a photon spends near gain atoms, raising the gain per unit length in proportion to n_g. The chain on the right is Dicke superradiance or superfluorescence (1954), and a laser held just below threshold can be tipped into lasing by a single photon.

REALITY CHECK Slow-light structures pay for the slowdown with narrow bandwidth, high loss and sensitivity to disorder, so extreme entries in the table are laboratory records, not something a crystal does easily. A near-threshold amplifier is also triggered by its own spontaneous emission, so a single photon does not produce a clean, deterministic burst.

Fig. 22 Maser-clocked microphone array, and the easiest route to a maser

MIXED



Easiest route to working maser:

1. Build woodpile crystal from diagram (copper + Teflon)
2. Source pentacene-doped para-terphenyl crystal
3. Place crystal at woodpile defect cavity centre
4. Pump with 532nm green laser at ~500mW
5. Monitor 2.4 GHz output with NanoVNA/spectrum analyser
6. Adjust coupling iris size until oscillation threshold reached

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WHAT IT SHOWS Top: a six-step build guide: a copper-rod woodpile cavity with a pentacene-doped crystal at the defect (532 nm pump, 2.4 GHz output); a clock chain that divides 2.4 GHz by 50 to 48 MHz, phase-locks it, then divides by 1000 to a 48 kHz word clock with under 1 ps RMS jitter; a ring of seven MEMS microphones spaced λ/2 apart; ADC wiring; and delay-and-sum beamforming on a Raspberry Pi. Bottom: a text list, “Easiest route to working maser”, ending in adjusting an iris until oscillation threshold is reached.

THE SCIENCE Room-temperature masers are real. Oxborrow, Breeze and Alford (Nature 488, 353, 2012) made one from pentacene-doped p-terphenyl crystal in a sapphire ring resonator, at about 1.45 GHz, using optical pumping; Breeze et al. (Nature 555, 493, 2018) followed with a continuous-wave diamond maser. The clock chain is standard synthesiser practice: 2.4 GHz ÷ 50 = 48 MHz and ÷ 1000 = 48 kHz. Delay-and-sum beamforming steers a microphone array by time-shifting channels, and half-wavelength spacing avoids grating lobes.

REALITY CHECK The maser adds no benefit here. An audio ADC sampling at 48 kHz needs a word clock with jitter of well under a nanosecond, which a few-dollar crystal oscillator already provides; a shared clock chip already synchronises all channels, and microphone noise, not clock phase noise, limits beamforming accuracy. The maser also has not been shown to work at 2.4 GHz with a 532 nm pump in this geometry, so the “easiest route” is an untested recipe.

CHAPTER 5

Non-Hermitian optics, phonons and magnons

Figures 23–29 · 7 entries

Fig. 23 PT-symmetric exceptional-point laser and power-scaling ladder

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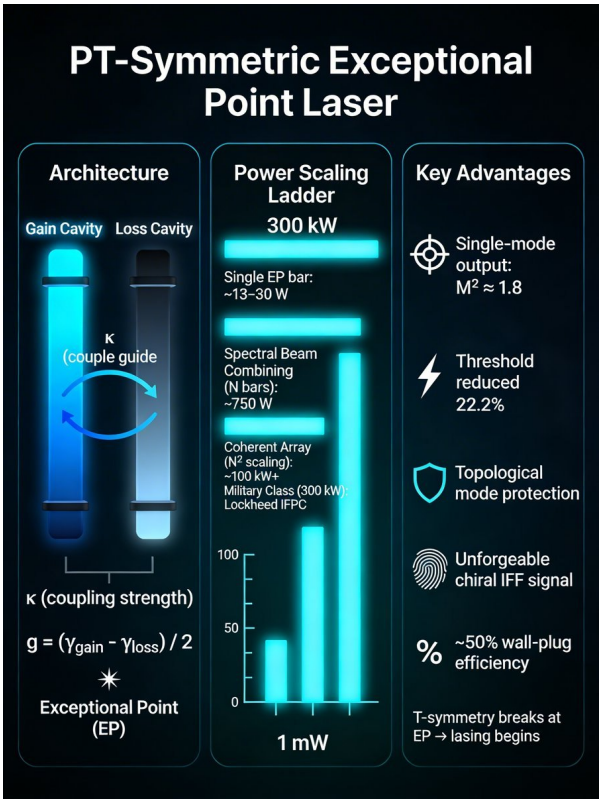


Files: 2029503925859361136-HCo-TGGagAAtRqn.jpg

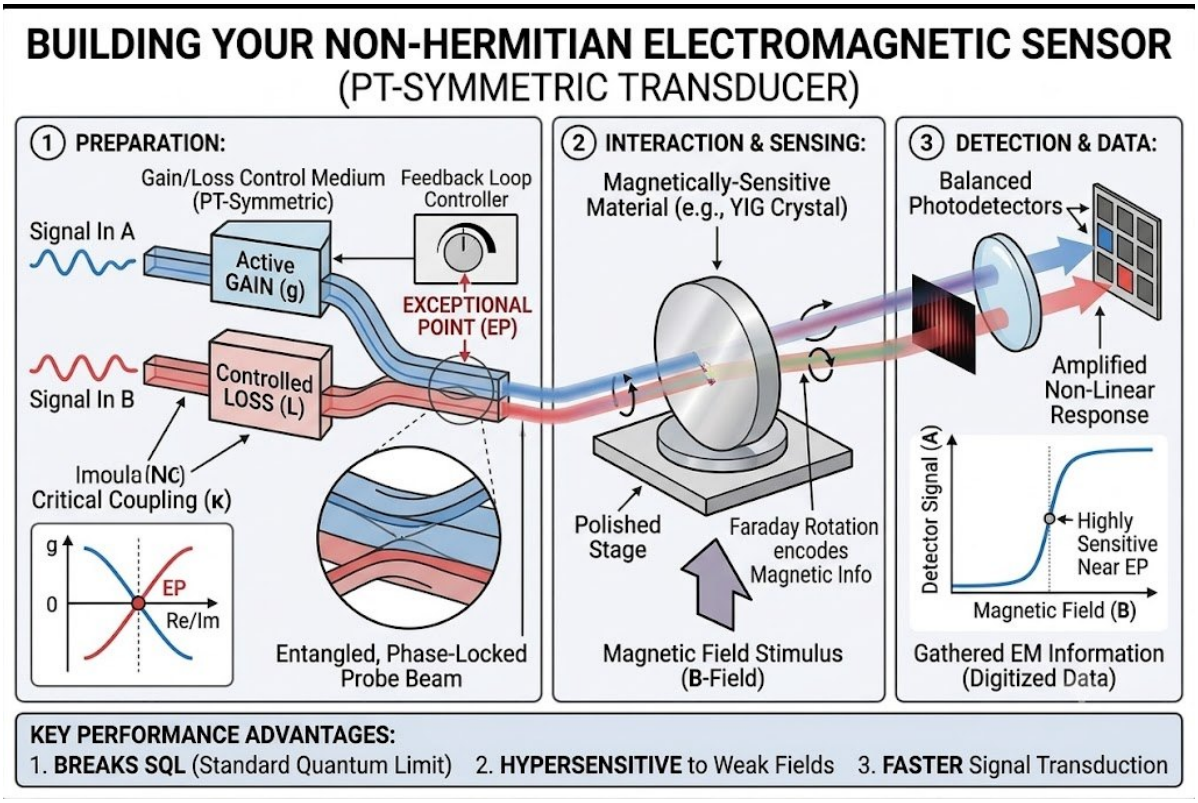
WHAT IT SHOWS An infographic of a gain cavity and a loss cavity coupled with strength κ, the equation $g = (\gamma_{gain} - \gamma_{loss})/2$, a bar chart of “power scaling” from 1 mW to about 300 kW, and advantages: $M^2 \approx 1.8$ single-mode output, 22.2 % lower threshold, topological protection, an “unforgeable chiral IFF signal”, and about 50 % wall-plug efficiency.

THE SCIENCE Parity–time (PT) symmetric lasers are real. Two coupled resonators, one with gain and one with equal loss, have supermodes $\omega = \pm\sqrt{(\kappa^2 - g^2)}$ that merge at an exceptional point (EP) where $\kappa = g$. Feng et al. (Science 2014) and Hodaei et al. (Science 2014) used this to force single-mode lasing in microrings.

REALITY CHECK The power ladder is the weak part. Coherently combining N lasers multiplies power by N, and on-axis intensity by N^2, so “N^2 scaling” of output power is wrong. Fibre lasers at 100 kW class exist for other reasons, and ~40–50 % wall-plug efficiency is typical of diode-pumped fibre lasers, not a PT-symmetry benefit. The threshold reduction, the IFF claim and the topological protection are unsourced.

Fig. 24 Building a non-Hermitian electromagnetic sensor (PT-symmetric transducer)

MIXED



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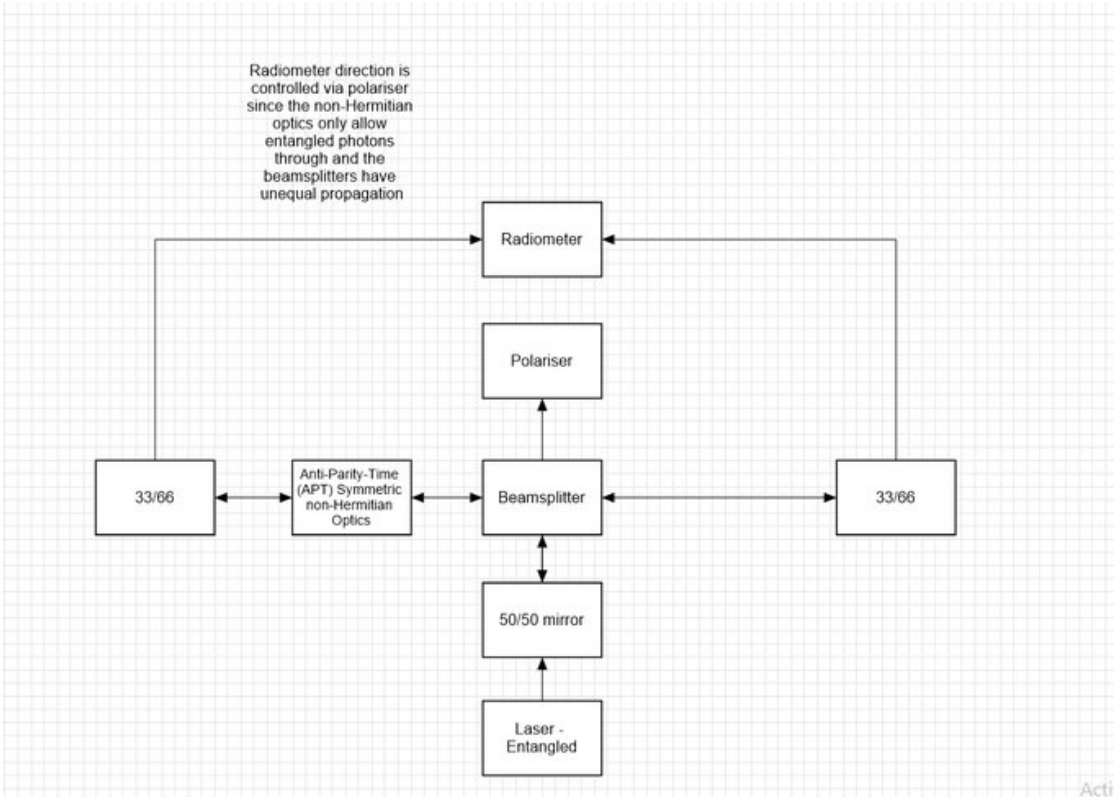
WHAT IT SHOWS A three-step illustration: a gain/loss control medium tuned to an exceptional point, a magnetically sensitive YIG crystal that rotates polarisation by the Faraday effect, and balanced photodetectors reading an amplified response; claims to break the standard quantum limit, be hypersensitive, and be faster.

THE SCIENCE Near an exceptional point the eigenvalue splitting goes as the square root of a small perturbation ε, a much steeper response than the linear dependence of a normal resonator (Hodaei et al., Nature 2017; Chen et al., Nature 2017). YIG (yttrium iron garnet) is a standard low-loss magnetic material with a large Faraday rotation. Balanced photodetection cancels common-mode laser noise.

REALITY CHECK A steeper response is not automatically a better sensor. Follow-up analyses (Langbein 2018; Lau and Clerk 2018; Mortensen et al. 2018) found that the extra noise near an EP, including the Petermann factor, often cancels the gain in signal, so “breaks the SQL” is not established.

Fig. 25 Anti-PT-symmetric radiometer controlled by a polariser

SPECULATIVE



Files: HE0gtSRbcAA5Kim.png

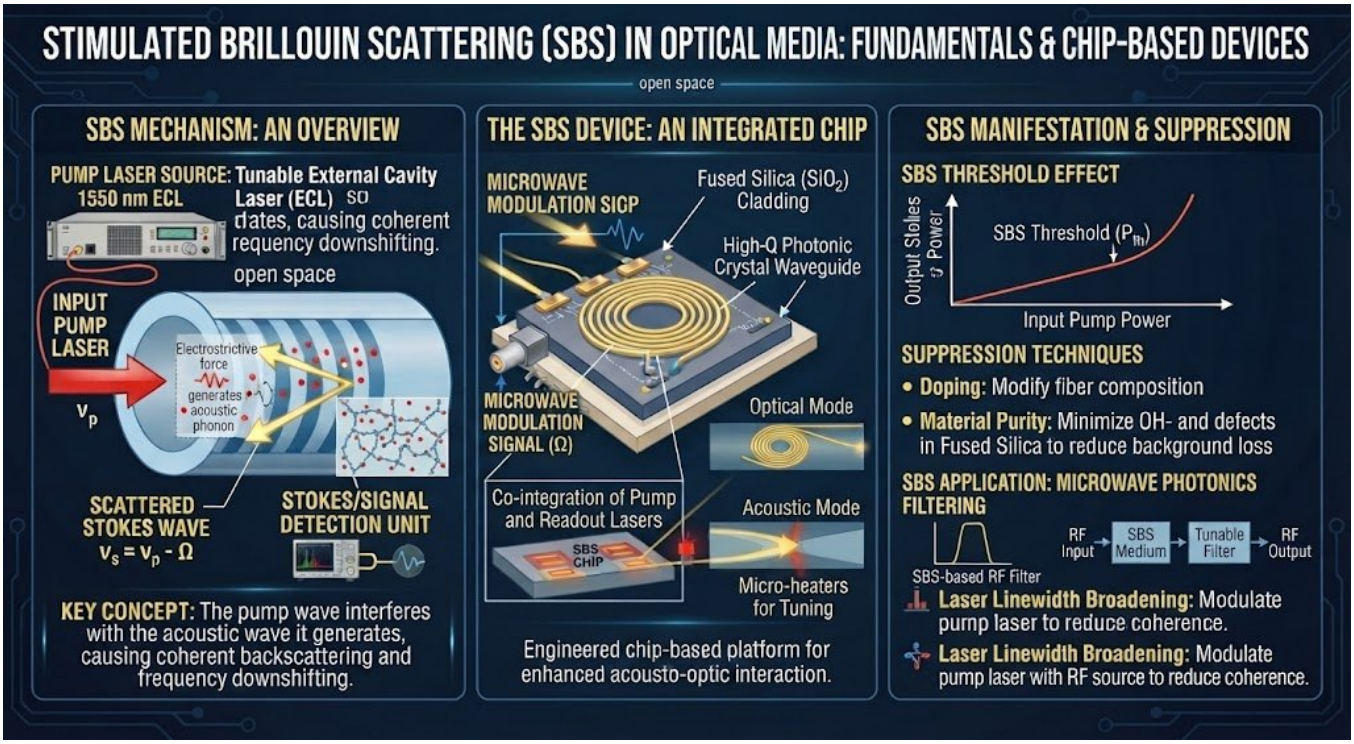
WHAT IT SHOWS Entangled laser → 50/50 mirror → beam splitter → radiometer, with a polariser branch, “33/66” elements and an “anti-parity-time (APT) symmetric non-Hermitian optics” block. Caption: radiometer direction is controlled by the polariser because the non-Hermitian optics only pass entangled photons and the beam splitters propagate unequally.

THE SCIENCE Anti-PT symmetry has been demonstrated (Peng et al., Nature Physics 2016, in coupled atomic vapour cells). A Crookes radiometer (1873) spins because gas at the edges of the vanes flows from the hot side to the cold side (thermal transpiration); the direction is set by which face absorbs more light and, in a good vacuum, only the tiny radiation pressure P/c remains.

REALITY CHECK A polariser changes how much light is absorbed, which can change torque size, but the radiometer’s sense of rotation is set by black versus reflective vane faces. The role of “entangled” photons in the drive is not physical: a radiometer responds to absorbed energy, not to entanglement.

Fig. 26 Stimulated Brillouin scattering in optical media and on a chip

MAINSTREAM



Files: f8401e0c-4d4f-4715-bb25-391965615bb8.jpg

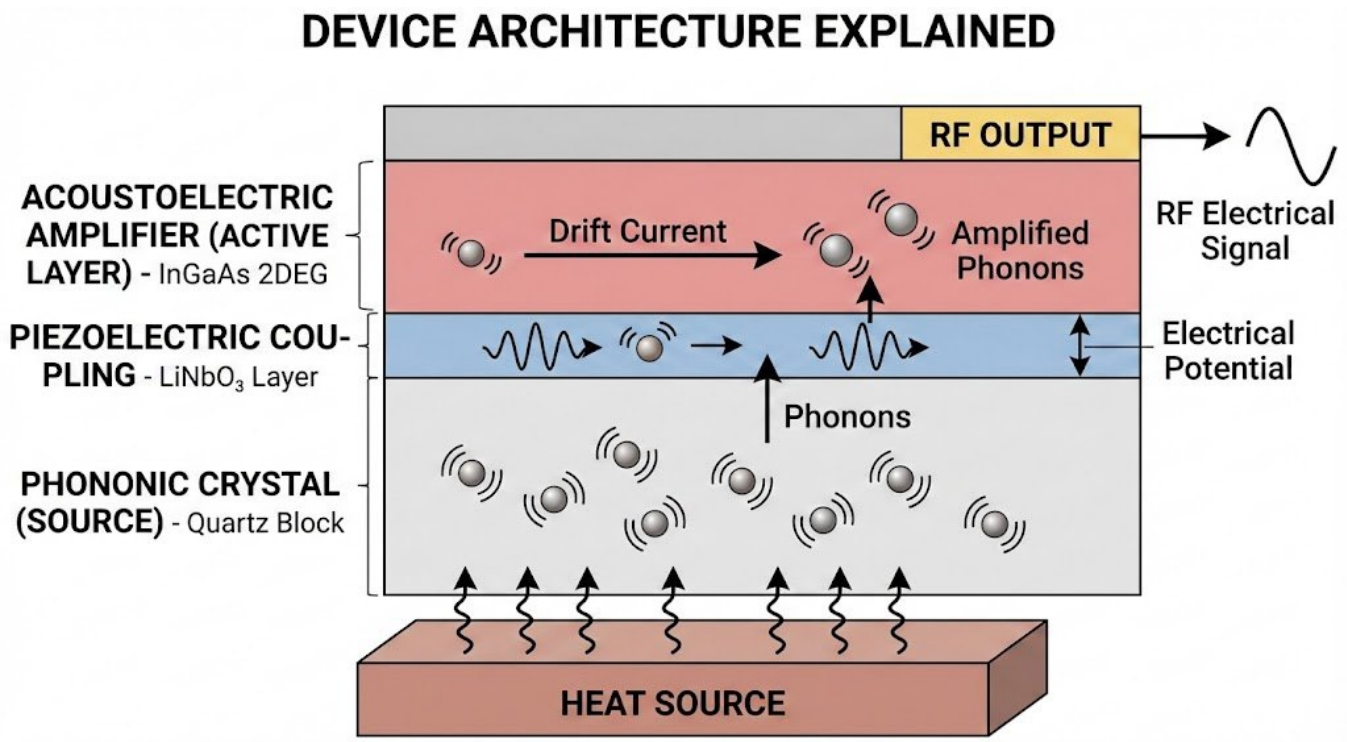
WHAT IT SHOWS A three-panel infographic: the SBS mechanism (a 1550 nm pump creates an acoustic wave by electrostriction, and a Stokes wave is scattered backwards at a lower frequency); an integrated chip with silica cladding, a photonic-crystal waveguide and microwave modulation; and the SBS threshold, suppression techniques (doping, linewidth broadening) and use in microwave-photonic filters.

THE SCIENCE This one is standard, correct physics. The pump (ω_p) scatters off a travelling acoustic phonon (Ω_B) to give a Stokes wave $\omega_s = \omega_p - \Omega_B$. In silica fibre at 1550 nm the shift is about 10.8 GHz and the gain bandwidth 30–50 MHz. The threshold is set by pump power, effective area and length (Smith, 1972). Chip-scale Brillouin lasers with sub-hertz linewidth have been demonstrated (Gundavarapu et al., Nature Photonics 2019).

REALITY CHECK Nothing to correct: this panel is a clean summary.

Fig. 27 Acoustoelectric amplifier: heat → phonons → RF

MIXED



Files: 1994079180154139062-G6xkcSjbQAAWXma.jpg

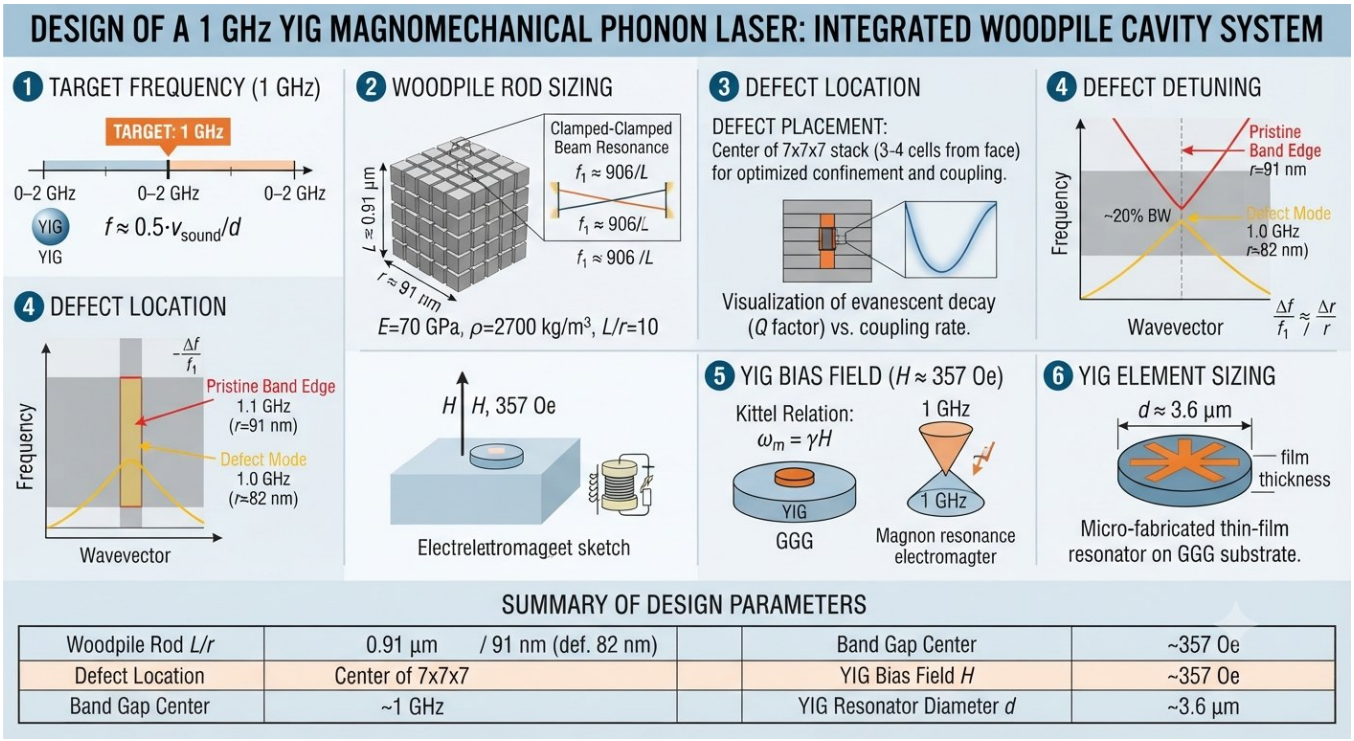
WHAT IT SHOWS A layered device: a heat source under a quartz phononic-crystal block, a $LiNbO_3$ piezoelectric coupling layer and an InGaAs two-dimensional electron gas layer where drift current amplifies phonons, with RF output at the top.

THE SCIENCE Acoustoelectric amplification is real: when carriers drift faster than the sound speed, they can transfer energy to an acoustic wave and amplify it (Hutson, McFee and White, 1961, in CdS). Surface acoustic waves in $LiNbO_3$ couple to a 2DEG in GaAs and InGaAs heterostructures.

REALITY CHECK The amplifier's gain comes from the DC bias that drives the drifting carriers, not from the heat. A single-temperature heat source cannot supply net work: thermal (Johnson–Nyquist) noise power is only $k_B T\Delta f$, about 4×10^{-21} W per hertz at 300 K, and the second law needs a colder sink to draw usable power.

Fig. 28 1 GHz YIG magnetomechanical phonon laser in a woodpile cavity

MIXED



SUMMARY OF DESIGN PARAMETERS

Woodpile Rod *L/r*	0.91 µm / 91 nm (def. 82 nm)		Band Gap Center	~357 Oe
Defect Location	Center of 7x7x7		YIG Bias Field *H*	~357 Oe
Band Gap Center	~1 GHz		YIG Resonator Diameter *d*	~3.6 µm

Files: u92nggu92nggu92n.png

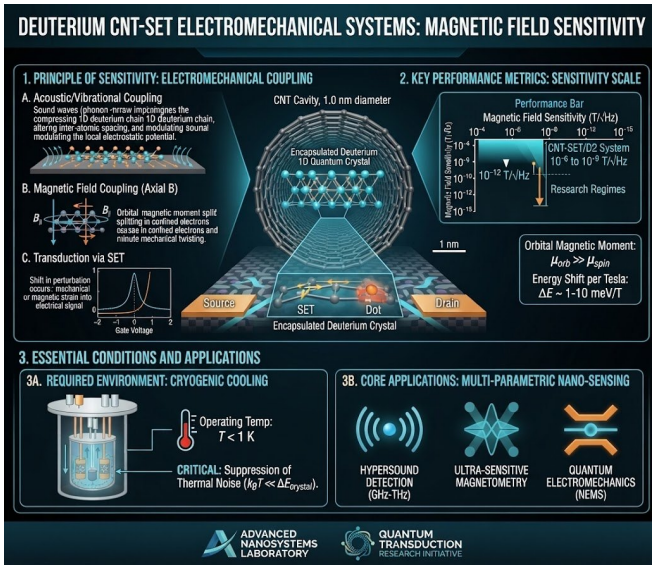
WHAT IT SHOWS A six-step design sheet for a 1 GHz phonon laser: target frequency, woodpile rod sizing (rods 0.91 µm long, L/r = 10, E = 70 GPa, ρ = 2700 kg/m³), defect placement in a 7×7×7 stack, band-edge detuning, a YIG bias field of about 357 Oe by the Kittel relation $\omega_m = \gamma H$, and a 3.6 µm YIG disc on GGG.

THE SCIENCE The magnetic step checks out: the Kittel relation gives f = γH with γ/2π = 2.8 MHz/Oe, so 1 GHz needs H = 1000 / 2.8 = 357 Oe (about 36 mT). The material numbers (E = 70 GPa, ρ = 2700 kg/m³) are aluminium, with a bar sound speed of √(E/ρ) ≈ 5.1 km/s. Magnon–phonon coupling in YIG is well established (magnomechanics, Zhang et al., Science Advances 2016), and phonon lasers have been made in optomechanical systems (Grudinin et al., PRL 2010).

REALITY CHECK This is a design sketch with no dissipation budget: reaching lasing needs the magnon–phonon gain to exceed phonon loss in a micrometre-scale structure, which has not been shown for this geometry.

Fig. 29 Deuterium in a 1 nm nanotube: phonon and magnetic-field sensing

MIXED



Yes, deuterium inside a 1.0 nm carbon nanotube can act as a detector for **high-frequency sound waves (phonons)** and mechanical vibrations, functioning essentially as an ultra-sensitive nanoscale microphone.

Rather than detecting audible sound through the air like a standard microphone, it detects mechanical pressure waves through the following mechanism:

1. Acoustic Phonon Coupling

- **Mechanical Strain:** Sound waves are propagating mechanical vibrations (phonons) through a medium. When an acoustic wave travels through the substrate or the carbon nanotube itself, it rhythmically compresses and expands the 1.0 nm channel.
- **Perturbing the Quantum Crystal:** Because the deuterium is tightly confined in a 1D quantum crystal, even sub-angstrom mechanical deformations of the nanotube wall alter the potential well spacing and disrupt the periodic density peaks.

2. Transduction via the SET

- **Position-to-Charge Modulation:** As the acoustic wave compresses the channel, the deuterium wave packets shift their spatial coordinates.
- **Electrical Conversion:** The adjacent Single-Electron Transistor (SET) detects this mechanical compression as a direct change in local electrostatic potential, converting the high-frequency acoustic wave (phonon) into a readable electrical current variation.

This setup is typically explored in **nano-electromechanics** for detecting ultra-high-frequency (GHz to THz) hypersound or thermal phonons at cryogenic temperatures.

Files: bca02b7d-ff34-4ac2-9496-88d53109a985.jpg, HSVpwfrboAAOUOo.png

WHAT IT SHOWS Left: an infographic for a 1 nm carbon-nanotube cavity holding a 1D deuterium quantum crystal read out by a single-electron transistor (SET); acoustic coupling, orbital magnetic coupling, cryogenic operation below 1 K, and quoted magnetic sensitivity of 10^{-6}–10^{-9} T/√Hz. Right: a text explanation of how the acoustic wave shifts the deuterium density peaks and the SET converts that to current.

THE SCIENCE SETs measure charge to about 10^{-5} e/√Hz and have detected the motion of nanomechanical resonators (Knobel and Cleland, Nature 2003). Hydrogen isotopes adsorbed in nanotubes form 1D chains at low temperature and behave as quantum solids. Nanotube orbital magnetic moments are large, about 20 Bohr magnetons or 1 meV/T (Minot et al., Nature 2004).

REALITY CHECK The energy shift of 1–10 meV/T on the panel is at the top of what nanotubes show (the high end is around 170 Bohr magnetons). The quoted 1 μT–1 nT/√Hz sensitivity is far behind SQUID magnetometers (about 1 fT/√Hz), so the device would be a curiosity, not a competitive sensor. All of it needs temperatures below 1 K.

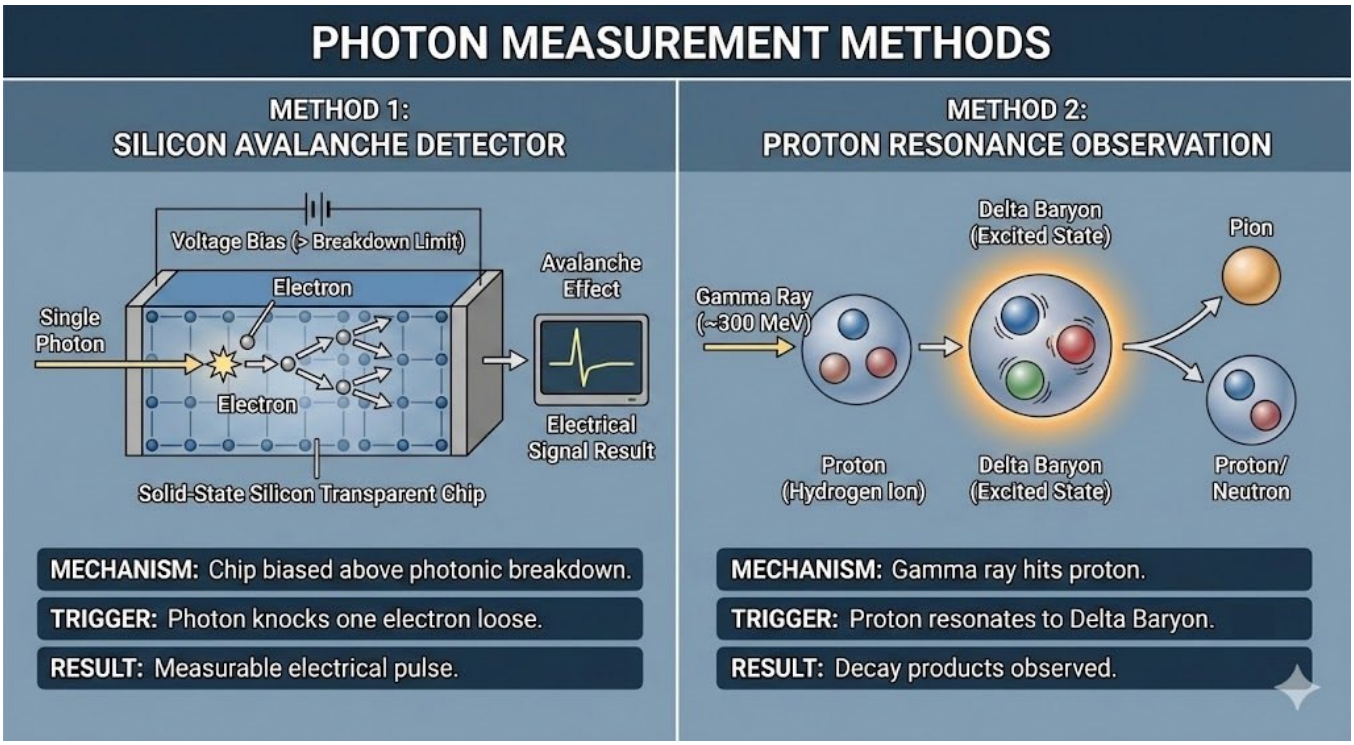
CHAPTER 6

Nuclear and particle physics

Figures 30–34 · 5 entries

Fig. 30 Photon measurement methods: silicon avalanche detector and Δ-baryon resonance

MAINSTREAM



Files: 2002165320807821714-G8keL23bQAAVmog.jpg · Also in repository: 2002352317262426426-G8nlySxbMAEDw..., 2032911925152198779-HDZanhwacAABF...

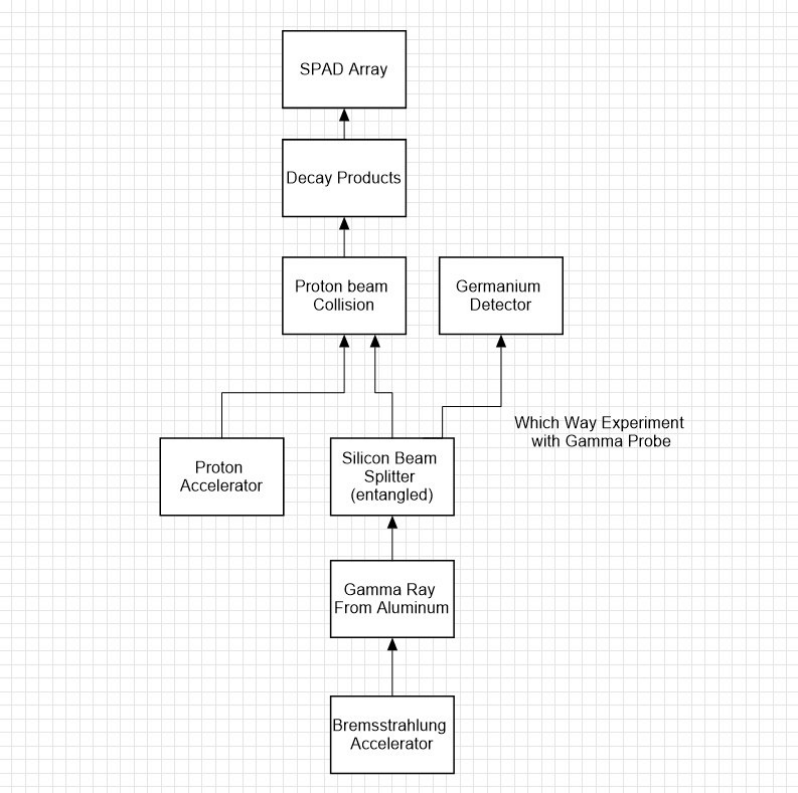
WHAT IT SHOWS Two side-by-side methods. Method 1: a solid-state silicon chip biased above breakdown, where one photon frees one electron that triggers an avalanche and a measurable pulse. Method 2: a ~300 MeV gamma ray excites a proton to a Δ baryon, which decays to a pion plus a proton or neutron. (Images 50 and 69 are copies.)

THE SCIENCE Method 1 is a single-photon avalanche diode (SPAD) operated in Geiger mode: above the breakdown voltage, one carrier triggers a self-sustaining avalanche, a quench circuit resets it, and timing jitter is tens of picoseconds. Method 2 is real hadron physics: photons of about 320–340 MeV excite the Δ(1232) resonance, which lives 5.6×10^{-24} s and decays to Nπ. The threshold photon energy is $(M_\Delta^2 - m_p^2)/(2m_p) \approx 340$ MeV.

REALITY CHECK Only method 1 measures an unknown photon in the ordinary sense. Method 2 needs a tagged bremsstrahlung beam from an electron accelerator (as at MAMI or ELSA) and it detects the hadronic decay products, destroying the photon.

Fig. 31 Which-way experiment with a gamma probe

MIXED



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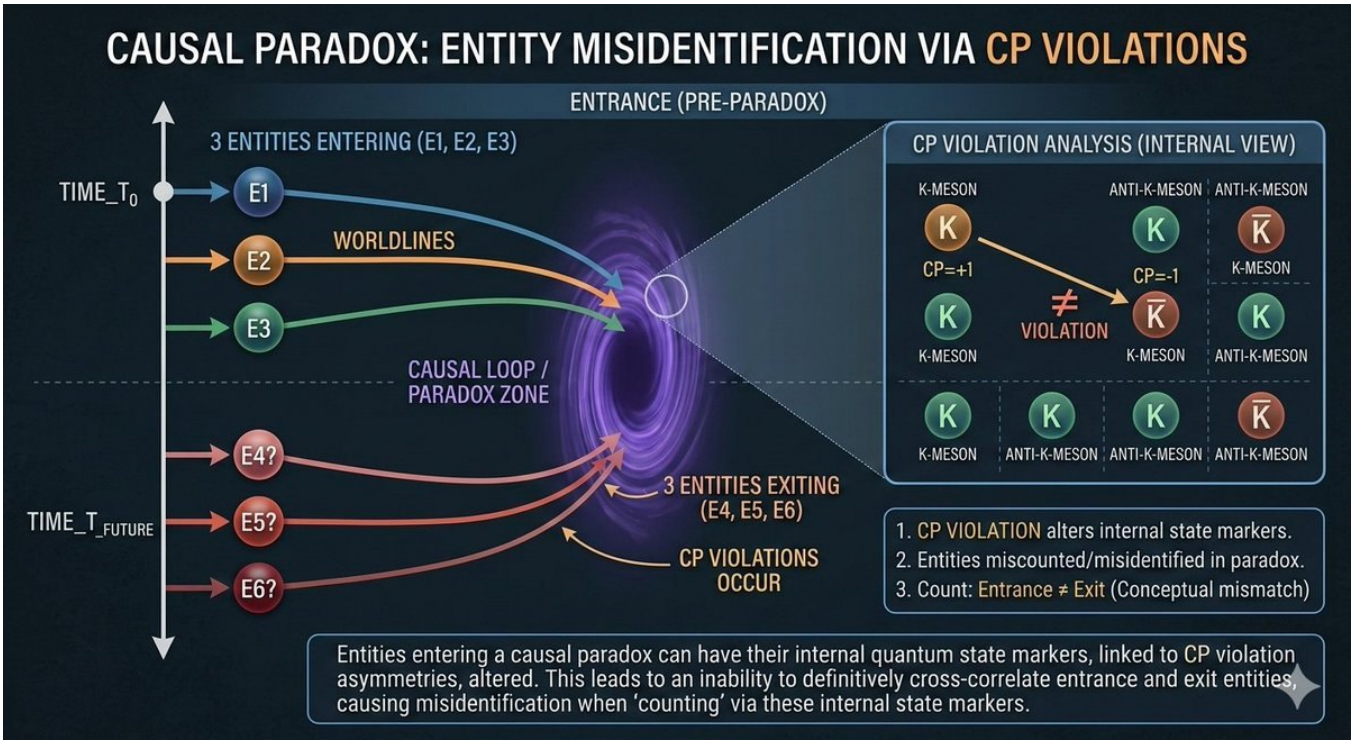
WHAT IT SHOWS A flowchart: bremsstrahlung accelerator → gamma ray from aluminium → silicon beam splitter (entangled) → proton beam collision (proton accelerator) → decay products → SPAD array, with a germanium detector on a side branch, titled “Which Way Experiment with Gamma Probe”.

THE SCIENCE A which-way experiment asks through which arm a particle went; recording that information destroys interference. Perfect silicon crystals do act as beam splitters for X-rays and neutrons through Laue diffraction (Bonse–Hart interferometers, neutron interferometry by Rauch et al., 1974). Germanium detectors are the standard for gamma spectroscopy, with energy resolution of ~2 keV at 1.3 MeV. Aluminium-27 has a well-known proton-capture line producing gammas near 1.8 MeV.

REALITY CHECK The chain mixes an interferometer, a nuclear reaction and a hadron detector with no explanation of what is meant to interfere or be measured. A coherent gamma interferometer would need crystal perfection to a fraction of an atomic spacing, and the proton collision step has no stated role.

Fig. 32 Causal paradox: entity misidentification via CP violation

SPECULATIVE



Files: 2027174968140935611-HCH425pbsAAzqyK.jpg

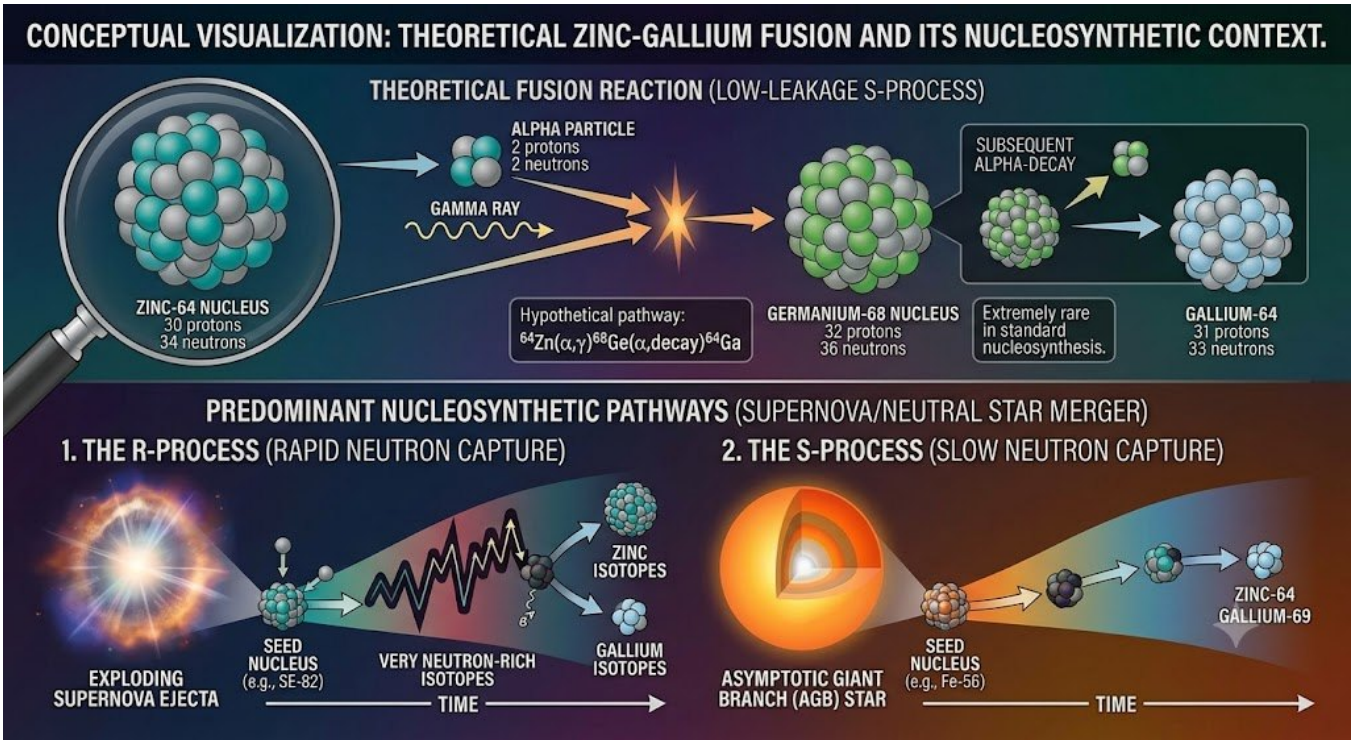
WHAT IT SHOWS A dark infographic: three entities (E1–E3) enter a closed timelike curve and three (E4–E6) exit, with a side panel on neutral K-meson CP violation suggesting that altered internal "markers" make the entrance and exit counts disagree.

THE SCIENCE CP violation in neutral kaons is real and Nobel-winning: Cronin and Fitch (1964) saw the long-lived K_L, which is mostly CP-odd, decay to two pions (CP-even) about 0.2 % of the time. By the CPT theorem, CP violation implies T violation, which was measured directly by CPLEAR (1998) and BaBar (2012). Closed timelike curves appear as solutions of general relativity, and the Novikov self-consistency principle says any events on them must be self-consistent.

REALITY CHECK The step from CP violation to a counting mismatch across a time loop is not physics: CP violation changes decay rates of kaons, not the identity or number of particles going around a loop, and no theory links the two.

Fig. 33 Zinc–gallium fusion versus the r- and s-processes

MIXED



Files: 628e0d17-1ee5-40b6-b814-838e9332acbf.jpg

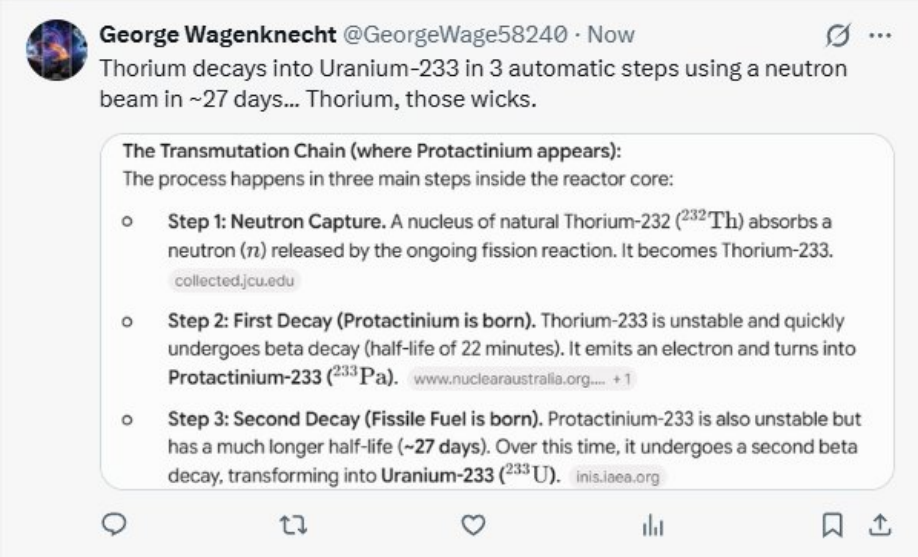
WHAT IT SHOWS An infographic that sketches a hypothetical alpha-capture chain: zinc-64 (30 protons, 34 neutrons) + α → germanium-68, followed by an “alpha decay” to gallium-64, and contrasts it with the r-process (supernova and neutron-star merger) and s-process (AGB stars) routes to zinc and gallium isotopes.

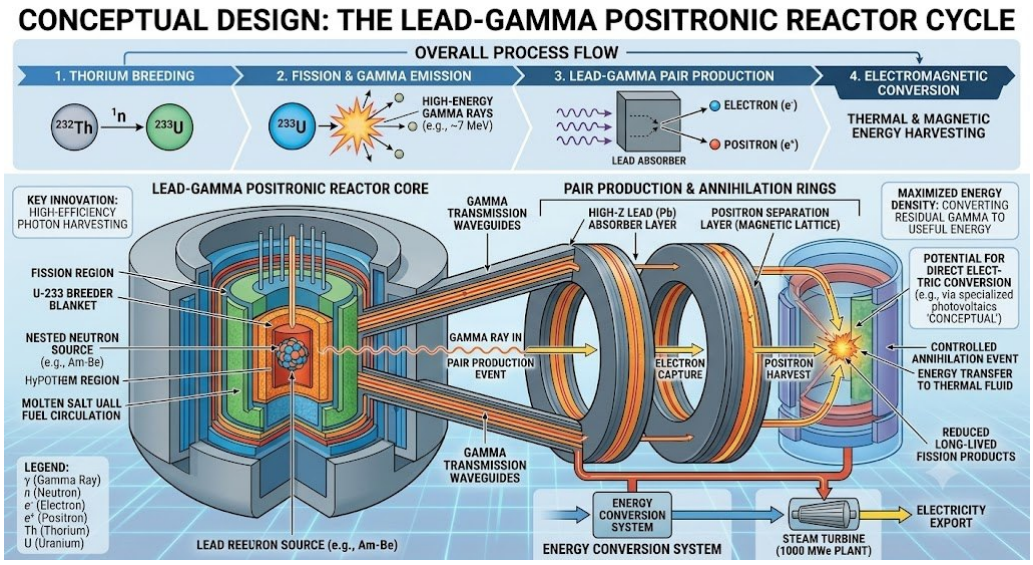
THE SCIENCE The r-process (rapid neutron capture) and s-process (slow neutron capture) are the two main ways to build nuclei heavier than iron; neutron-star mergers were confirmed as an r-process site by the 2017 kilonova. Zinc-64 is stable and is mainly made in explosive nucleosynthesis. The reaction ^{64}Zn(α,γ)^{68}Ge is allowed.

REALITY CHECK The chain’s last step is wrong. Germanium-68 (half-life 271 days) decays by electron capture to gallium-68, which then decays by β^+ to stable zinc-68. Alpha decay of Ge-68 would give zinc-64, not gallium-64, and alpha decay of Ge-68 is not observed.

Fig. 34 Thorium fuel cycle and the “lead-gamma positronic reactor”

MIXED





Files: c5b58dad-e11e-4fa0-9894-dad90a8a8f49.png, 85d8486d-1368-41c5-974c-506910bb7379.jpg

WHAT IT SHOWS Left: a post summarising the thorium decay chain (neutron capture on Th-232 gives Th-233, which decays in about 22 minutes to Pa-233, which decays in about 27 days to U-233). Right: an infographic, “Conceptual design: the lead-gamma positronic reactor cycle”, that routes fission gammas through lead pair-production layers to positron/electron annihilation and a 1000 MWe steam turbine.

THE SCIENCE The chain on the left is textbook: $^{232}Th + n \rightarrow {}^{233}Th \rightarrow (\beta^-, 22\ min)\ {}^{233}Pa \rightarrow (\beta^-, 27\ d)\ {}^{233}U$, the basis of the thorium fuel cycle studied in India’s three-stage programme and the Oak Ridge molten-salt experiment. On the right, gamma rays with energy above 1.022 MeV can convert to electron–positron pairs in a nucleus’s field, with a cross-section that grows roughly as Z^2, so lead is the standard absorber; a positron annihilates with an electron to give two 511 keV photons.

REALITY CHECK The infographic does not identify an energy path that beats an ordinary heat cycle. Prompt fission gammas are only about 7 MeV of the roughly 200 MeV released per fission, pair production followed by annihilation just turns gamma energy into gamma energy again, and everything that is absorbed ends up as heat in the lead. Designing or building a reactor is outside this handbook.

CHAPTER 7

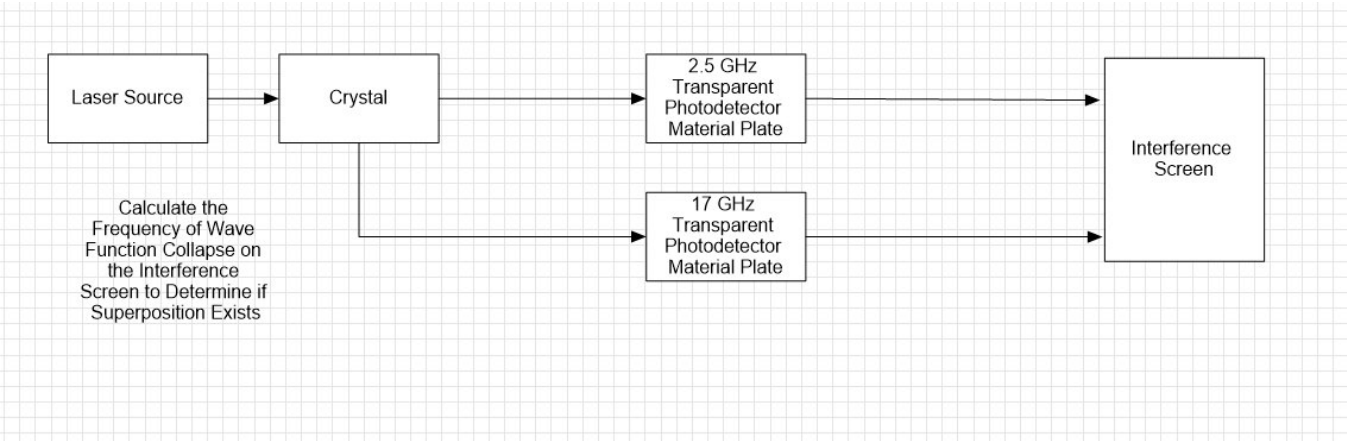
Entanglement, interference and “retrocausal” set-ups

Figures 35–51 · 17 entries

Background for this whole chapter. Entanglement is real and heavily tested: measurements on two entangled particles can be correlated more strongly than any classical theory allows (Bell tests, Nobel Prize 2022). But the no-signalling theorem guarantees that what you do to one half does not change the statistics of the other half; the correlation only appears when the two sets of records are brought together. Also, “entangled laser” is not a standard component: laser beams are classical-like coherent states, and entangled photon pairs are usually made by spontaneous parametric down-conversion (SPDC) with an efficiency around one pair per million pump photons.

Fig. 35 Two photodetector plates at 2.5 GHz and 17 GHz: does “wavefunction collapse” have a frequency?

SPECULATIVE



Files: 1979813599460930005-G3m19YZWcAA7e0B.png · Also in repository: 1979814380842934643-G3m2oTPXEAAJK...

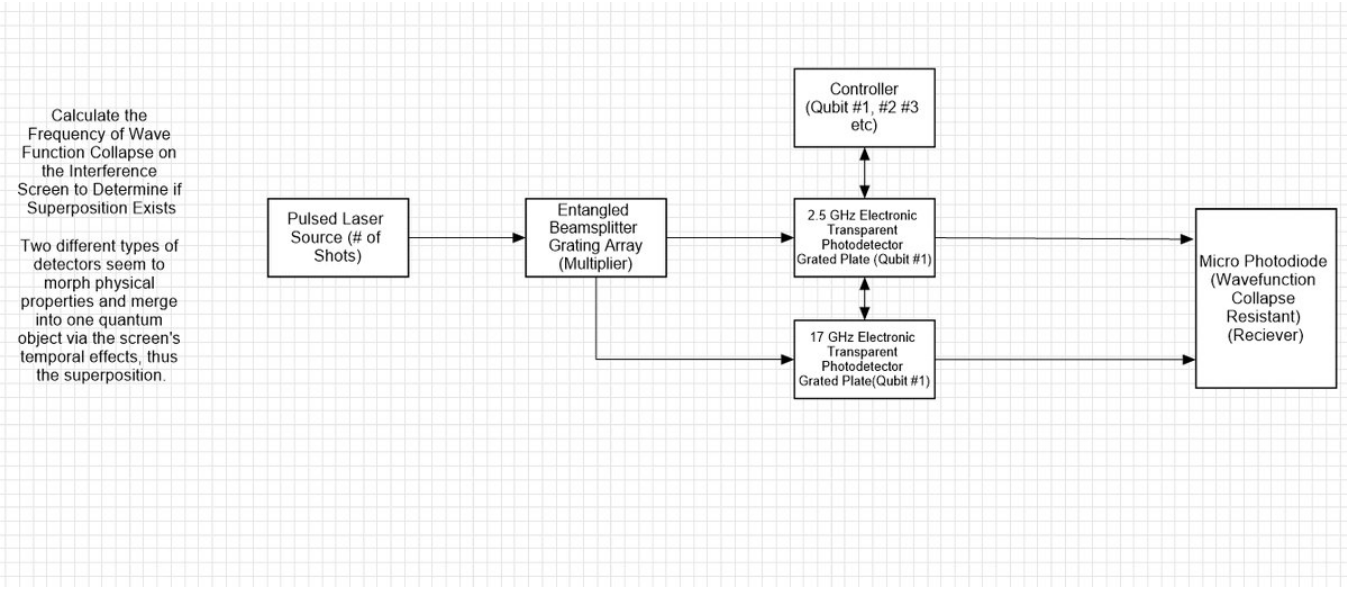
WHAT IT SHOWS Laser → crystal → two branches, one through a 2.5 GHz transparent photodetector plate and one through a 17 GHz plate, both ending at an interference screen. Note: “Calculate the frequency of wave function collapse on the interference screen to determine if superposition exists.” (Image 23 is a duplicate.)

THE SCIENCE Fast photodiodes with 2.5 GHz and 17 GHz bandwidth are ordinary, and transparent graphene photodetectors with tens of GHz bandwidth have been built (Xia et al., Nature Nanotech. 2009). In a double-slit or two-path experiment, superposition is tested by the visibility of interference fringes, $V = (I_{max} - I_{min})/(I_{max} + I_{min})$; which-path information reduces V.

REALITY CHECK A detector bandwidth sets how quickly it can register events; it is not a “collapse frequency”, and collapse is not a periodic event that can be counted. A partially absorbing plate in one arm would act as a which-path marker and simply reduce fringe visibility.

Fig. 36 Pulsed laser, grating array and qubit controller

SPECULATIVE



Files: 1979824073401307641-G3m_fz_WwAAThgz.png

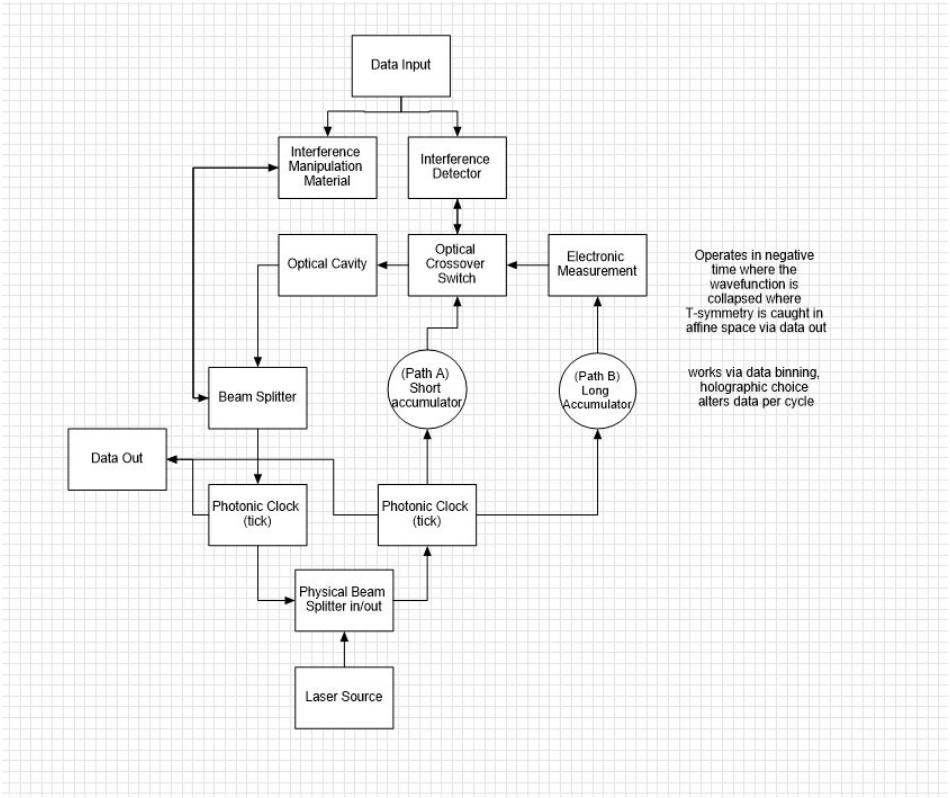
WHAT IT SHOWS A refined version of the previous flowchart: pulsed laser → entangled beam-splitter grating array (a multiplier) → two electronic transparent photodetector grated plates at 2.5 and 17 GHz, each labelled qubit #1, plus a controller for qubits 1–3, all feeding a micro-photodiode (“wavefunction-collapse-resistant receiver”). Text: the two types of detectors “morph physical properties and merge into one quantum object via the screen’s temporal effects”.

THE SCIENCE Time-resolved detection of pulsed light is standard practice (time-correlated single-photon counting). Grating arrays can split a beam into many copies, and a photodiode can be placed behind partially transparent detectors.

REALITY CHECK Detectors in series remove light and add noise; nothing in optics makes two detectors “merge” into a single quantum object. Whether superposition survives depends on whether the detector reveals which path the photon took.

Fig. 37 Photonic computer with short and long accumulators (“negative time”)

SPECULATIVE



Files: 1984589343802155251-G4qtfa8bQAINq_v.png

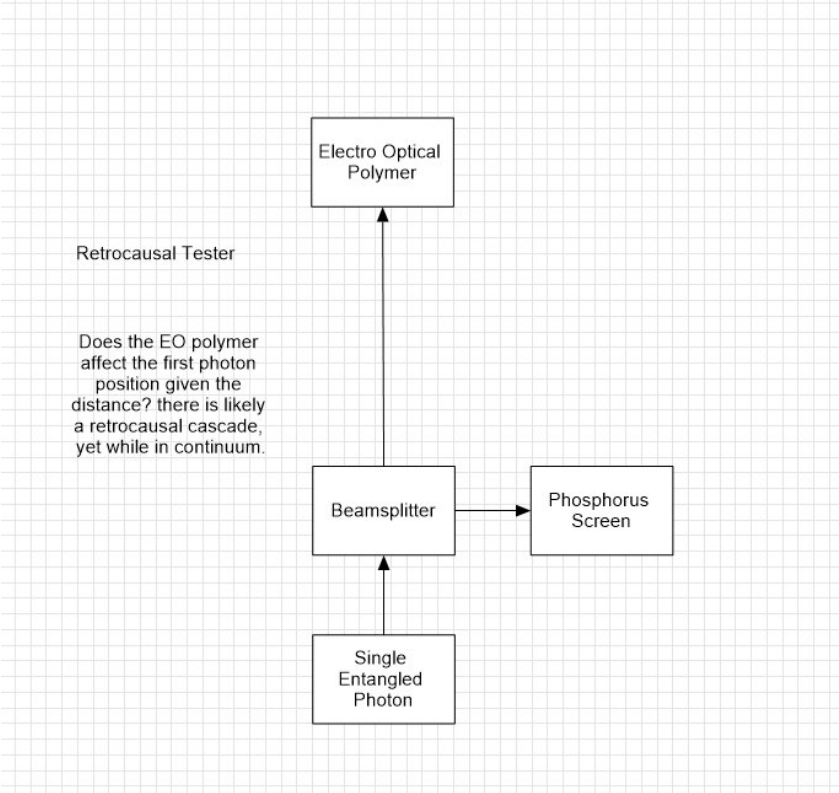
WHAT IT SHOWS A large block diagram: laser source, beam splitters, two photonic clocks (“tick”), a short accumulator (path A) and a long accumulator (path B), an optical crossover switch, an interference detector, electronic measurement, an optical cavity, and data in/out. Notes: it operates “in negative time where the wavefunction is collapsed where T-symmetry is caught in affine space via data out”, and works via data binning with a “holographic choice” per cycle.

THE SCIENCE The parts are real. Two paths of unequal length (short and long delay lines) are the basis of time-bin interferometers (Franson, 1989) and delay-line reservoir computers; optical switches route pulses, and time-tagging electronics bin arrival times.

REALITY CHECK “Negative time” and “affine space” have no defined meaning in the diagram; running an interferometer with a time-reversed clock does not make the wavefunction collapse earlier. The useful reading is a delay-line interferometer with time-binned readout.

Fig. 38 Retrocausal tester with an electro-optic polymer

SPECULATIVE



Files: 1985093643733189010-G4x4JlOboAAcrPs.png

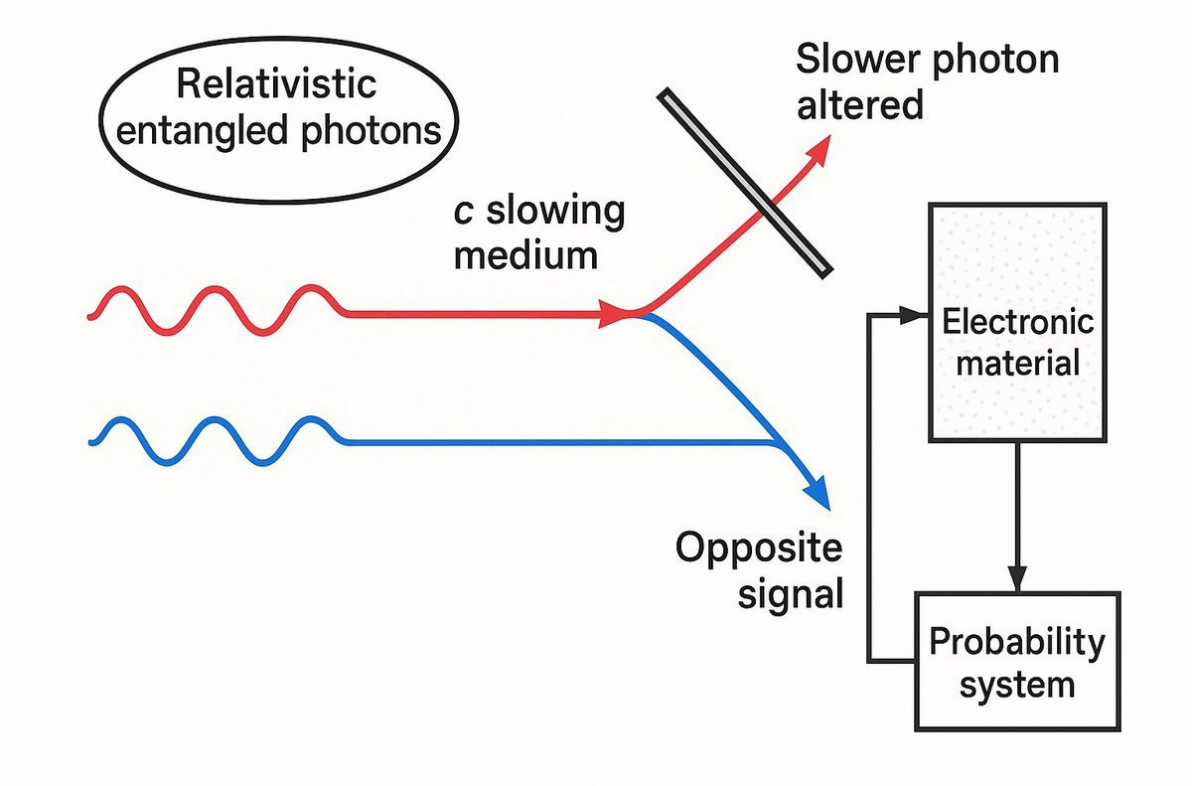
WHAT IT SHOWS A single entangled photon enters a beam splitter; one branch goes to a phosphor screen and the other to an electro-optic (EO) polymer. Note: “Does the EO polymer affect the first photon position given the distance? There is likely a retrocausal cascade, yet while in continuum.”

THE SCIENCE This is a delayed-choice or quantum-eraser layout (Wheeler, 1978; Scully and Drühl, 1982; Kim et al., 2000). EO polymers change their refractive index with voltage (Pockels effect, r_{33} of order 100 pm/V) and are used in fast optical modulators.

REALITY CHECK In real quantum-eraser experiments the pattern on the first screen never changes when you act on the second photon; interference appears only after sorting the screen hits by the second photon’s outcome. The no-signalling theorem forbids using this to send or receive messages backwards in time.

Fig. 39 Relativistic entangled photons in a slow-light medium

SPECULATIVE



Files: 1987716783990767623-G5XJ4aEbQAA8Uya.jpg · Also in repository: 1987834150221750587-G5Y0dEQbgAEC5..., 2032915625920835884-HDZdbavaMAMNI..., 2032924659516403722-HDZmJmCagAA0g...

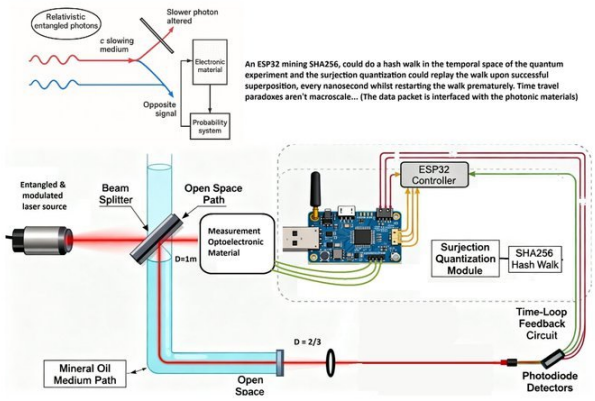
WHAT IT SHOWS A schematic in which entangled photons enter a “c-slowing medium”; the slower photon is altered at a mirror, the “opposite signal” goes to an electronic material feeding a “probability system” that loops back. (Images 31, 70 and 71 are copies.)

THE SCIENCE Slow light is real: electromagnetically induced transparency lowers the group velocity, and entanglement can be stored in and retrieved from atomic vapours (Choi et al., Nature 2008). Every photon travels at c in vacuum, so “relativistic photons” describes any light.

REALITY CHECK Slowing one photon does not change its partner or let the pair carry information faster than light. The “probability system” feedback loop has no defined operation; whatever is done to one photon leaves the other’s statistics unchanged.

Fig. 40 ESP32 hash walk in the temporal space of a quantum experiment

SPECULATIVE



Files: 1988739842013089856-G5lsVyibwAA7mPU.jpg

WHAT IT SHOWS A small annotated schematic: entangled, modulated laser → beam splitter → mineral-oil medium path and open-space path → measurement optoelectronic material → time-loop feedback circuit → photodiode detectors, with an ESP32 controller and a “surjection quantisation module” running a SHA-256 hash walk. Text: the walk could be replayed “upon successful superposition”.

THE SCIENCE SHA-256 is a cryptographic hash; a hash walk uses repeated hashing as a pseudo-random sequence. An ESP32 microcontroller manages a few tens of kilohashes per second. Mineral oil ($n \approx 1.47$) is a convenient transparent liquid.

REALITY CHECK A hash function’s output is deterministic; hashing more does not make a measurement more likely to show superposition, and a single-shot outcome cannot be “replayed”. The diagram does not define what a “successful superposition” signal is.

Fig. 41 Early stop-signal detection in time-of-flight measurement

MIXED

Files: 1990370560640266644-G583ePtakAAbGpK.png · Also in repository: 1990535159238840799-G5_NJlFbMAEAt...

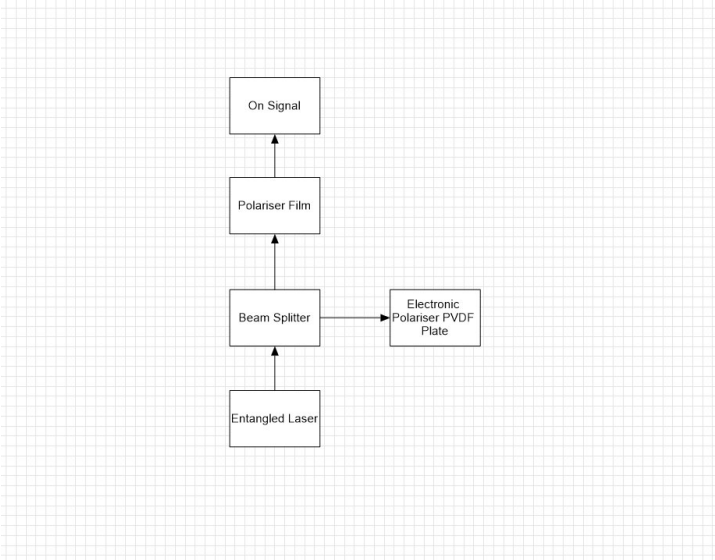
WHAT IT SHOWS A timeline of total path time from photon A to photon B with a stop signal at T = 0.4 and “time saved” shaded; check-marks list time savings, greater precision and reduced measurement noise. (Image 34 is a duplicate.)

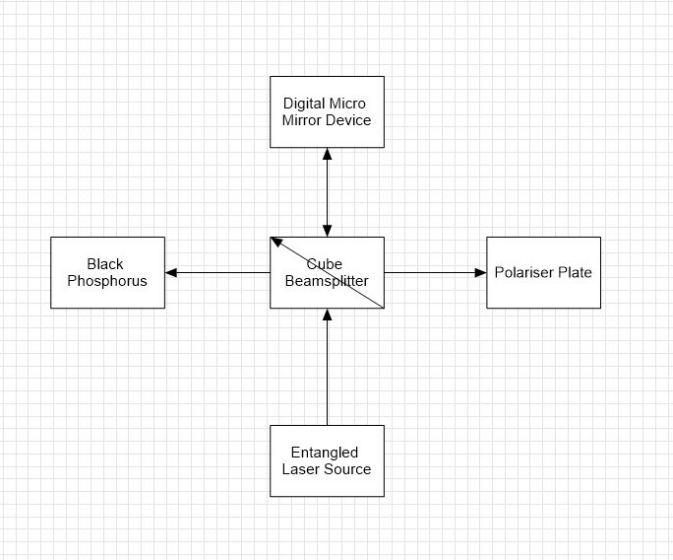
THE SCIENCE Time-of-flight and time-correlated single-photon counting measure the delay between a start and a stop signal. Stopping a measurement early is a real optimisation for sequential decision rules. In TCSPC, however, using only the first stop event biases the histogram towards early arrivals (pile-up).

REALITY CHECK Stopping at T = 0.4 does save time, but it collects fewer photons, and shot-noise error grows as 1/√N. “Greater precision” and “reduced noise” are not automatic consequences and generally go the other way for the same signal.

Fig. 42 Entangled laser with polariser film, PVDF plate and DMD/black-phosphorus variant

MIXED





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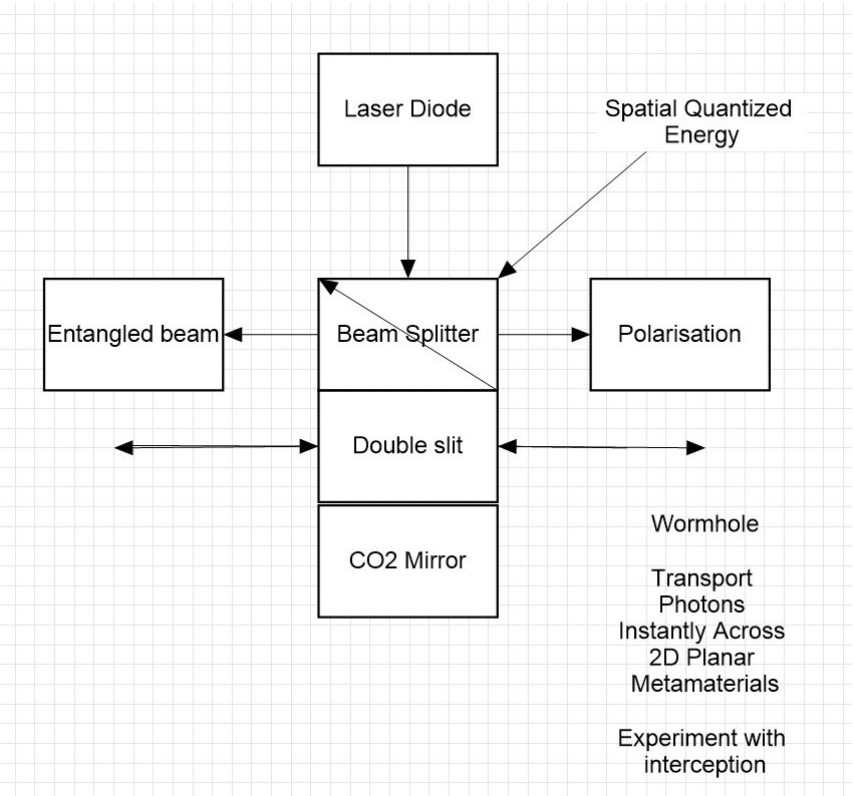
WHAT IT SHOWS Two sparse flowcharts. Left: entangled laser → beam splitter → polariser film → “On signal”, with a branch to an electronic polariser PVDF plate. Right: entangled laser source → cube beam splitter with branches to a digital micromirror device, a black-phosphorus plate and a polariser plate.

THE SCIENCE PVDF is a piezoelectric and pyroelectric polymer used in film sensors. A digital micromirror device (DMD) is a chip of individually tilting mirrors used in projectors and structured illumination. Black phosphorus has in-plane optical anisotropy, so it absorbs linearly polarised light differently along its armchair and zigzag axes (Xia et al., Nature Communications 2014), and it makes a compact polarisation-sensitive infrared detector.

REALITY CHECK The diagrams list components without a quantity to be measured. As drawn they are polarisation-analysis stages; the role of entanglement is unspecified.

Fig. 43 “Wormhole”: transporting photons instantly across 2D metamaterials

SPECULATIVE



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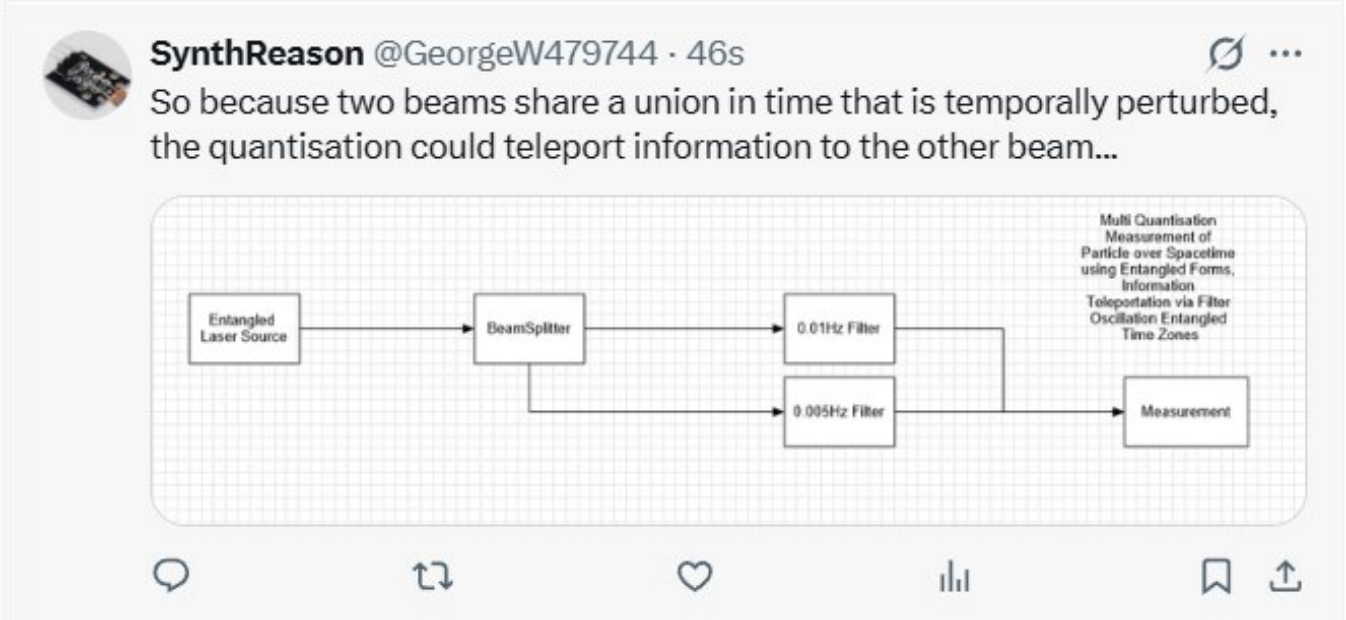
WHAT IT SHOWS Laser diode → beam splitter (with “spatial quantised energy”) → entangled beam, polarisation, double slit and CO_2 mirror, with a note: “Wormhole – transport photons instantly across 2D planar metamaterials. Experiment with interception.”

THE SCIENCE Metasurfaces (2D arrays of sub-wavelength scatterers) shape phase and polarisation, and transformation optics can build electromagnetic “cloaks” and even wormhole analogues that guide rays from one region to another (Greenleaf et al., PRL 2007). CO_2 laser mirrors (for 10.6 μm light) are ordinary optics.

REALITY CHECK Optical wormhole analogues route light, at or below the speed of light, through a designed medium; they do not teleport photons. Phase or group velocities that exceed c never carry information faster than c.

Fig. 44 Two-filter “time zones” and information teleportation

SPECULATIVE



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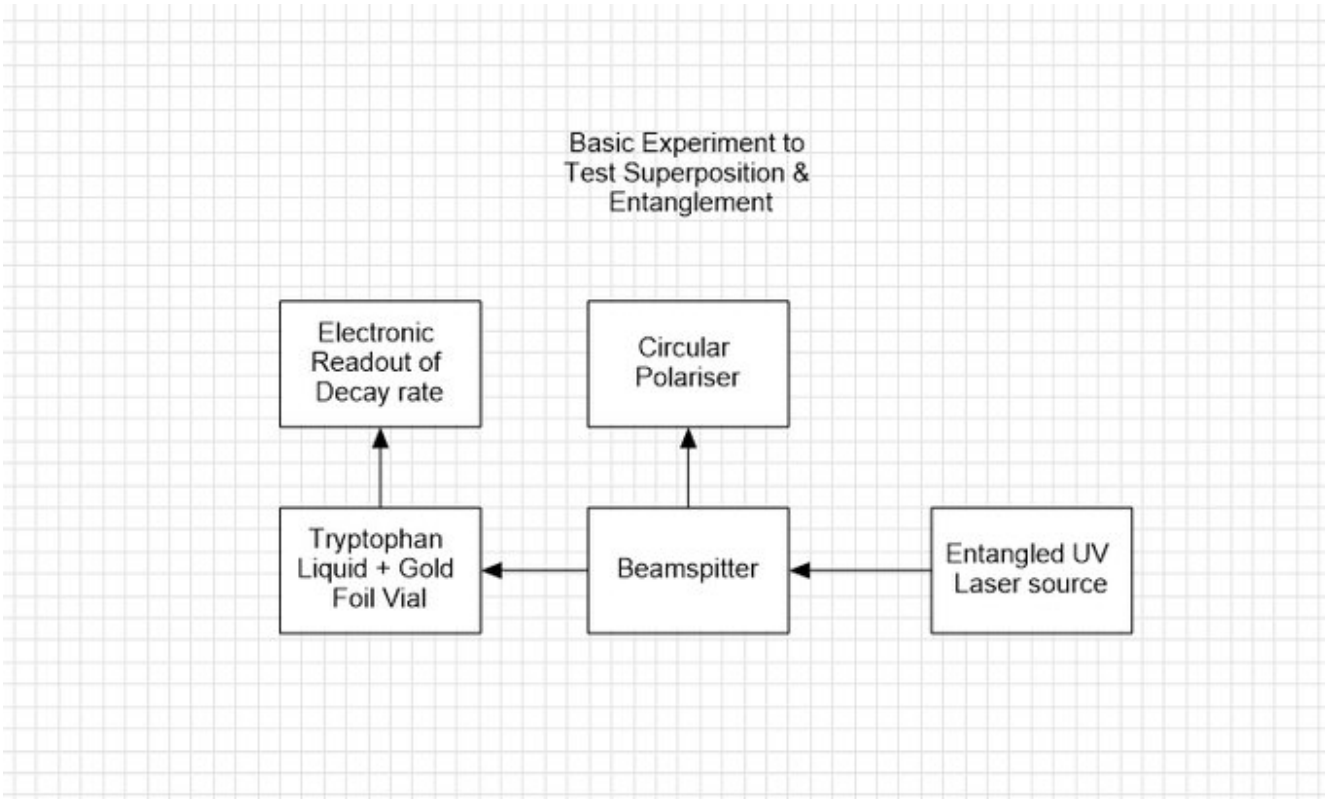
WHAT IT SHOWS A post: “because two beams share a union in time that is temporally perturbed, the quantisation could teleport information to the other beam”, with a diagram of an entangled laser source, a beam splitter, a 0.01 Hz filter and a 0.005 Hz filter, and a “multi-quantisation measurement” block.

THE SCIENCE Quantum teleportation is real (Bennett et al., 1993; Bouwmeester et al., 1997), but it needs a classical communication channel to complete: without the classical bits, the receiver sees random results. Narrow spectral filters (here at 0.01 Hz, far narrower than any optical filter) select long coherence times.

REALITY CHECK The scheme shows no classical channel, so any “information” it transfers would be faster-than-light signalling, which is forbidden. Filters at 5–10 mHz would also need a source coherent over 100–200 seconds, far beyond ordinary lasers.

Fig. 45 Tryptophan and gold-foil vial: testing superposition and entanglement

MIXED



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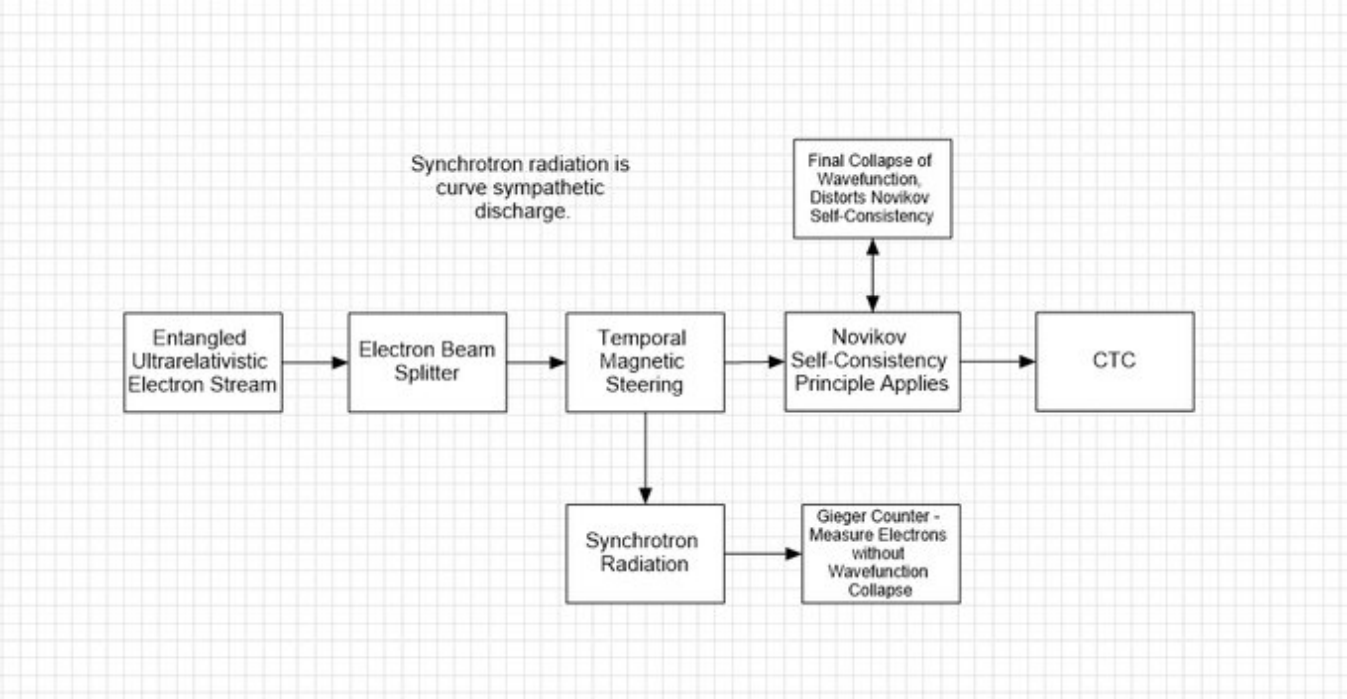
WHAT IT SHOWS Entangled UV laser source → beam splitter; one branch goes to a vial of tryptophan liquid plus gold foil with an electronic readout of decay rate, and the other goes through a circular polariser. Title: “Basic experiment to test superposition and entanglement”.

THE SCIENCE Tryptophan absorbs near 280 nm and fluoresces with a lifetime of about 2.5–3 ns in water. Plasmonic metals like gold change an emitter’s decay rate through the Purcell effect (enhancement at 10–50 nm, quenching closer). A 2024 study (Babcock et al., J. Phys. Chem. B) argued that networks of tryptophan show collective (superradiant) emission enhancement.

REALITY CHECK A changed decay rate shows collective or environment-modified emission; it does not on its own demonstrate entanglement or superposition. Entangled UV pairs at 280 nm are also difficult: they would need a deep-UV pump and non-linear crystals, and pair rates are low.

Fig. 46 Entangled electron stream, Novikov self-consistency and a CTC

SPECULATIVE



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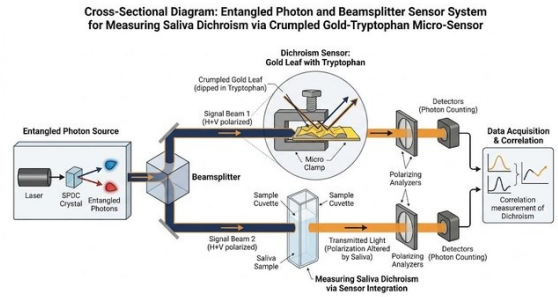
WHAT IT SHOWS A flowchart: entangled ultrarelativistic electron stream → electron beam splitter → temporal magnetic steering → Novikov self-consistency principle applies → CTC; a branch takes synchrotron radiation to a Geiger counter that "measures electrons without wavefunction collapse"; a note reads "Synchrotron radiation is curve sympathetic discharge".

THE SCIENCE Synchrotron radiation is light emitted by relativistic charges bent by a magnetic field; storage rings produce it routinely. The Novikov principle (1980s) says a closed timelike curve can only host self-consistent histories, and Deutsch's model (1991) treats CTCs in quantum mechanics.

REALITY CHECK Closed timelike curves are hypothetical, and magnets do not create them. A Geiger counter that registers electrons is by definition a measurement, so it cannot count them "without collapse".

Fig. 47 Saliva dichroism with entangled photons and a crumpled gold-tryptophan sensor

MIXED



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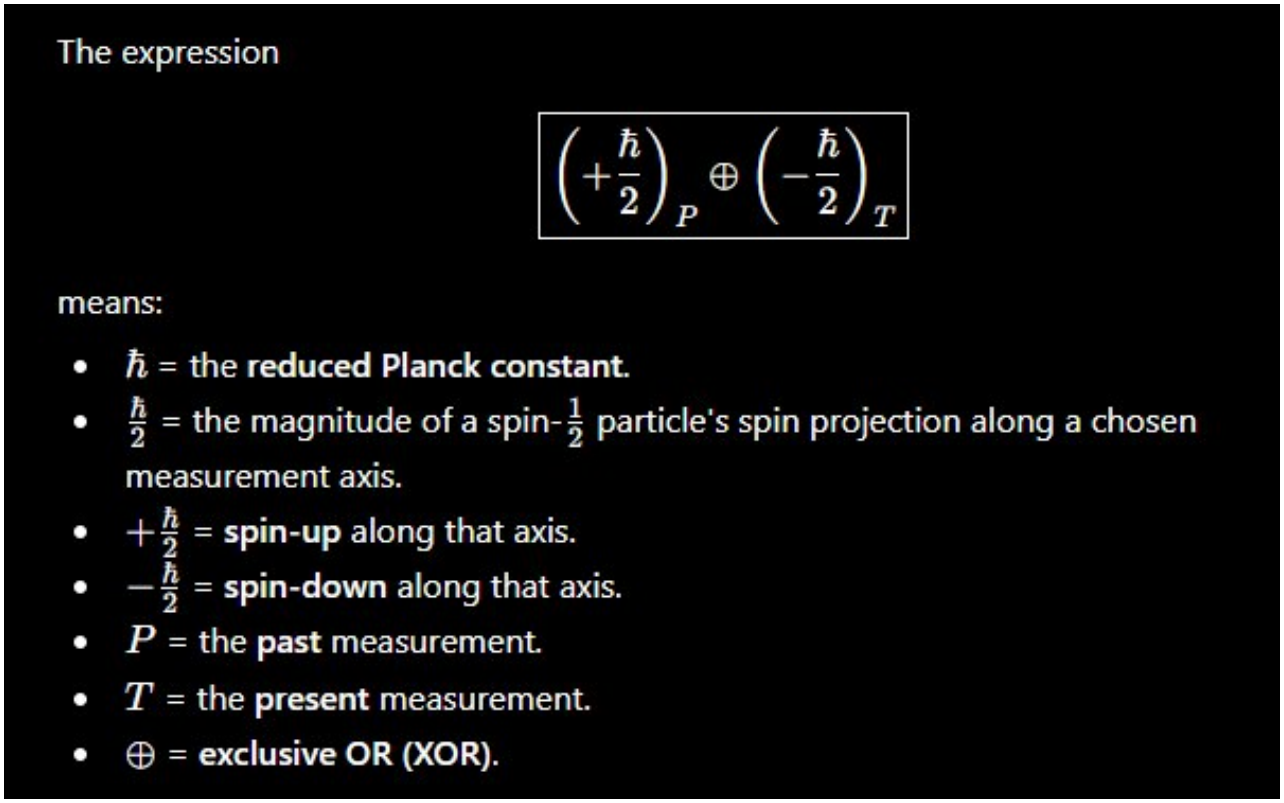
WHAT IT SHOWS A cross-section: a laser and SPDC crystal make entangled photons; a beam splitter sends one beam onto a crumpled gold leaf dipped in tryptophan (held in a micro-clamp) and the other through a cuvette of saliva; polarising analysers and single-photon counters feed a coincidence correlator that infers dichroism.

THE SCIENCE Dichroism is the differential absorption of two polarisations. Twin-photon (SPDC) beams can measure a small absorption below the shot-noise limit of classical light, a real result (Brida et al., Nature Photonics 2010). Gold leaf and tryptophan are cheap, chiral-sensitive materials.

REALITY CHECK Saliva is turbid and absorbs deep-UV strongly, so few photons survive, wiping out any sub-shot-noise gain, which needs low loss. Correlated-photon measurements also add cost and complexity compared with a standard spectrophotometer.

Fig. 48 Spin up in the past XOR spin down in the present

MAINSTREAM



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WHAT IT SHOWS A short glossary for the expression $(+\hbar/2)_P \oplus (-\hbar/2)_T$, defining ħ (reduced Planck constant), ħ/2 (spin-½ projection along a chosen axis), + and − (up and down), P (past measurement), T (present measurement) and ⊕ (exclusive OR).

THE SCIENCE The definitions are correct. For a spin-½ particle, the projection on any axis is +ħ/2 or −ħ/2 (Stern–Gerlach, 1922). Measuring the same axis twice gives the same answer, so “up then down” on one axis does not happen for a single particle. For the two halves of a singlet pair, however, outcomes on the same axis are always opposite, which is the XOR-like relation the expression encodes.

REALITY CHECK The expression is a logical statement, not a dynamical law: it says nothing about how or why the past and present outcomes differ.

Fig. 49 The spacetime interval

MAINSTREAM

Spacetime interval equation

For two events separated by a time difference Δt and a spatial distance Δx along one spatial axis, the Lorentzian "distance" (interval) is:

$$s^2 = c^2(\Delta t)^2 - (\Delta x)^2$$

where c is the speed of light. phys.libretexts +1

- If $s^2 > 0$, the separation is time-like (a slower-than-light signal can connect them). publish.obsidian +1
- If $s^2 = 0$, it is light-like (only light can connect them). phys.libretexts +1
- If $s^2 < 0$, it is space-like (no causal connection by any signal). publish.obsidian +1

Files: 1996132482673197545-G7Ov6f2bgAA-tVB.jpg

WHAT IT SHOWS A chat-assistant answer: $s^2 = c^2(\Delta t)^2 - (\Delta x)^2$, with $s^2 > 0$ time-like, $s^2 = 0$ light-like and $s^2 < 0$ space-like.

THE SCIENCE This is the invariant of special relativity in the (+,−,−,−) sign convention: all inertial observers agree on s^2. Time-like separated events can be causally connected by slower-than-light signals, light-like by light, and space-like events cannot influence each other. It underpins why entanglement cannot carry messages: the two ends of a Bell test are usually space-like separated.

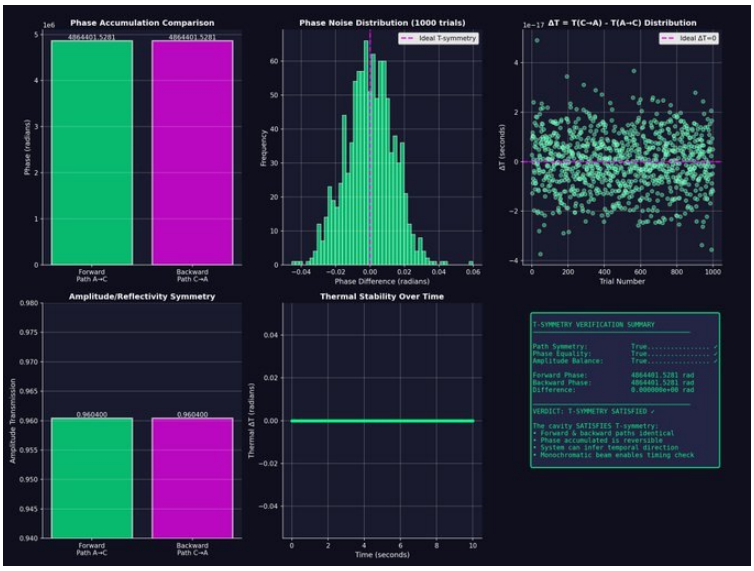
REALITY CHECK Some texts use the opposite sign convention, so a negative sign does not mean "wrong".

Fig. 50 T-symmetry “verification” simulation: results and plots

MIXED

Key Results:

1. **T-Symmetry is Satisfied ✓**
   - Forward path (A→B→C) and backward path (C→B→A) maintain **identical phase accumulation**
   - Both paths: **4.864×10^{6} radians** (perfect match)
   - Amplitude transmission is **symmetric** (0.9604 both directions)
2. **Timing Analysis ✓**
   - $\Delta T = T(C \to A) - T(A \to C) = \mathbf{0\ ps}$ (within measurement precision)
   - Perfect reversibility for monochromatic beam
   - **No timing mismatch detected**
3. **Noise Robustness ✓**
   - Tested with 1000 Monte Carlo trials of phase noise
   - Mean ΔT: -1.06×10^{-19} s (effectively zero)
   - Phase noise standard deviation: **< 0.1 rad** ✓ ROBUST
   - System maintains T-symmetry despite realistic noise
4. **Thermal Stability ✓**
   - Tested over 10-second timescale
   - Thermal drifts affect both paths equally
   - **No breaking of T-symmetry**



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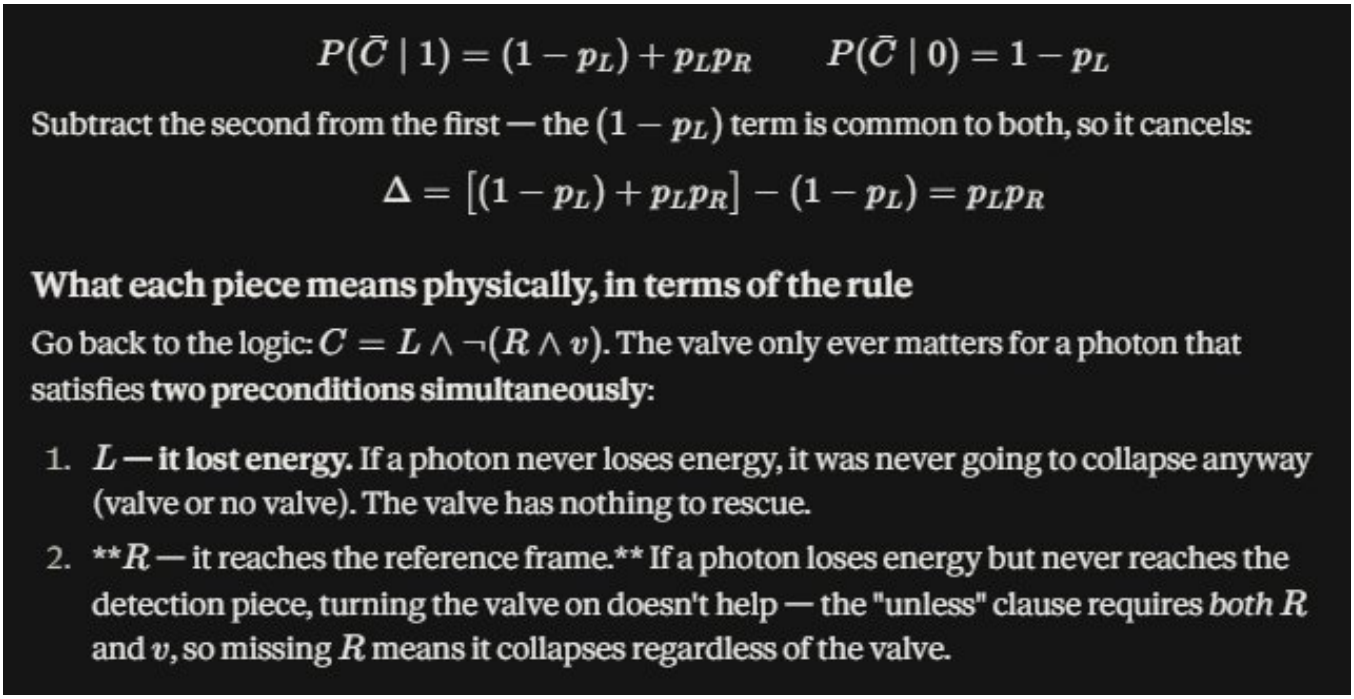
WHAT IT SHOWS A results summary (left) and a plot panel (right). Forward and backward paths both accumulate 4.864×10^{6} rad; amplitude transmission is 0.9604 both ways; $\Delta T = 0$ ps; 1000 Monte Carlo runs of phase noise ($\sigma < 0.1$ rad, mean $\Delta T \approx -1 \times 10^{-17}$ s); thermal drift is flat; all checks are marked “satisfied”.

THE SCIENCE For a linear, passive, non-magnetic medium Lorentz reciprocity guarantees that forward and backward transmission are equal, so the same phase and amplitude in both directions is expected. Breaking that symmetry needs a magnetic bias, a nonlinearity, or time-modulation (Caloz et al., 2018).

REALITY CHECK The code computes phase as the same formula for both directions, so identical numbers are guaranteed, and random noise that is symmetric by construction gives $\Delta T \approx 0$. The simulation confirms its own assumptions; it is not a test of nature.

Fig. 51 A valve that rescues photons: the probability derivation

MAINSTREAM



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WHAT IT SHOWS A dark-theme derivation: P(C | 1) = (1 − p_L) + $p_L p_R$ and P(C | 0) = 1 − p_L; subtracting, the common term cancels and Δ = $p_L p_R$. The rule is C = L ∧ ¬(R ∧ v): a photon "collapses" if it lost energy (L) unless it both reaches the reference frame (R) and the valve is on (v). The valve only matters for a photon that satisfies both preconditions.

THE SCIENCE This is straightforward probability. For independent events, P(L ∧ R) = $p_L p_R$. With the valve on, the photon survives if it did not lose energy (probability 1 − p_L) or if it lost energy but reached the reference ($p_L p_R$); with the valve off it survives only in the first case. The difference is $p_L p_R$; for p_L = 0.3 and p_R = 0.5 the valve adds 0.15 to the survival probability.

REALITY CHECK The algebra is right if L and R are independent. "Collapse" here is just a label for one outcome of a classical probability model; the calculation says nothing about quantum wavefunction collapse.

CHAPTER 8

Computation, complexity and optical solvers

Figures 52–65 · 14 entries

Fig. 52 Experimental set-up for quantum complexity growth in a high-loss cavity

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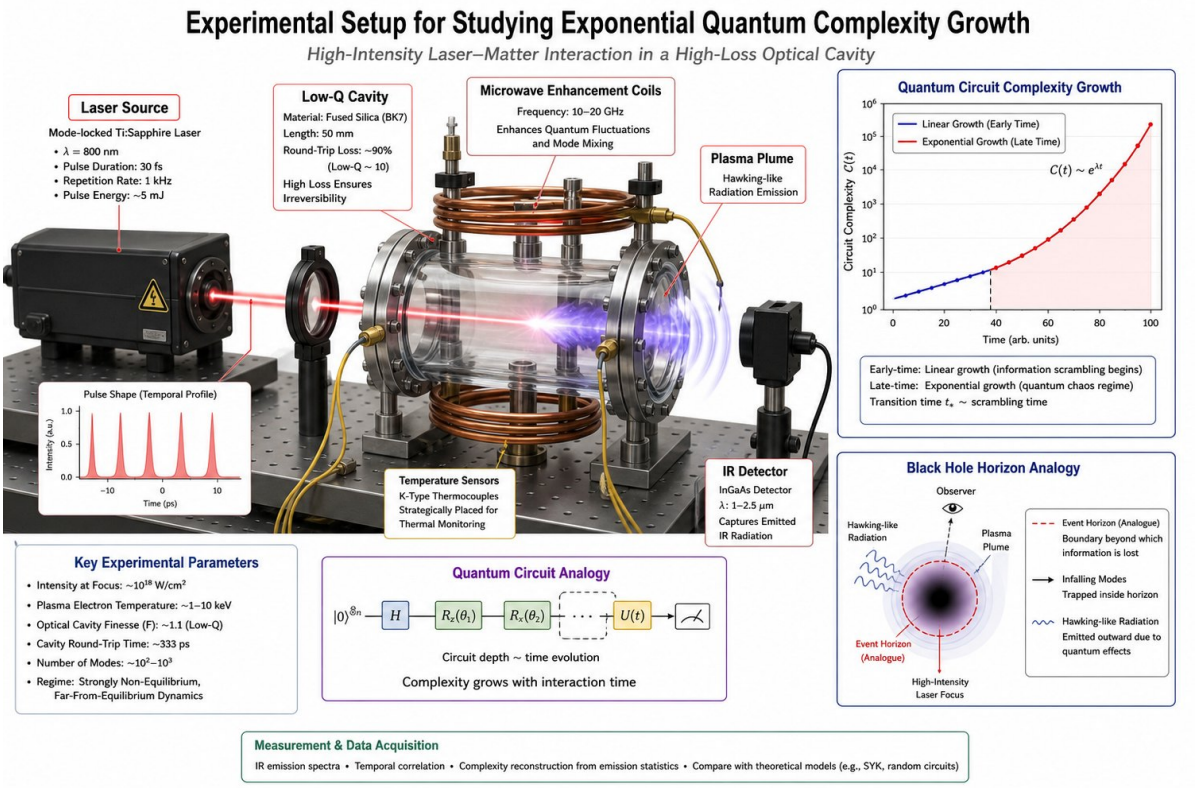

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WHAT IT SHOWS An infographic: a mode-locked Ti:sapphire laser (800 nm, 30 fs, 1 kHz, ~5 mJ) drives a low-Q fused-silica cavity (10–20 GHz microwave coils) producing a plasma plume with “Hawking-like radiation” read by an InGaAs infrared detector. Insets: circuit complexity C(t) ~ e^{λt}, a quantum-circuit analogy, and a black-hole horizon analogue.

THE SCIENCE “Complexity” in quantum chaos is measured several ways. Krylov complexity grows exponentially at early times in chaotic systems, bounded by λ ≤ 2α (Parker et al., PRX 2019), and Brown–Susskind circuit complexity grows linearly for a time of order e^S before saturating. Analogue Hawking radiation has been studied in Bose–Einstein condensates (Steinhauer, 2016) and claimed, controversially, in laser filaments (Belgiorno et al., 2010). A 30 fs, 5 mJ pulse is 1.7×10^{11} W and can reach 10^{18} W/cm² when focused tightly, producing keV-temperature plasma.

REALITY CHECK The inset reverses the usual order: the infographic labels “early-time linear, late-time exponential”. In standard treatments, exponential growth comes first and saturation last. A cavity with finesse ≈ 1.1 is hardly a cavity, and no observable linking a laser plasma’s IR emission to circuit complexity is defined.

Fig. 53 Krylov complexity and microwave-comb driving

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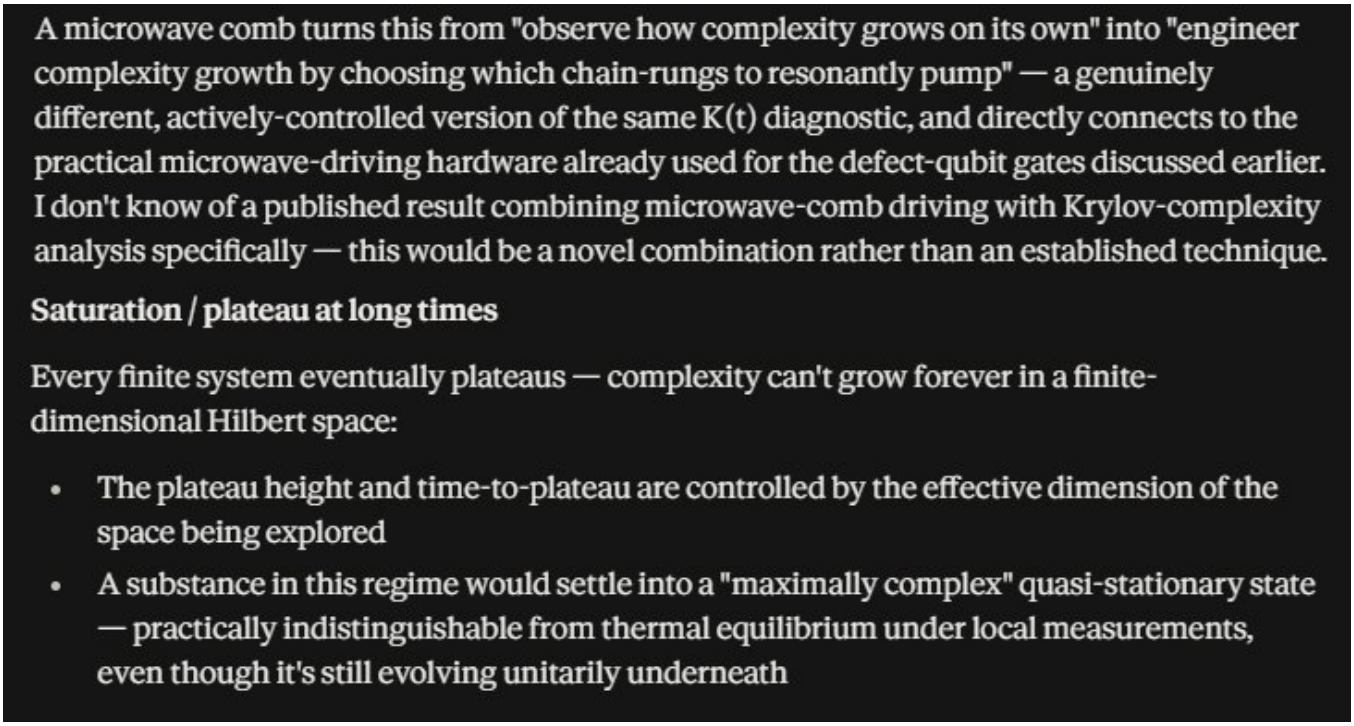

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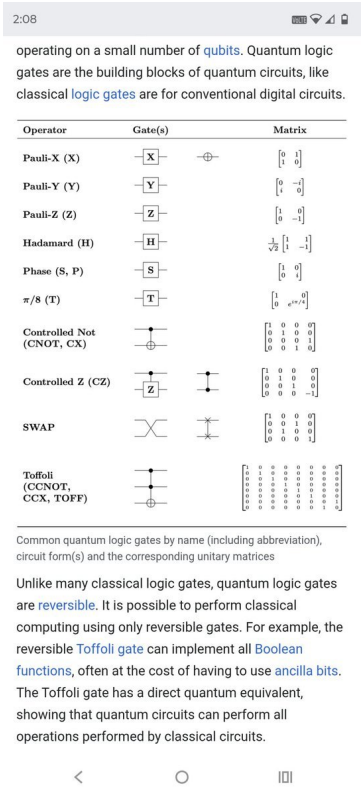
WHAT IT SHOWS A chat-assistant answer describing how a microwave comb could “engineer complexity growth by choosing which chain-rungs to resonantly pump”, calling this a novel combination rather than an established technique, and noting that complexity saturates at long times in a finite Hilbert space to a quasi-stationary “maximally complex” state.

THE SCIENCE Krylov complexity K(t) measures how far an operator spreads along the Krylov basis of repeated commutators with the Hamiltonian. In a finite-dimensional Hilbert space it must plateau; the plateau height and time depend on the effective dimension. A maximally complex state looks thermal to local measurements although it still evolves unitarily.

REALITY CHECK The answer itself flags that combining microwave driving with Krylov analysis is not a published result. That is honest, and it means the idea is an open proposal.

Fig. 54 Common quantum logic gates

MAINSTREAM



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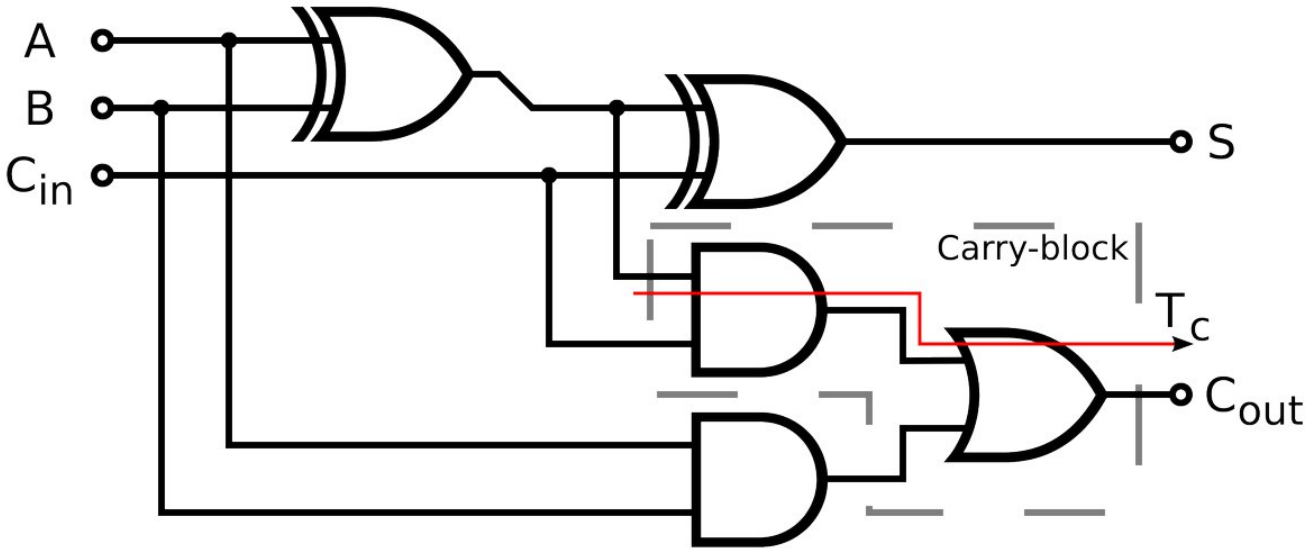
WHAT IT SHOWS A mobile screenshot of the Wikipedia table of quantum logic gates: Pauli-X, Y, Z, Hadamard, phase (S, P), π/8 (T), controlled-NOT, controlled-Z, SWAP and Toffoli, each with its circuit symbol and unitary matrix, and text explaining that quantum gates are reversible.

THE SCIENCE Every gate is a unitary matrix, so it preserves probability and can be undone. Single-qubit gates rotate the state on the Bloch sphere; CNOT and CZ entangle qubits; Toffoli (CCNOT) is universal for reversible classical computation, so a quantum computer can do everything a classical one can. H, T and CNOT together are approximately universal (Solovay–Kitaev).

REALITY CHECK Nothing to correct: this is reference material.

Fig. 55 One-bit full adder with carry-block critical path

MAINSTREAM



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WHAT IT SHOWS A logic-gate schematic: inputs A, B and C_{in} feed two XOR gates to give the sum S, and AND/OR gates form the carry-out C_{out}; a red arrow marks the carry block and its delay T_c.

THE SCIENCE A full adder computes $S = A \oplus B \oplus C_{in}$ and $C_{out} = AB + C_{in}(A \oplus B)$. Chaining n of them gives a ripple-carry adder whose delay is about $n \times T_c$, the carry path being the "critical path" that limits clock speed; carry-lookahead adders shorten it.

REALITY CHECK The diagram is a standard textbook circuit and needs no correction.

Fig. 56 Repeated-shot error math for a probabilistic algorithm

MAINSTREAM

For $p = 2/3$:

- $\delta = 10^{-3}$:
  $k \geq \frac{\ln 10^{-3}}{\ln(1/3)} \approx \frac{-6.9078}{-1.0986} \approx 6.3 \rightarrow$ **7 shots.**
- $\delta = 10^{-6}$:
  $k \geq \frac{\ln 10^{-6}}{\ln(1/3)} \approx \frac{-13.8155}{-1.0986} \approx 12.6 \rightarrow$ **13 shots.**
- $\delta = 10^{-9}$:
  $k \approx 19 \rightarrow$ **19 shots.**
- $\delta = 10^{-12}$:
  $k \approx 25 \rightarrow$ **25 shots.**

So for a textbook BQP algorithm on a 13-variable combinatorial problem:

- **~13 shots** gets you to one-in-a-million error.
- **~25 shots** gets you to one-in-a-trillion error.

Files: Fermion.jpg

WHAT IT SHOWS A chat-assistant answer for a success probability p = 2/3: with k repeated shots, failure probability $\delta \leq (1/3)^k$, so $k \geq \ln \delta / \ln(1/3)$ gives 7 shots for $\delta = 10^{-3}$, 13 for 10^{-6}, 19 for 10^{-9} and 25 for 10^{-12}.

THE SCIENCE Check: $\ln(10^{-3}) / \ln(1/3) = -6.908 / -1.0986 = 6.29$, so 7 shots. Amplifying success by repetition is how bounded-error quantum (BQP) algorithms are made reliable.

REALITY CHECK $(1/3)^k$ is the right failure probability when a single correct answer can be recognised, for example a factor you can multiply out to check. For an algorithm whose output is a yes/no answer that cannot be checked, you take a majority vote, and the needed shots follow a binomial (Chernoff-type) bound, which is much larger: about 81 shots for $\delta = 10^{-3}$ and about 421 for 10^{-12} at $p = 2/3$.

Fig. 57 Entangled beam, ITO grating and spatial-to-temporal mapping for a combinatorial problem

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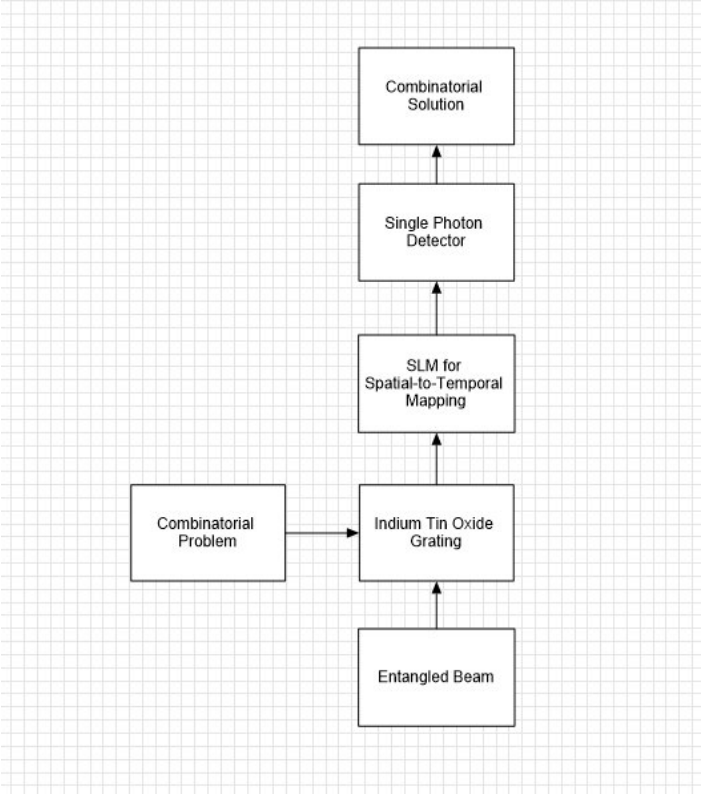

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WHAT IT SHOWS A vertical flowchart: combinatorial problem and entangled beam feed an indium tin oxide (ITO) grating → spatial light modulator (SLM) for spatial-to-temporal mapping → single-photon detector → “combinatorial solution”.

THE SCIENCE Light can be used to search: Oltean and Muntean (2009) proposed a light-based device for the subset-sum problem, and optical Ising machines using SLMs and coherent light have solved optimisation problems of hundreds to thousands of variables (Inagaki et al., Science 2016; Pierangeli et al., PRL 2019). ITO is a transparent conductor that lets you place electrodes in a light path.

REALITY CHECK These optical solvers exchange time for light: an exponentially large search space needs exponentially many light paths or an exponentially small signal per path, so they have not shown a scaling advantage on NP-hard problems.

Fig. 58 Hole plate, light valves and spatial photon counter

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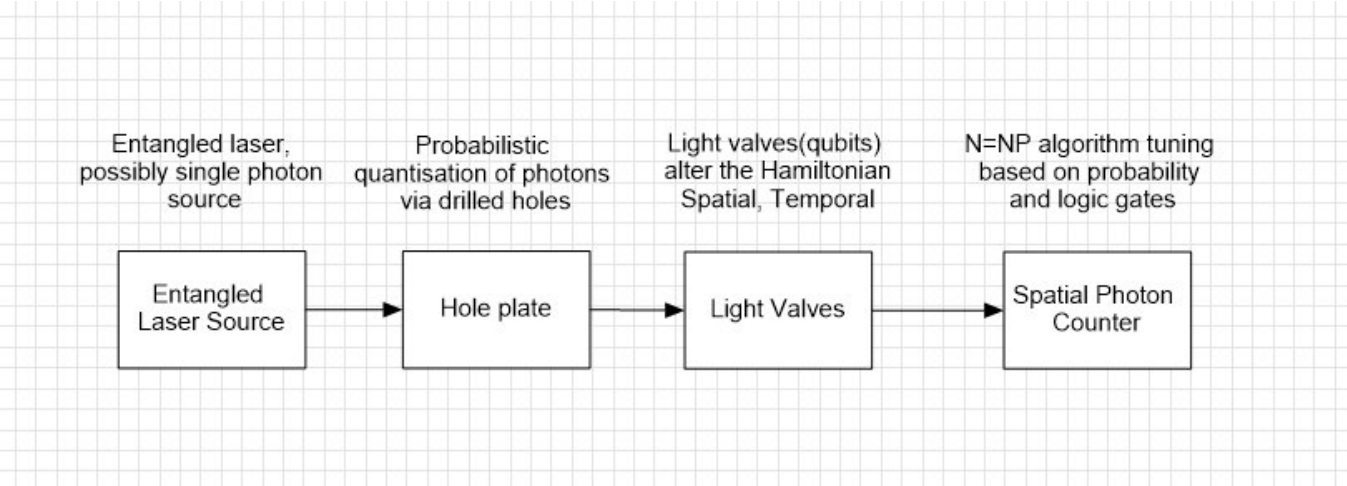

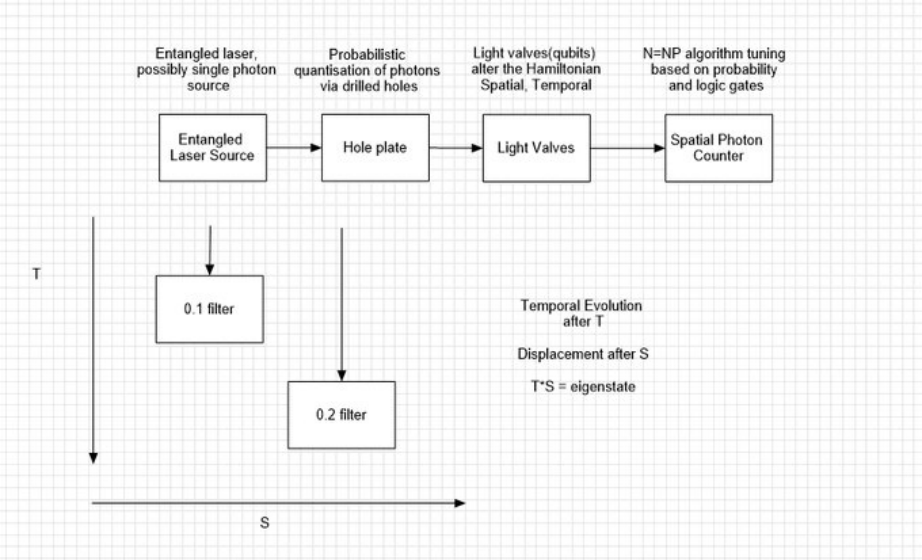

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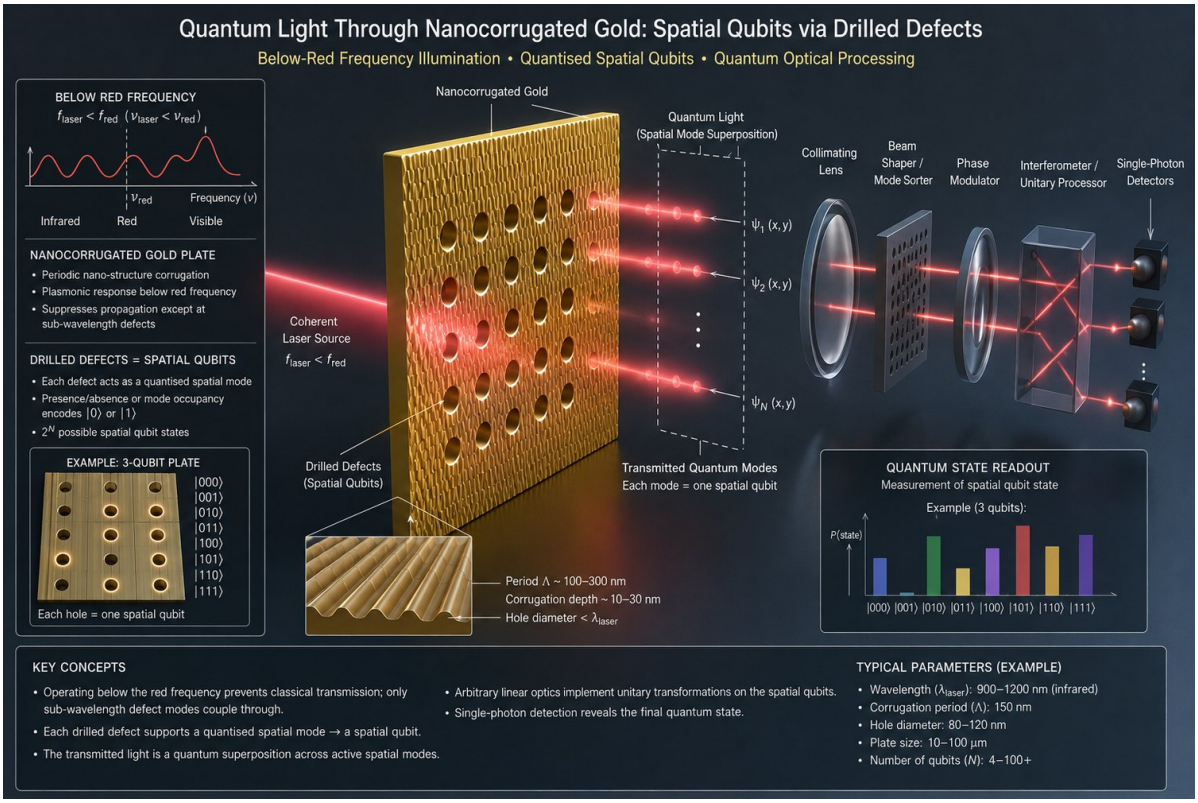
WHAT IT SHOWS Left: entangled laser (possibly single-photon) → hole plate (“probabilistic quantisation of photons via drilled holes”) → light valves (“qubits alter the Hamiltonian, spatial, temporal”) → spatial photon counter, labelled “N=NP algorithm tuning based on probability and logic gates”. Right: the same chain with 0.1 and 0.2 filters on a T–S plane, marked “temporal evolution after T, displacement after S, T*S = eigenstate”.

THE SCIENCE A hole plate is an aperture array; each hole is a spatial mode, and a spatial photon counter (a camera such as an EMCCD or SPAD array) reads which one fired. Spatial light modulators and valves can imprint phases to encode a problem’s couplings, which is how photonic Ising machines work. “N=NP” is presumably a typo for the P versus NP question.

REALITY CHECK Whether the problem is solved with fewer resources than a classical machine depends on the mapping, which the diagram does not give. The “T·S = eigenstate” note does not follow from any stated operators.

Fig. 59 Nanocorrugated gold with drilled holes as spatial qubits

MIXED



Files: 6dc53d8b-fc99-4e22-8a3e-986dc56947a5.png · Also in repository: QPU.jpg

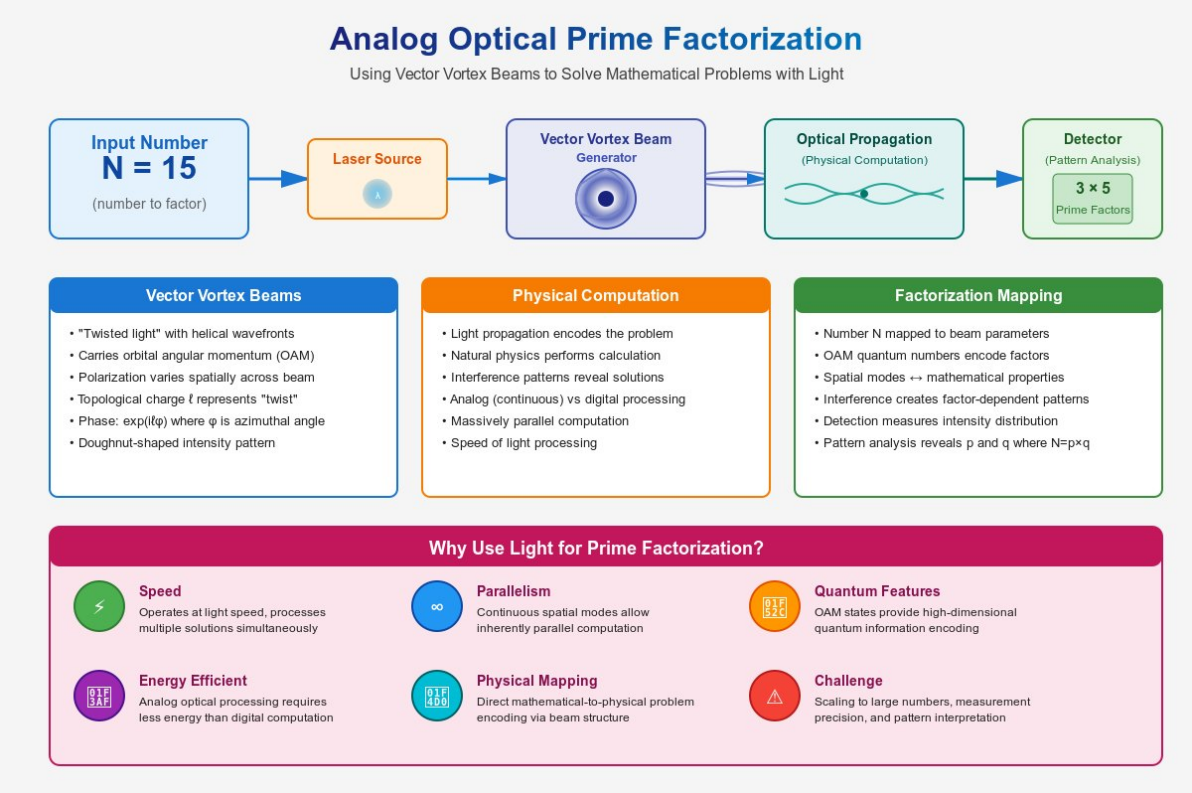
WHAT IT SHOWS An infographic and a post (“Latest quantum computer hardware…”): coherent light below the “red” cut-off frequency passes a nanocorrugated gold plate with drilled defect holes; each hole is a spatial qubit, giving 2^N possible states for N holes, followed by beam shaping, a phase modulator, an interferometer and single-photon detectors, with a bar chart of a 3-qubit state.

THE SCIENCE Light transmits through sub-wavelength holes in metal films far more than simple theory predicts (extraordinary optical transmission; Ebbesen et al., Nature 1998). Spatial modes of light can encode quantum information, and linear optics plus photon counting can implement unitary transformations (Reck et al., 1994).

REALITY CHECK A single photon spread across N holes lives in an N-dimensional space, holding $\log_2 N$ qubits’ worth, not N qubits or 2^N states. To reach 2^N you would need N independent, addressable photons. The algorithms “that can be run in N=NP” need much more than a hole array.

Fig. 60 Analog optical prime factorisation with vector-vortex beams

MIXED



Files: 2001536794773590338-G8bjGqRasAACMN6.png

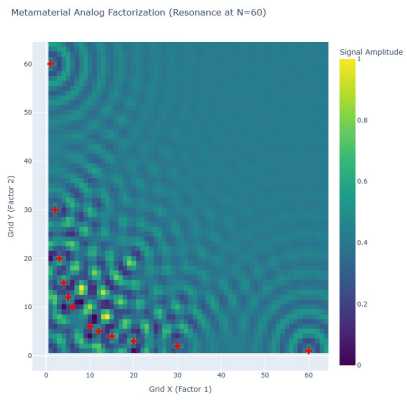
WHAT IT SHOWS An infographic for factoring N = 15 into 3 × 5: an input number, a laser source, a vector-vortex beam generator, optical propagation and a detector reading an interference pattern; benefits listed as speed, parallelism, energy efficiency and physical mapping, with the challenge of scaling.

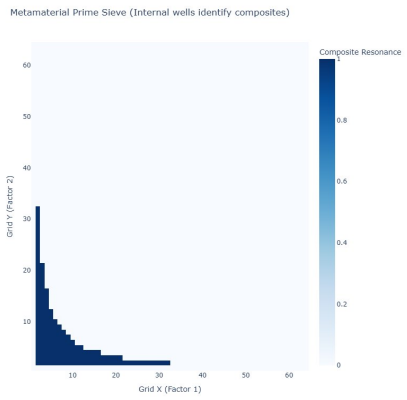
THE SCIENCE Vector-vortex beams carry orbital angular momentum (OAM), and interference of OAM modes can encode Gauss sums that reveal factors (a proposal related to Gauss-sum factorisation experiments; Mehring et al., 2007, in NMR). The smallest demonstration of Shor's algorithm factored 15 = 3×5 on a 7-qubit NMR machine (Vandersypen et al., Nature 2001).

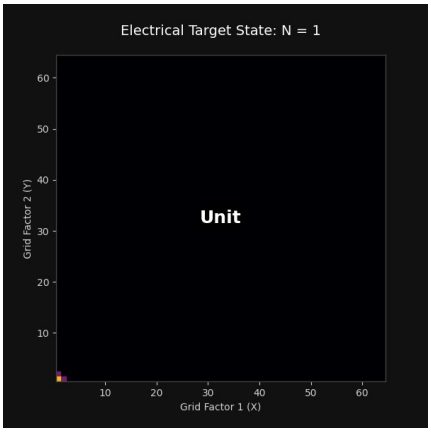
REALITY CHECK Analog Gauss-sum schemes need precision and signal that grow exponentially with the number of digits, so they do not scale. Factoring 15 is a demonstration size that trial division does in an instant.

Fig. 61 Metamaterial factorisation and prime-sieve heat maps, and the prime animation

CONTEXT







Files: metamaterial_factorization.png, metamaterial_sieve.png, prime_animation.gif

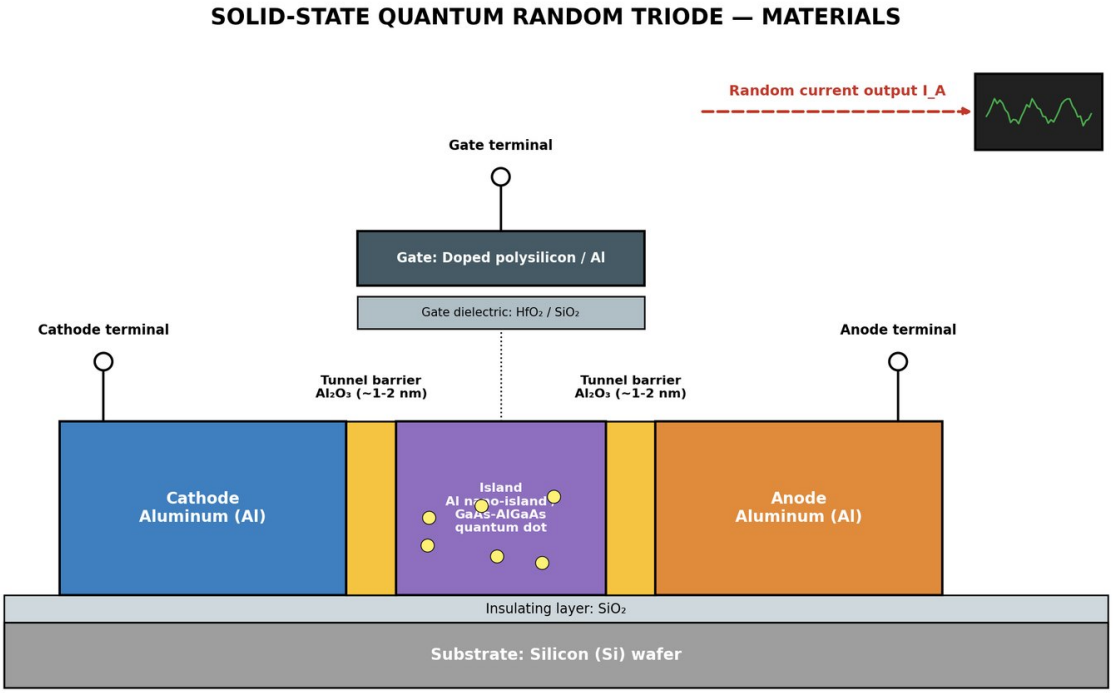
WHAT IT SHOWS Three computer-generated figures. (1) "Metamaterial Analog Factorization (resonance at N=60)": a heat map with red crosses at the divisor pairs of 60. (2) "Metamaterial Prime Sieve": a staircase region of composites. (3) The first frame of an animation, "Electrical Target State: N = 1", labelled "Unit".

THE SCIENCE The crosses lie on the hyperbola xy = 60: (1,60), (2,30), (3,20), (4,15), (5,12), (6,10) and their mirror images. A composite number is exactly one that can be written as a product of two factors greater than 1, so the “sieve” marks every product xy ≤ 64, and the leftover empty cells are primes, which is the sieve of Eratosthenes drawn as an image.

REALITY CHECK These are visualisations of divisor structure, not experiments; a real physical “resonance” would need hardware that computes xy for every cell in parallel, and the hardware’s size grows with N.

Fig. 62 Solid-state quantum random triode: materials

MIXED



Random bits derive from stochastic single-electron tunneling (shot noise) across the Al_2O_3 barriers.

Files: quantum_random_triode_materials.png

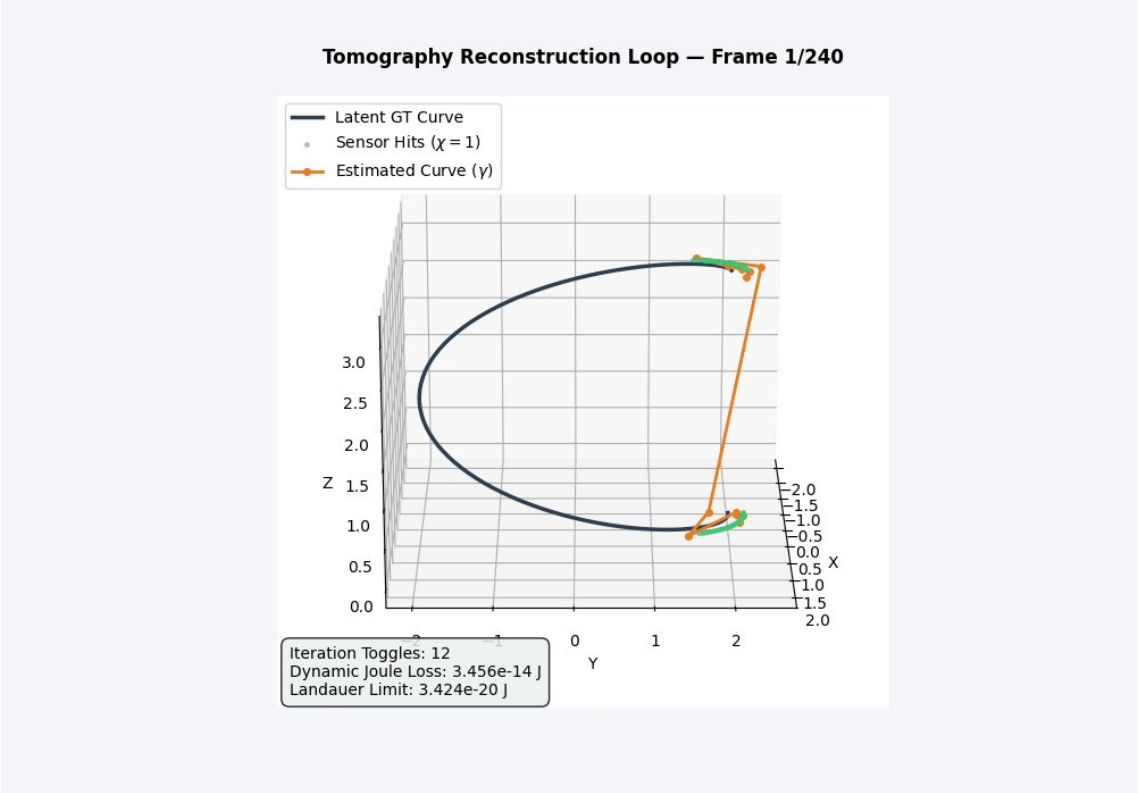
WHAT IT SHOWS A cross-section: an aluminium cathode and anode, a central island (aluminium nano-island / GaAs–AlGaAs quantum dot) separated by 1–2 nm Al_2O_3 tunnel barriers, a doped polysilicon or aluminium gate over an HfO_2/SiO_2 dielectric, all on an SiO_2-insulated silicon wafer. Random bits come from stochastic single-electron tunnelling (shot noise).

THE SCIENCE This is a single-electron transistor (SET). Al/AlO_x/Al tunnel junctions are the standard fabrication (Fulton and Dolan, 1987). Coulomb blockade occurs when the charging energy $e^2/2C$ exceeds k_BT, so aluminium SETs work at millikelvin temperatures. Tunnelling is a Poisson process, so shot noise is a genuine quantum-random source.

REALITY CHECK The island is labelled both aluminium and a GaAs/AlGaAs quantum dot, which are different material systems that cannot be grown on the same aluminium-oxide junctions on silicon. The device would need a dilution refrigerator, whereas a reverse-biased transistor is a far simpler random source (see the random-number generator chapter).

Fig. 63 Tomography reconstruction loop with a Landauer-limit counter

MIXED



Files: tomography_reconstruction, complex naturalisa...

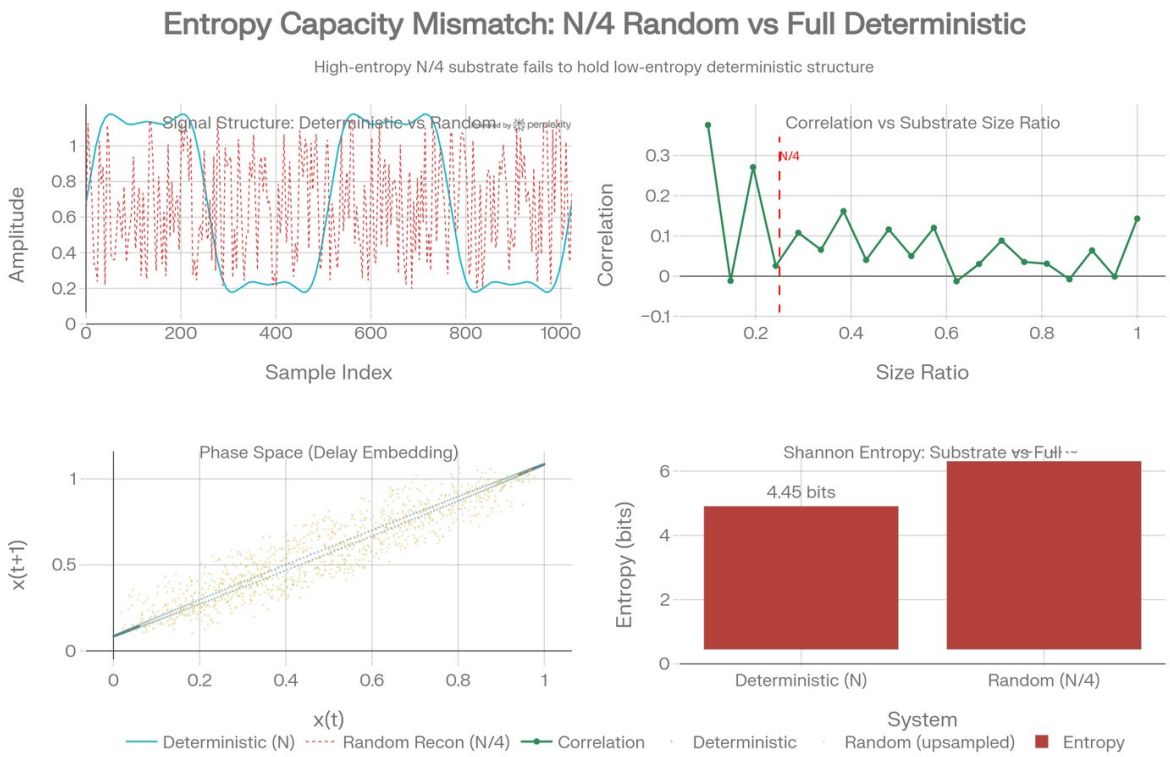
WHAT IT SHOWS The first frame (1/240) of an animation: a 3D latent curve, sensor hits and an estimated curve, with a status box reading “Iteration toggles: 12; dynamic Joule loss: 3.456×10^{-14} J; Landauer limit: 3.424×10^{-20} J”.

THE SCIENCE Landauer’s principle: erasing one bit dissipates at least $k_B T \ln 2$, which is 2.87×10^{-21} J at 300 K. Twelve toggles give $12 \times 2.87\times10^{-21} \approx 3.4\times10^{-20}$ J, matching the box. Tomography reconstruction estimates an unknown curve from sensor hits by iterating.

REALITY CHECK The Joule loss shown is six orders of magnitude above the Landauer bound, which is typical: real electronics today dissipates roughly 10^5–10^6 times the limit per switching operation.

Fig. 64 Entropy-capacity mismatch: N/4 random versus fully deterministic

CONTEXT



Files: af63119c.png

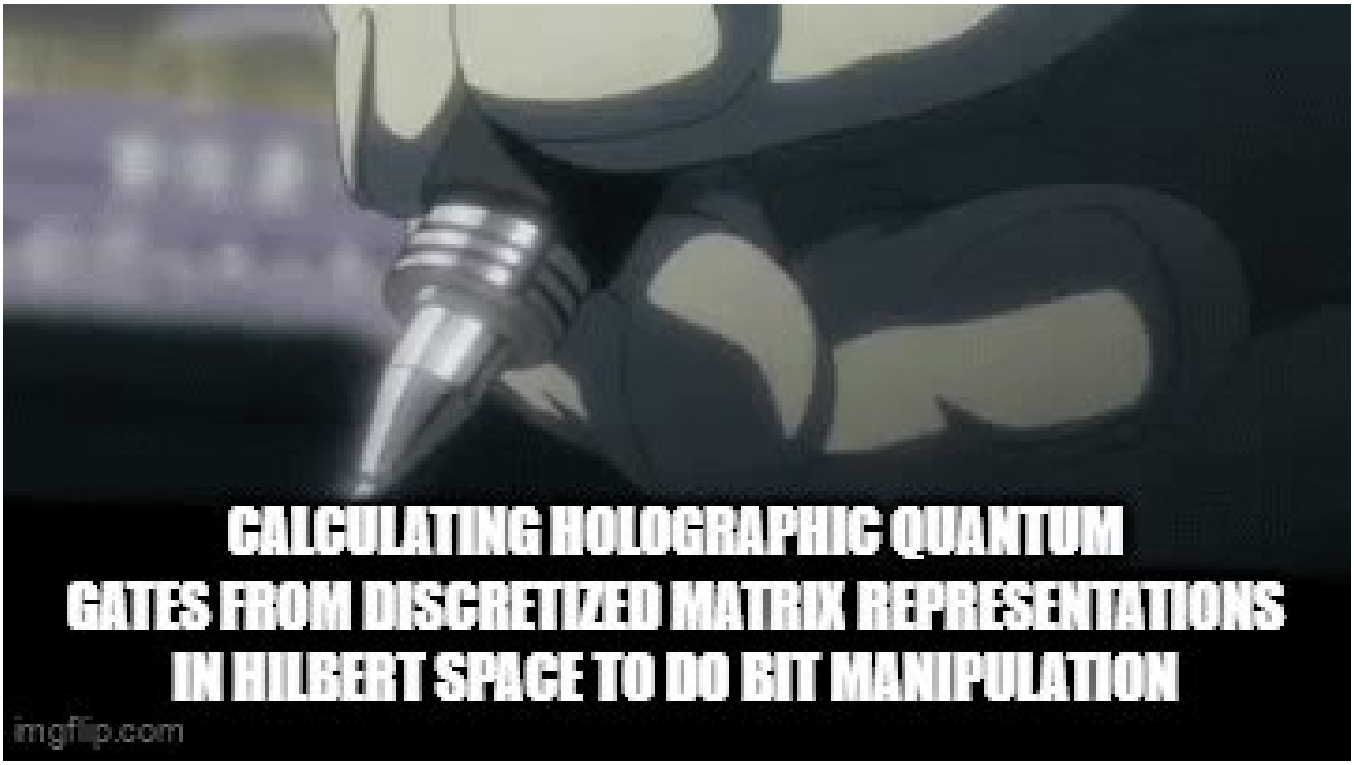
WHAT IT SHOWS Four plots on a dark background: a deterministic signal against an N/4 random reconstruction; correlation versus size ratio (noisy, low); a phase-space delay embedding; and Shannon entropy (4.45 bits for the deterministic substrate versus about 6 bits for random).

THE SCIENCE Shannon entropy $H = -\Sigma\ p \log_2 p$ measures the uncertainty of a source. A smooth deterministic signal has low entropy and can be predicted from few samples, while a random signal cannot. Compressed sensing shows a structured signal can be recovered from far fewer random measurements than samples, but only if it is sparse in some basis and enough measurements (about $K \log(N/K)$) are taken.

REALITY CHECK The plots illustrate that a quarter-size random substrate does not capture a full-size deterministic structure, which is not surprising; the correlation stays low until the sizes match.

Fig. 65 “Holographic quantum gates” meme

CONTEXT



Files: ayxs3o.gif

WHAT IT SHOWS A still from an animated meme (first frame shown) with the caption “Calculating holographic quantum gates from discretized matrix representations in Hilbert space to do bit manipulation”.

THE SCIENCE Quantum gates are unitary matrices acting on state vectors in a Hilbert space (see the gate table). Bit manipulation in the ordinary sense is done with classical reversible or irreversible gates.

REALITY CHECK It is a joke image, offering no calculation.

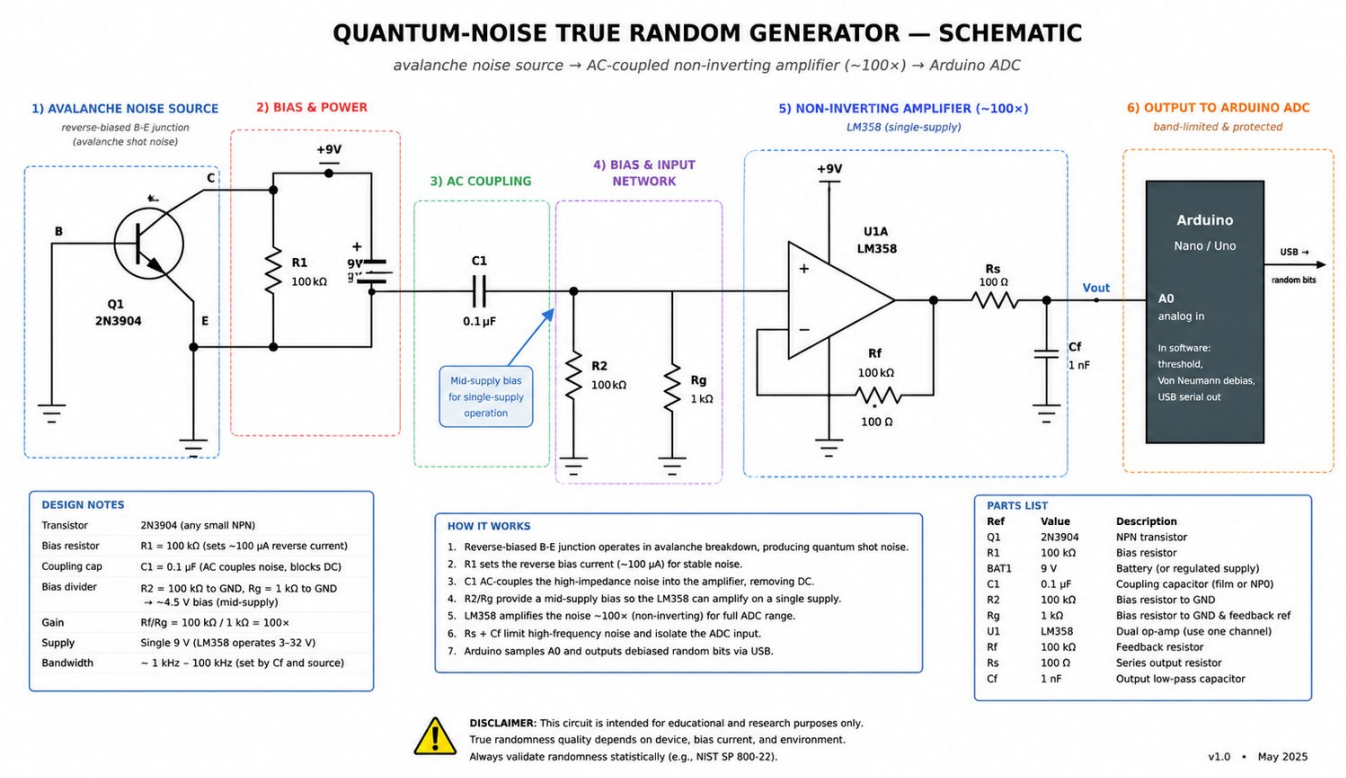
CHAPTER 9

Randomness, hashing and gambling

Figures 66–71 · 6 entries

Fig. 66 Quantum-noise true random generator (avalanche noise → Arduino)

MAINSTREAM



Files: 6bb6f964-2090-419b-837c-faf3a585f864.png

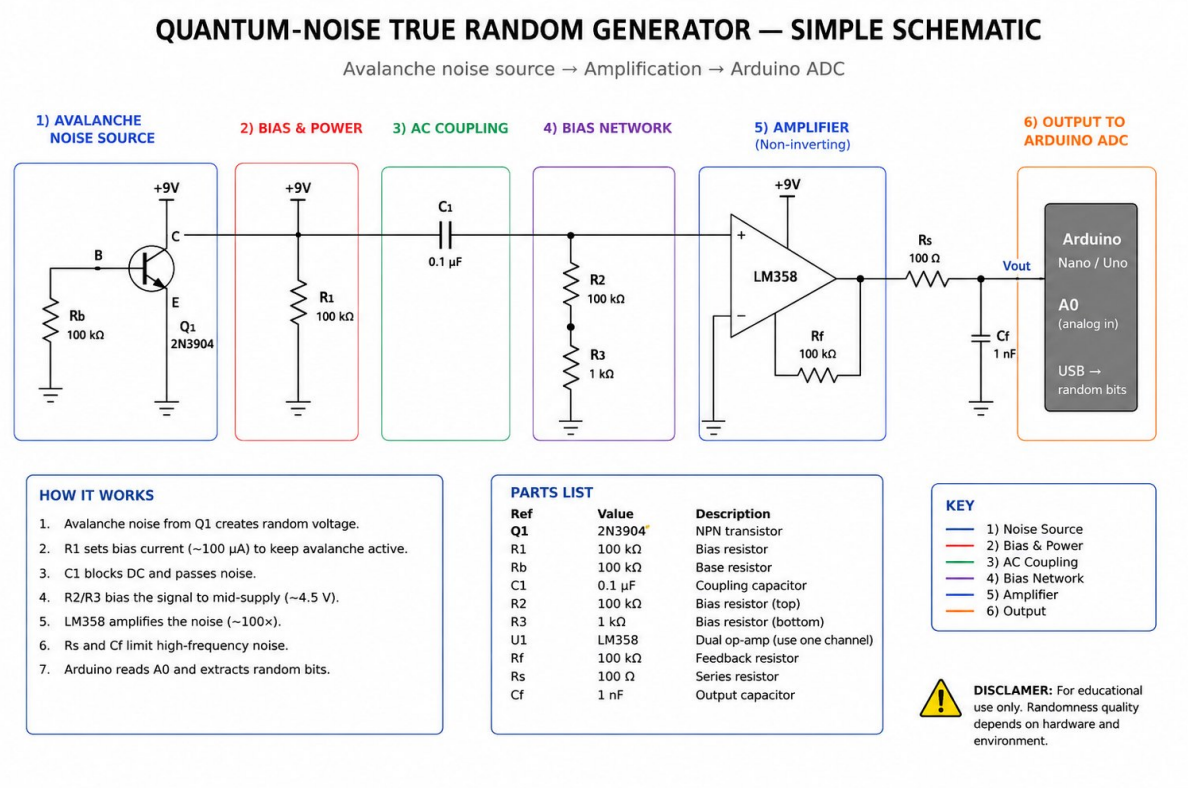
WHAT IT SHOWS A full schematic: a 2N3904 transistor with reverse-biased base–emitter junction as an avalanche-noise source, bias resistors, a 0.1 µF AC-coupling capacitor, an LM358 non-inverting amplifier (gain ≈ 100), an RC output filter and an Arduino ADC input, with a parts list and a note to validate randomness with NIST SP 800-22.

THE SCIENCE A reverse-biased silicon junction breaks down at about 6–7 V (the emitter–base junction of a 2N3904), and avalanche multiplication produces broadband shot noise from the quantum-random arrival of carriers. AC coupling removes the DC level, the op-amp amplifies about 100×, and the Arduino's ADC samples the noise. Debiasing (for example von Neumann whitening) and statistical tests are always required.

REALITY CHECK With a single 9 V supply the amplifier needs its input biased to about mid-supply (4.5 V), which needs two equal resistors from the supply to ground. The note says the 100 kΩ and 1 kΩ pair sets this bias, but 100 kΩ over 1 kΩ gives only about 0.09 V, so as drawn the divider values likely set the gain instead. Check the polarity and bias before building.

Fig. 67 Quantum-noise random generator: simple schematic

MAINSTREAM



Files: ccfb8d66-87b1-4cdc-81d7-64bc5a64f31f.png

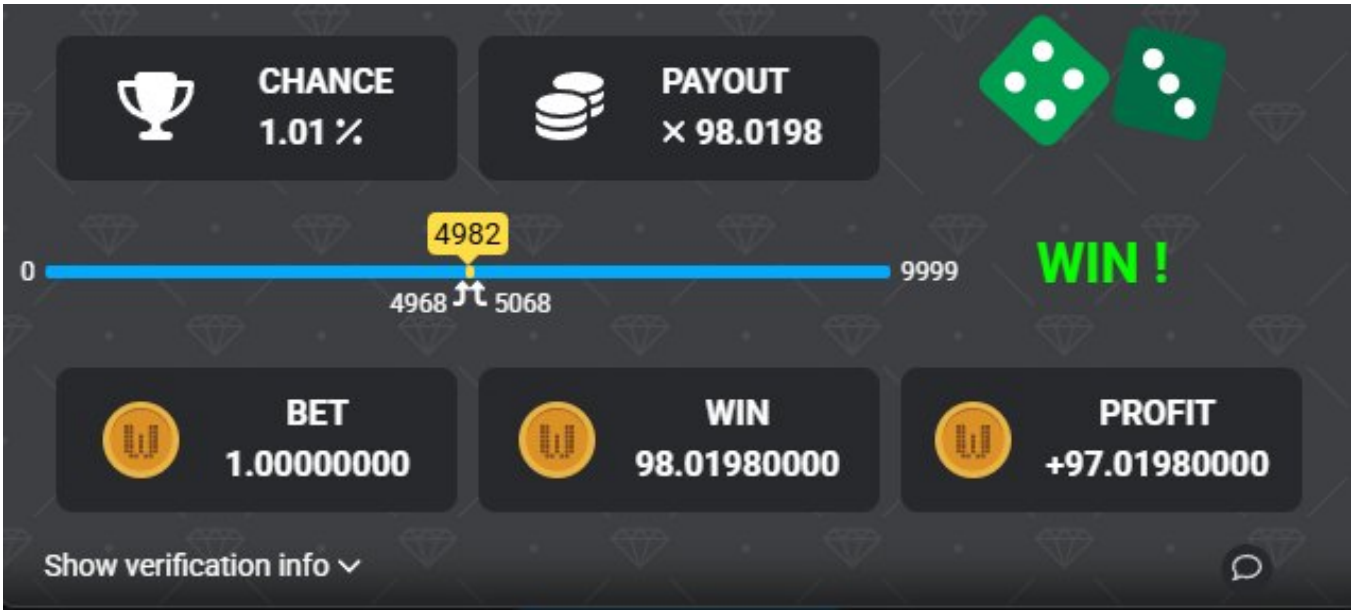
WHAT IT SHOWS A cleaner version of the previous circuit with six labelled stages (noise source, bias and power, AC coupling, bias network, amplifier, Arduino ADC output), a “how it works” list, a parts list and a disclaimer.

THE SCIENCE Same principle: avalanche noise of a reverse-biased junction, amplified and sampled. The steps are: avalanche noise from Q1, bias resistor sets a current near 100 µA, capacitor blocks DC, resistors bias the signal to mid-supply, LM358 amplifies about 100×, and the Arduino reads A0.

REALITY CHECK This version says R2/R3 (100 kΩ and 1 kΩ) bias the signal to about 4.5 V; that needs a 1:1 divider between the supply and ground, so the values need adjusting to avoid a floating input.

Fig. 68 Crypto dice-game result

CONTEXT



Files: 1968906714452774945-G1L2OAQakAAj4jy.png

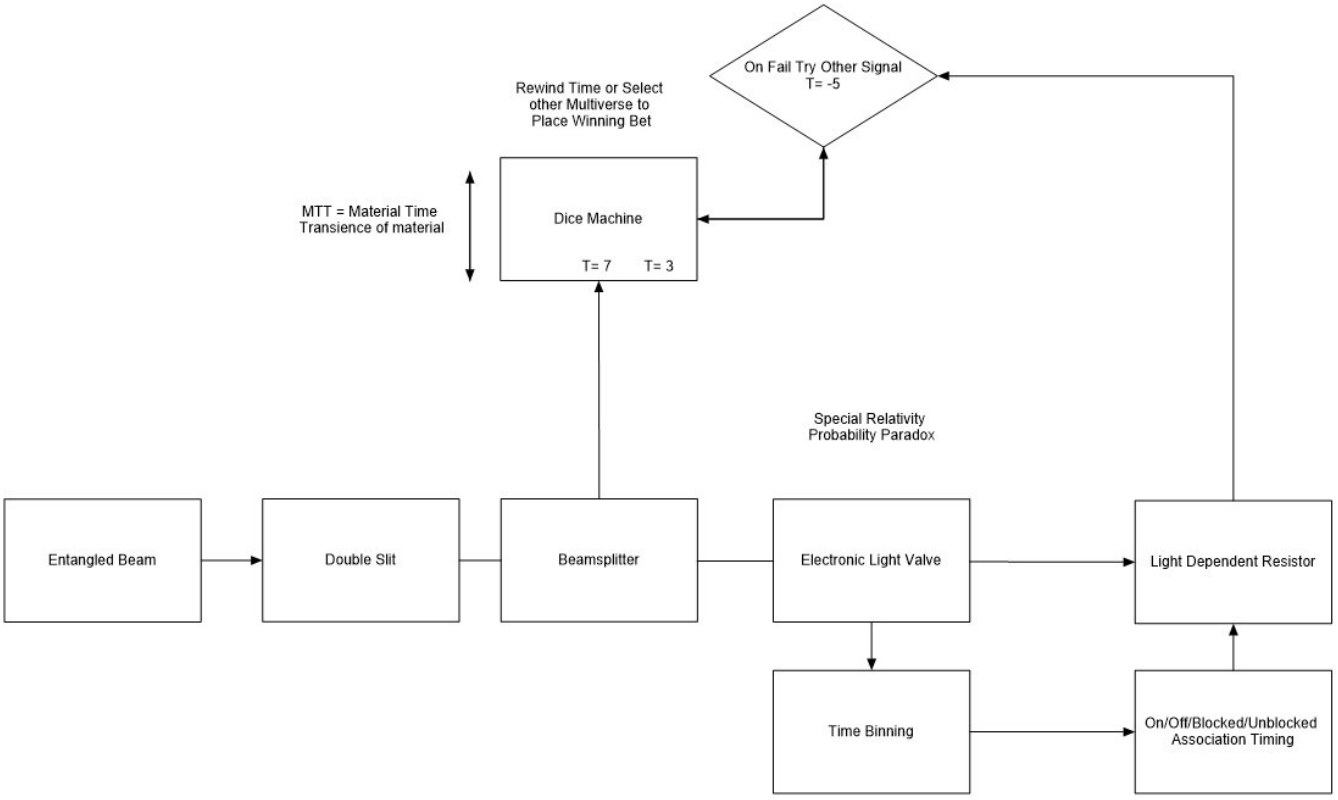
WHAT IT SHOWS A screenshot from a “provably fair” crypto dice game: chance 1.01 %, payout ×98.0198, roll 4982 in a 0–9999 range (win zone 4968–5068), bet 1.0000, win 98.0198, profit +97.0198.

THE SCIENCE Payout multiplier = 99 / chance % → 99 / 1.01 = 98.0198, so the house keeps 1 % on average: the expected return per unit bet is 0.0101 × 98.0198 = 0.99. A single win says nothing about an edge, since one win in 100 attempts is exactly what a 1.01 % chance predicts. Provably-fair schemes commit to a hashed server seed (SHA-256) before the roll and reveal it afterwards.

REALITY CHECK A win here is a lucky draw of a fair-but-taxed game, and the rest of this handbook’s dice diagrams do not change that.

Fig. 69 Dice machine with entangled beam, light valve and “rewind time”

SPECULATIVE



Files: 2021660310319641021-HA5hVUca8AEVXKM.png · Also in repository: MTT.png

WHAT IT SHOWS Entangled beam → double slit → beam splitter → electronic light valve → light-dependent resistor, with time binning and an on/off/blocked association timer; a dice machine (T = 7, T = 3) feeds a decision box, “On fail try other signal, T = −5”, with the note “Rewind time or select other multiverse to place winning bet” and “MTT = Material Time Transience of material”, titled “Special Relativity Probability Paradox”. (Image 130 is a copy.)

THE SCIENCE The optics are conventional (double slit, light valve, LDR). Retrocausal and many-worlds language is borrowed from foundations of quantum theory (delayed-choice experiments; Everett’s interpretation).

REALITY CHECK No interpretation of quantum mechanics lets an experimenter see a future dice result or rewind time; the no-signalling theorem forbids using entanglement to learn a distant random outcome early, and a game seeded by a hash of a secret server seed is not correlated with your optics. The device cannot beat the 1 % house edge.

Fig. 70 Light travel distance per hash count

CONTEXT

Unit	Distance (m)	Distance (mm)	Hashes (at 1 MH/s)
1	99.83	99,830.9	0.3330
2	199.66	199,661.8	0.6660
3	299.49	299,492.7	0.9990
4	399.32	399,323.6	1.3320
5	499.15	499,154.4	1.6650
6	598.99	598,985.3	1.9980
7	698.82	698,816.2	2.3310
8	798.65	798,647.1	2.6640
9	898.48	898,478.0	2.9970
10	998.31	998,308.9	3.3300

Files: 1999258749652074816-G77LPBQbsAArz8C.jpg

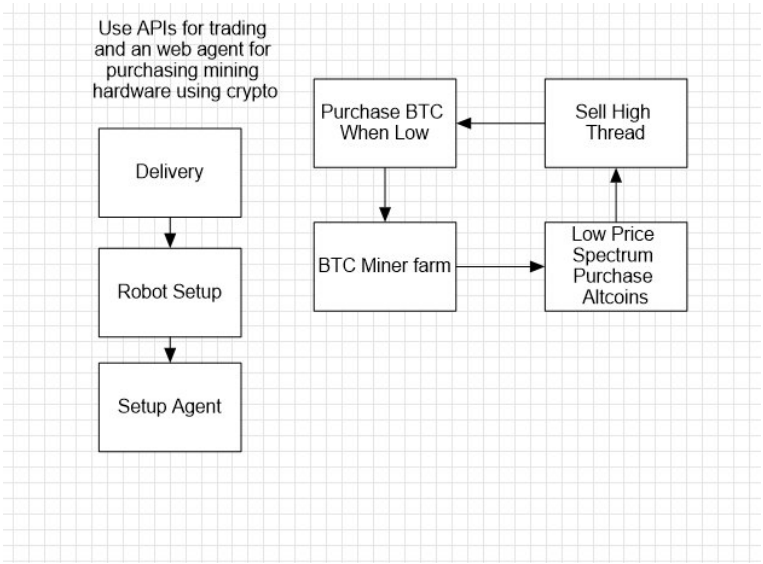
WHAT IT SHOWS A table with columns Unit, Distance (m), Distance (mm) and Hashes at 1 MH/s: Unit 1 = 99.83 m = 0.3330 hashes, Unit 2 = 199.66 m = 0.6660 hashes, up to Unit 10 = 998.31 m = 3.3300 hashes.

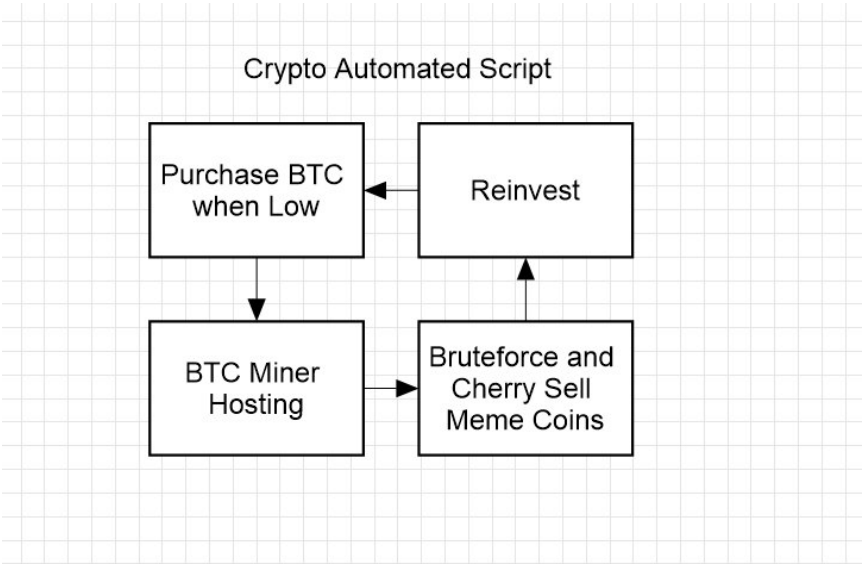
THE SCIENCE At 1 megahash per second one hash takes 1 µs, during which light travels 299.79 m. Unit 1 corresponds to 0.333 µs, and 299.79 m × 0.333 = 99.83 m, so the table is a unit conversion from light-travel distance to hash count.

REALITY CHECK The table is arithmetically correct. It is a ruler that measures distance in fractions of a hash interval.

Fig. 71 Automated crypto trading and mining flowcharts

CONTEXT





Files: 1986589854441087471-G5HI8LwacAEmgkW.png, 2024723916338655507-HBlDqpkbsAAtrMP.png

WHAT IT SHOWS Two flowcharts. Left: use APIs for trading and a web agent to buy mining hardware with crypto; delivery → robot set-up → set-up agent; a loop of buying BTC when low, running a miner farm, buying low-priced altcoins and selling high. Right: “Crypto Automated Script”: purchase BTC when low → BTC miner hosting → “Bruteforce and cherry sell meme coins” → reinvest.

THE SCIENCE These are process diagrams with no physics. Proof-of-work mining computes SHA-256 hashes; its expected profit is set by hash rate, electricity cost and network difficulty, and buy-low/sell-high loops carry price risk.

REALITY CHECK Not investment or trading advice. Automated strategies that assume they will always sell high can lose money quickly.

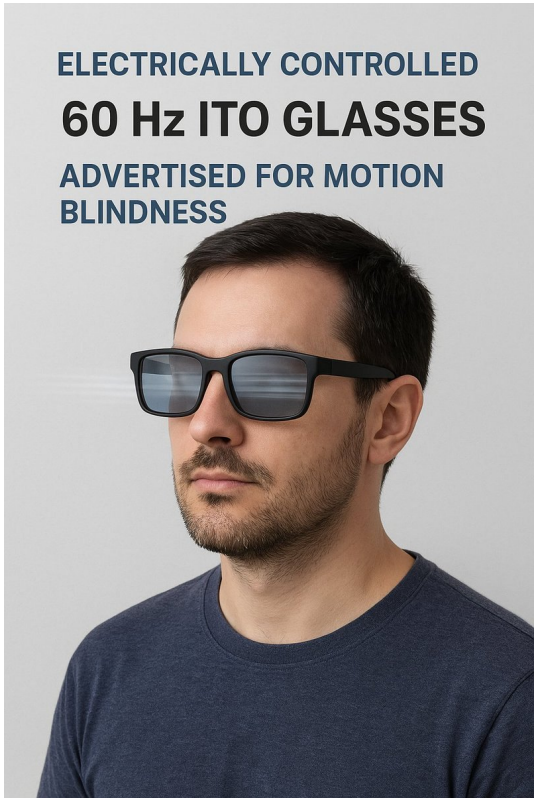
CHAPTER 10

Light, eyes and bodies: sensing and biophysics

Figures 72–82 · 11 entries

Fig. 72 Electrically controlled 60 Hz ITO glasses “for motion blindness”

MIXED



Files: 1972210249093599660-G16ywgEb0AQixz1.jpg

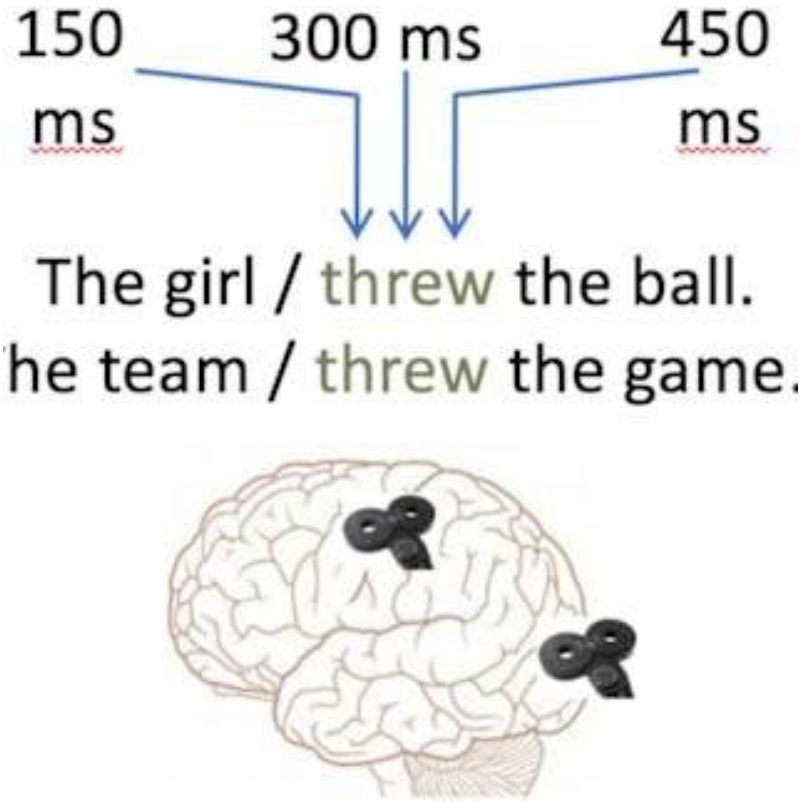
WHAT IT SHOWS A generated photo of a man in dark glasses with the caption “Electrically controlled 60 Hz ITO glasses advertised for motion blindness”.

THE SCIENCE Indium tin oxide (ITO) is a transparent conductor used as the electrode on liquid-crystal shutter lenses. Electronically shuttered “strobe” glasses flicker between clear and dark, giving the eye a series of freeze-frames; sports-vision and research products use rates of a few Hz to tens of Hz. Akinetopsia (“motion blindness”) is a rare disorder of the motion area V5/MT of the cortex, in which moving objects appear as a series of stills.

REALITY CHECK Strobed vision does not restore normal motion perception in akinetopsia. It is more often used to imitate the condition or to train sports reaction. Claims of therapeutic benefit are unproven.

Fig. 73 Time course of language processing: “threw the ball” versus “threw the game”

MAINSTREAM



Files: 2020281450168480005-HAI7RFfbcAAia8M.jpg

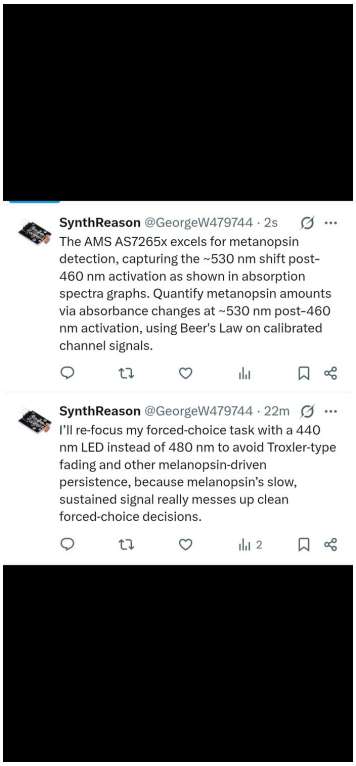
WHAT IT SHOWS A brain diagram with timing arrows at 150 ms, 300 ms and 450 ms above two sentences: “The girl threw the ball.” and “The team threw the game.”, with probe markers on the brain.

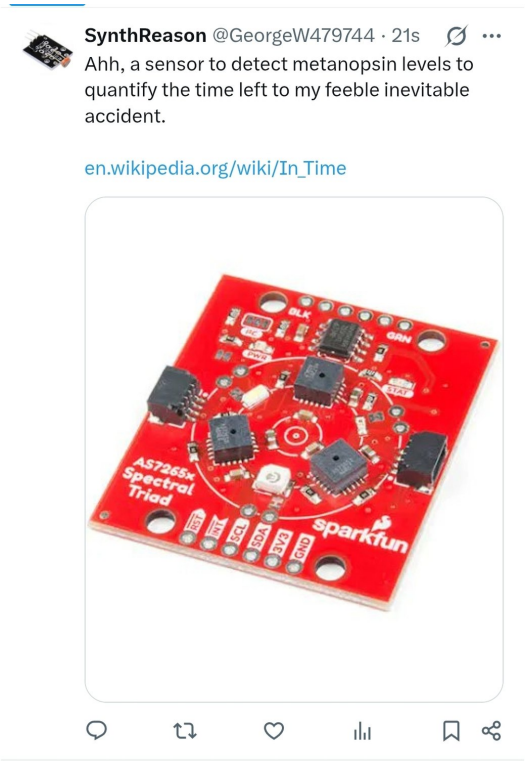
THE SCIENCE Electrical and magnetic recordings show that word recognition begins about 150–200 ms after a word appears, semantic integration peaks near 400 ms (the N400 response; Kutas and Hillyard, Science 1980), and structural re-analysis around 600 ms (the P600). A literal use of “threw” (a ball) and a figurative use (a game, meaning to lose deliberately) differ in how much context has to be integrated, which shows up in these components.

REALITY CHECK The three timings are typical rather than exact and vary with reading speed and experimental design.

Fig. 74 Melanopsin and the AS7265x spectral sensor

MIXED





Files: 3488.jpg, 3493.jpg

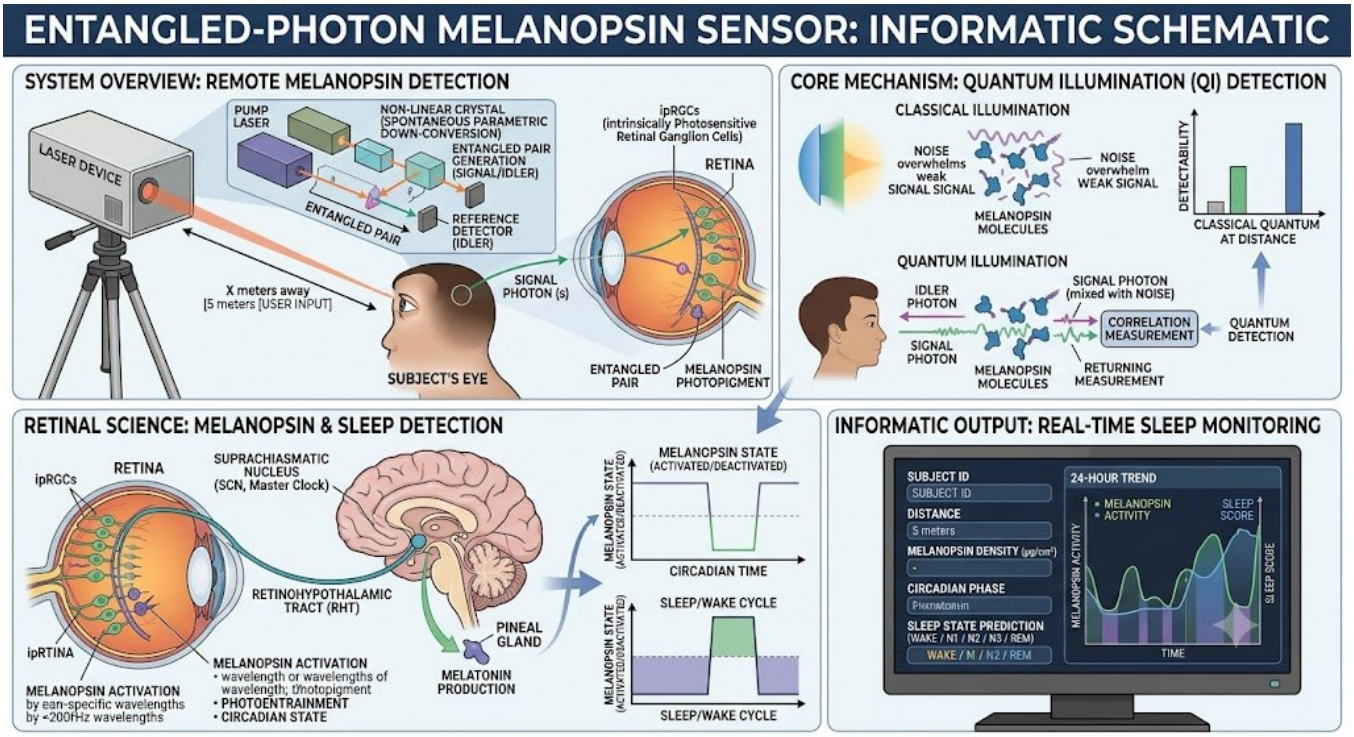
WHAT IT SHOWS Three posts and a product photo. They discuss using an AS7265x spectral triad sensor to detect melanopsin, measuring a ~530 nm absorbance shift after 460 nm activation with Beer's law, and switching a forced-choice task from a 480 nm to a 440 nm LED to avoid Troxler-type fading and melanopsin's slow persistent signal. A separate post shows a SparkFun AS7265x board as the proposed detector.

THE SCIENCE Melanopsin is the photopigment of intrinsically photosensitive retinal ganglion cells (ipRGCs) with peak sensitivity near 480 nm; it drives circadian entrainment and the sustained pupil constriction after blue light (the post-illumination pupil response). It is bistable or tristable, with states absorbing between roughly 470 and 580 nm (Emanuel and Do, Neuron 2015). Beer's law, $A = \varepsilon c l$, relates absorbance to concentration. Troxler fading is the disappearance of a steady peripheral stimulus. The AS7265x is an 18-channel spectral sensor covering 410–940 nm.

REALITY CHECK The AS7265x measures the spectrum of light that reaches the sensor; it cannot read the state of a pigment inside someone's retina. The 530 nm figure is not a standard melanopsin value. Also 440 nm still activates melanopsin substantially, so it reduces but does not remove the effect.

Fig. 75 Entangled-photon melanopsin sensor for remote sleep monitoring

SPECULATIVE



Files: 8b6839ab-6d22-498f-ad92-cee711ad51d4.jpg

WHAT IT SHOWS An informatic schematic: a laser device several metres from a subject's eye emits SPDC signal and idler photons, and "quantum illumination" is used to detect the signal returning from melanopsin in the retina; panels show ipRGCs, the suprachiasmatic nucleus, melatonin production and a sleep-monitoring dashboard.

THE SCIENCE Quantum illumination (Lloyd, 2008; Tan et al., 2008) is a real protocol: a signal photon is sent out, its entangled partner is kept, and correlation measurements can detect a weak reflecting target in bright noise, with up to a 6 dB error-exponent advantage. Melanopsin, the SCN and melatonin are the real circadian pathway.

REALITY CHECK A retina reflects a tiny fraction of light, and eye-safety limits cap the power that can be sent. The melanopsin absorbance change is small, and sleep state is already measured far more simply by EEG, actigraphy, pupillometry or radar. No experiment has detected pigment states in a live eye at a distance with entangled light.

Fig. 76 Direct-detection hierarchy of body movements

MAINSTREAM

Direct Detection Hierarchy

What becomes immediately available without a wall:

Signal	Amplitude	Detection Difficulty
Breathing	~5–10mm chest movement	Trivial
Heartbeat (chest)	~1–3mm	Trivial
Arterial pulse (neck)	~0.5–2mm	Easy
Skull surface pulse	~50–200μm	Straightforward
Scalp micromovement	~10–50μm	Achievable
Cortical vessel pulse	~1–10μm	Requires array
Capillary bed	~10–100nm	Challenging
Neural swelling	~1–10nm	Very challenging
Dendritic activity	~0.1–1nm	Extreme

Files: HEULSVobYAYyR_F.png

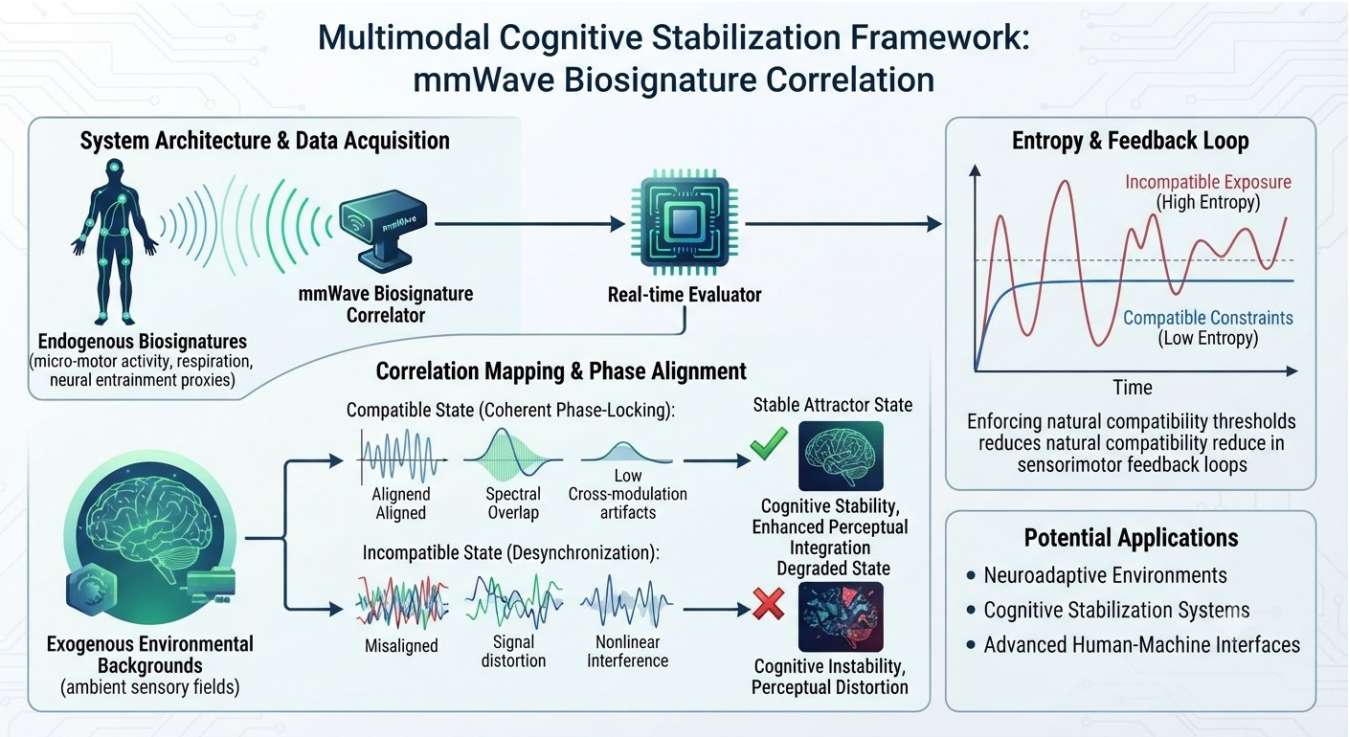
WHAT IT SHOWS A table of body signals by amplitude and detection difficulty: breathing 5–10 mm (trivial), heartbeat 1–3 mm (trivial), arterial pulse 0.5–2 mm (easy), skull surface pulse 50–200 μm (straightforward), scalp micromovement 10–50 μm (achievable), cortical vessel pulse 1–10 μm (requires array), capillary bed 10–100 nm (challenging), neural swelling 1–10 nm (very challenging), dendritic activity 0.1–1 nm (extreme).

THE SCIENCE These ranges are realistic. Ballistocardiography, laser Doppler vibrometry and radar detect mm-scale chest motion. Nerve fibres swell by nanometres during activity (Tasaki; Cohen, 1968), and phase-sensitive interferometry and OCT have detected this in tissue.

REALITY CHECK The ordering is sensible, since each row needs about 10× finer displacement sensitivity than the one above.

Fig. 77 Millimetre-wave biosignature correlation (“cognitive stabilization”)

SPECULATIVE



Files: wr1o4cwr1o4cwr1o.png

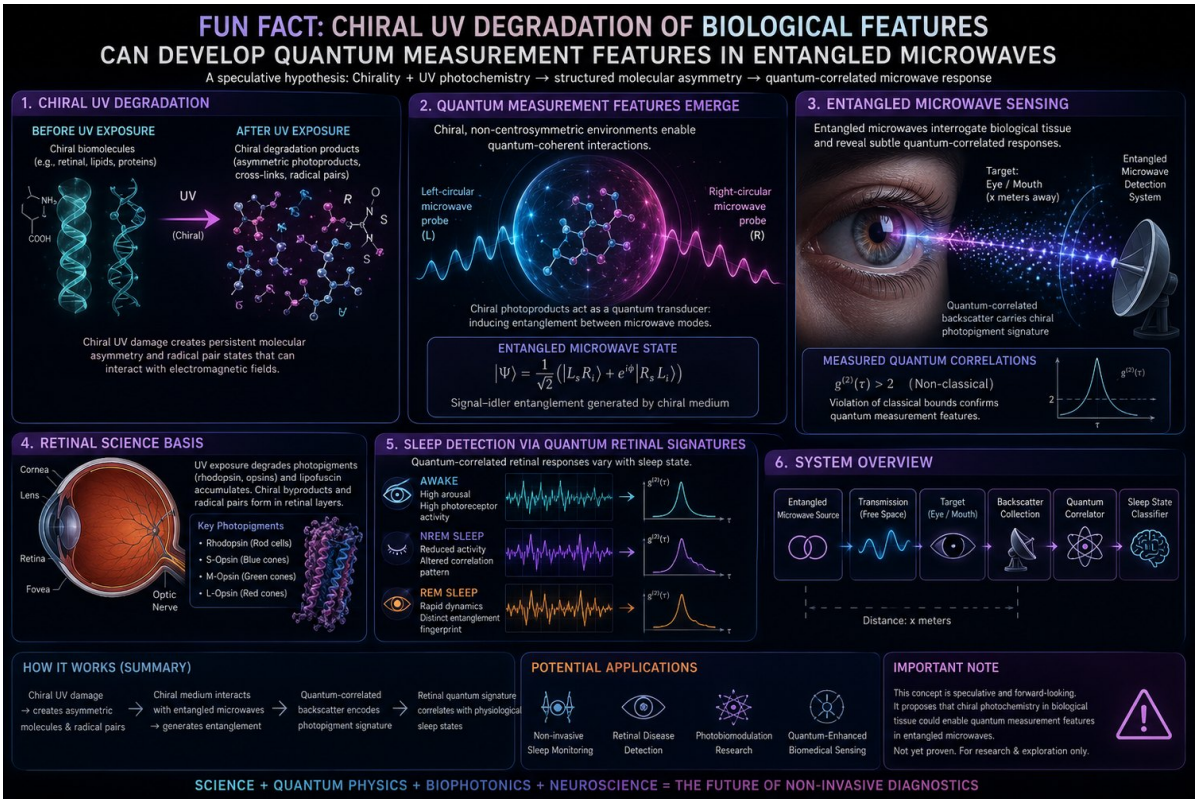
WHAT IT SHOWS An infographic: an mmWave radar reads a person’s micro-motor activity, respiration and neural-entrainment proxies, a real-time evaluator correlates them with ambient sensory fields, and “compatible” (phase-locked) states are said to give stable cognition while “incompatible” ones cause distortion, with an entropy and feedback loop.

THE SCIENCE Millimetre-wave FMCW radars at 60–77 GHz measure phase, so displacement resolution is a fraction of the 4–5 mm wavelength (tens of micrometres), enough to see breathing and heartbeat contactlessly (Droitcour et al., 2004). Phase-locking of brain rhythms to rhythmic stimuli (neural entrainment) is also real.

REALITY CHECK The step from radar-measured chest motion to “cognitive stability” is not supported by evidence; the panel names applications, not experiments.

Fig. 78 Chiral UV degradation and entangled-microwave sensing of the retina

SPECULATIVE



Files: df058dda-9fe6-4cea-8eaf-457a0659cacd.png

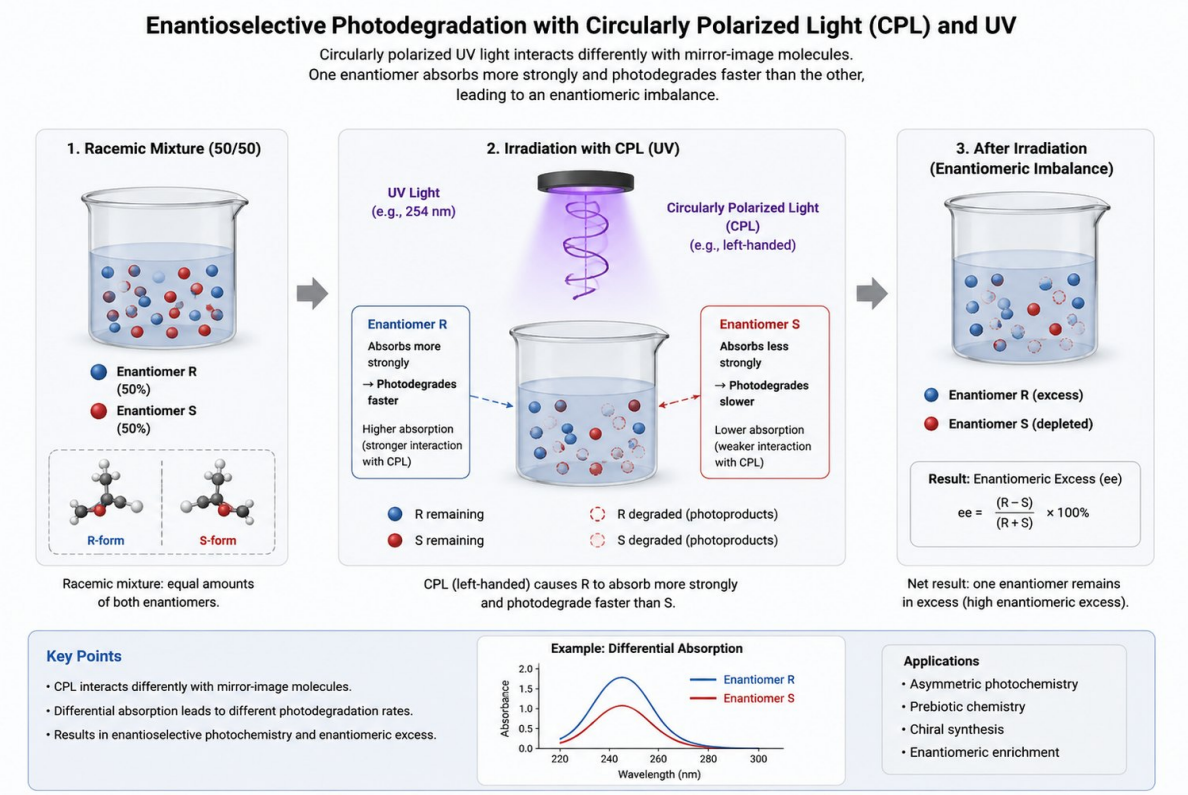
WHAT IT SHOWS A “fun fact” infographic marked “speculative hypothesis”: UV exposure turns biomolecules into chiral photoproducts that could act as quantum transducers for entangled microwaves, which would read the eye from metres away, measure quantum correlations ($g^2(\tau) > 2$, labelled non-classical) and classify sleep stages.

THE SCIENCE UV photochemistry of tryptophan and other biomolecules does produce photoproducts (photo-oxidation, cross-links, radical pairs), and microwave quantum illumination has been demonstrated with superconducting circuits (Barzanjeh et al., Sci. Adv. 2020).

REALITY CHECK Two problems. Non-classical light shows up as $g^2(0)$ below 1 (antibunching); thermal light already has $g^2 = 2$, and classical intensity noise can push g^2 above 2, so $g^2 > 2$ does not prove a quantum effect. And microwave photons (wavelengths of centimetres) cannot resolve retinal photochemistry at metre distances. The panel itself labels the idea speculative.

Fig. 79 Enantioselective photodegradation with circularly polarised light

MAINSTREAM



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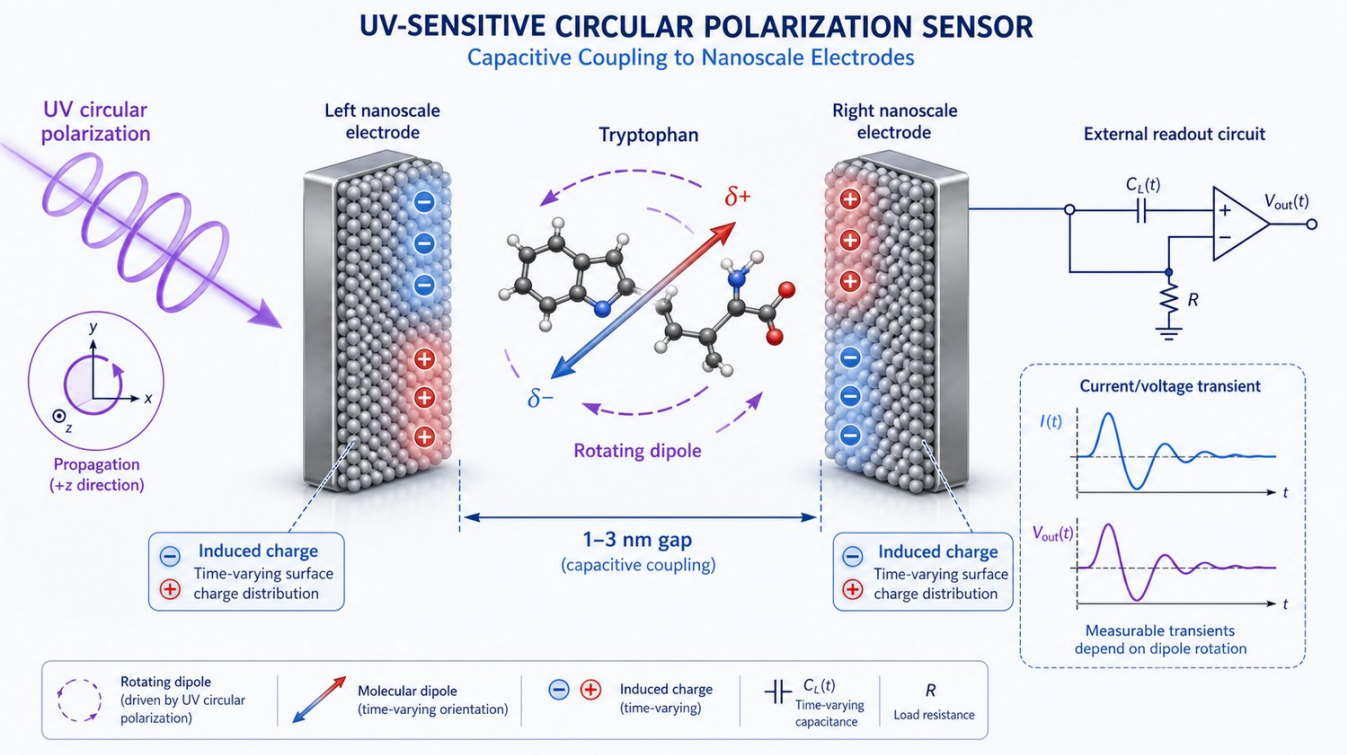
WHAT IT SHOWS A three-step scheme: a racemic (50/50) mixture; irradiation by left-handed circularly polarised UV (254 nm), which the R enantiomer absorbs more strongly and degrades faster; an enantiomeric excess left behind. A key box lists asymmetric photochemistry, prebiotic chemistry, chiral synthesis and enrichment.

THE SCIENCE This is established. Chiral molecules absorb left- and right-circularly polarised light differently (circular dichroism), so CPL destroys one enantiomer preferentially. Kuhn and Knopf (1930) demonstrated asymmetric photolysis, Meierhenrich et al. (Angew. Chem. 2010) produced small enantiomeric excesses of amino acids with synchrotron CPL, and CPL in star-forming regions has been proposed as an origin of homochirality (Bailey et al., Science 1998). Enantiomeric excess is ee = (R − S)/(R + S).

REALITY CHECK Excesses are small (a few percent) and require most of the material to be destroyed, so this is a demonstration of principle, not a route to bulk products.

Fig. 80 Circular-polarisation UV sensor with a tryptophan “rotating dipole” between nanoelectrodes

SPECULATIVE



Files: tryptophan-sensor.png

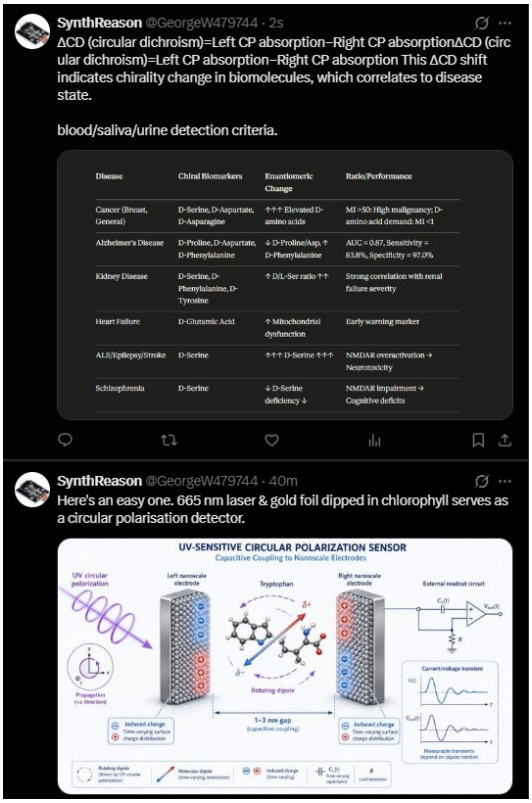
WHAT IT SHOWS Left- and right-hand nanoscale electrodes separated by a 1–3 nm gap, with a tryptophan molecule between them; UV circularly polarised light is said to drive a rotating dipole that induces time-varying charge on the electrodes, with a readout circuit (C_c(t), R) and current and voltage transients.

THE SCIENCE Circularly polarised light detectors are an active field (chiral perovskites, plasmonic metasurfaces). Tryptophan absorbs UV near 280 nm and is intrinsically chiral. Nanogap electrodes are standard in molecular electronics.

REALITY CHECK Electronics cannot follow the dipole. A 280 nm optical field oscillates at about 10^{15} Hz, and a readout amplifier has GHz bandwidth at most, so a “rotating dipole” at optical frequency would average out. The much larger absorption difference of a circular-dichroism sensor comes from chiral absorbers, and the tryptophan’s anisotropy factor is tiny (g ~ 10^{-4}–10^{-3}).

Fig. 81 Chirality as a disease marker, and a chlorophyll/gold circular-polarisation detector

MIXED



Files: unnamed (1).png

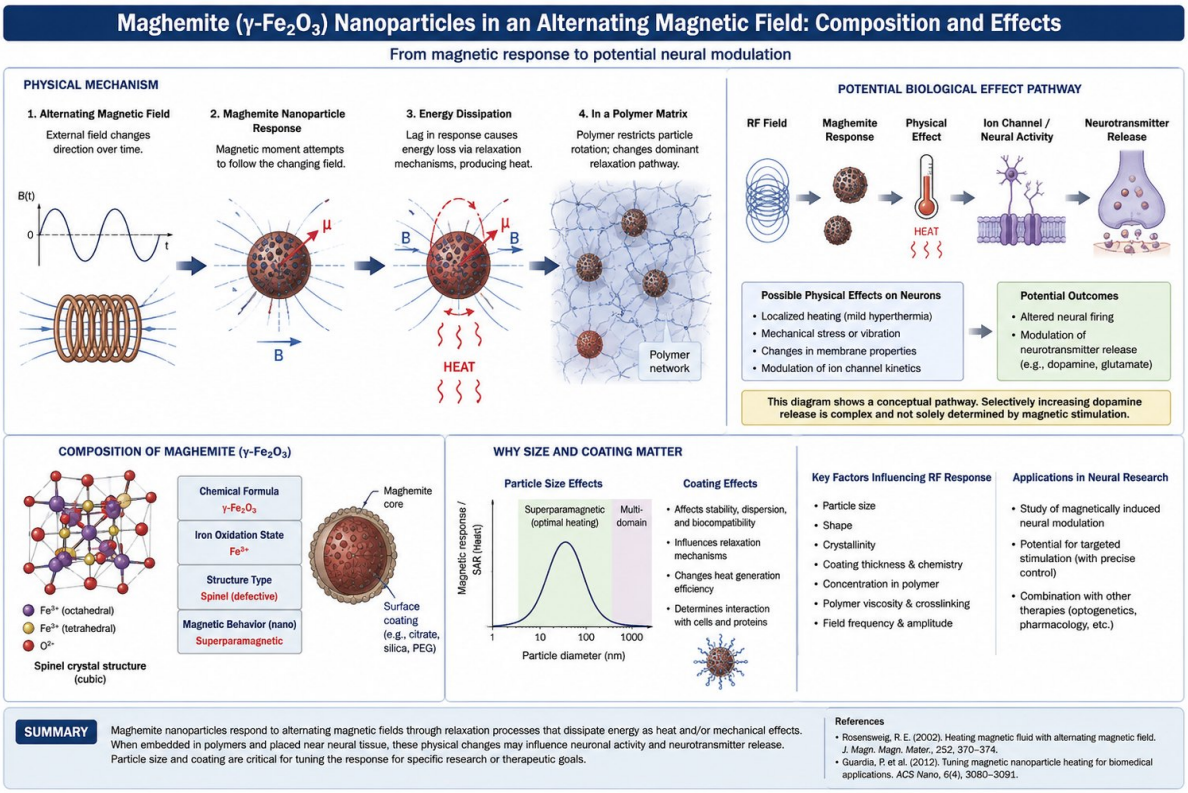
WHAT IT SHOWS A post defining ΔCD = left-CP absorption − right-CP absorption and claiming that chirality shifts in biomolecules correlate with disease, with a table of conditions (cancer, Alzheimer's, kidney disease, heart failure, ALS/stroke, schizophrenia), D-amino acid biomarkers and performance figures; a second post says a 665 nm laser and gold foil dipped in chlorophyll can act as a circular-polarisation detector.

THE SCIENCE D-amino acids are real biomarkers. D-serine is a co-agonist at the NMDA receptor; it is reduced in schizophrenia (Hashimoto et al., 2003), raised in kidney disease (Kimura et al., Sci. Rep. 2016) and altered in ALS (Sasabe et al., 2012). Chlorophyll a has its red absorption band near 665 nm, and is chiral.

REALITY CHECK Circular dichroism of blood or saliva is dominated by proteins; D-amino acids at micromolar levels are measured with chiral chromatography and mass spectrometry, not CD spectrometers. The performance figures in the table are unsourced and should be treated as unverified. A dip-coated foil's signal would be very weak.

Fig. 82 Maghemite nanoparticles in an alternating magnetic field

MAINSTREAM



Files: 9c46ce2e-ce91-4ca3-ae81-85abfd54b391.png

WHAT IT SHOWS A dense infographic: an alternating field flips the moment of γ-Fe_2O_3 nanoparticles; the lag produces relaxation heating; a polymer matrix changes relaxation; heat and mechanical effects may modulate neurons and neurotransmitter release; panels cover particle composition, size and coating.

THE SCIENCE Magnetic hyperthermia is well established (Rosensweig, JMMM 2002; Guardia et al., ACS Nano 2012). Superparamagnetic nanoparticles below about 20–30 nm reverse their moment by Néel relaxation, $\tau_N = \tau_0 \exp(KV/k_BT)$, or by physical rotation (Brownian relaxation), and the loss of energy per cycle heats the particle. Magnetothermal stimulation of neurons has been demonstrated (Huang et al., Nature Nanotechnology 2010; Chen et al., Science 2015).

REALITY CHECK The panel itself notes that selectively raising dopamine release is complex. Heating a few degrees is measurable, but specific neurotransmitter control is still a research goal.

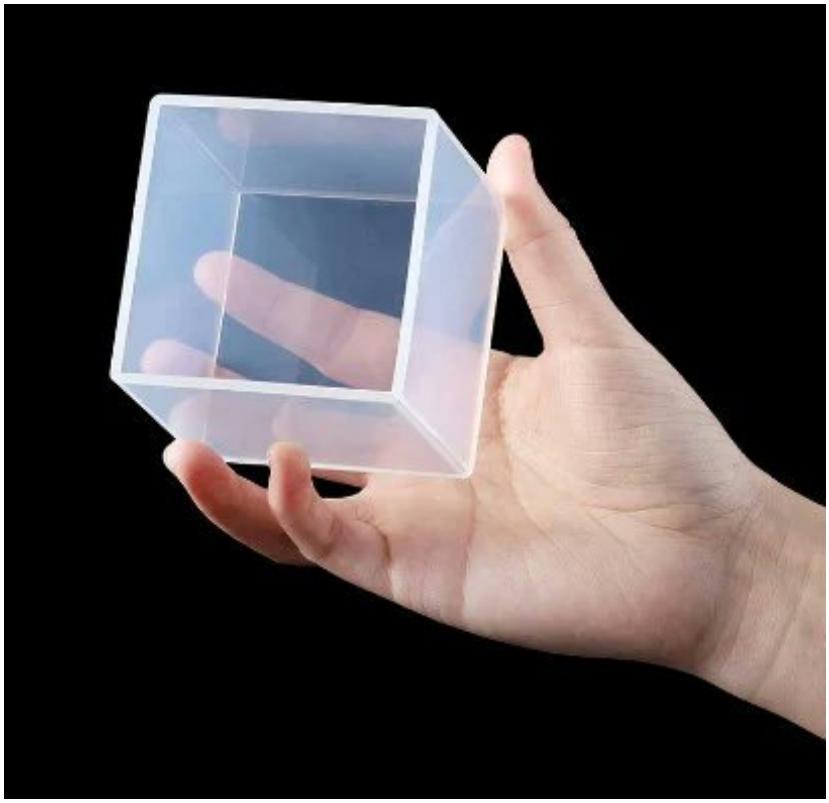
CHAPTER 11

Materials, devices and batteries

Figures 83–101 · 19 entries

Fig. 83 Clear resin cube held in a hand

CONTEXT



Files: 1970060842189443291-G1cOJKEacAAUH-y.jpg

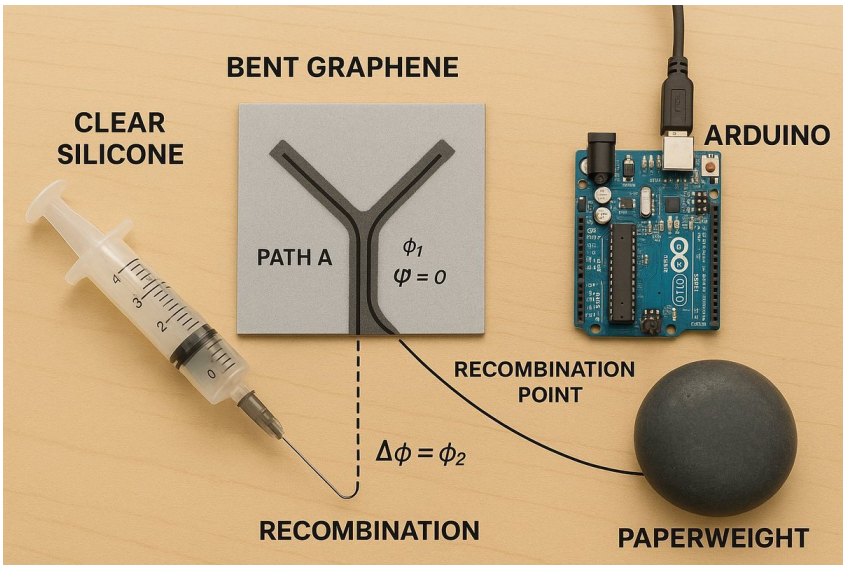
WHAT IT SHOWS A photograph of a transparent, square resin or silicone block held in a hand against a black background.

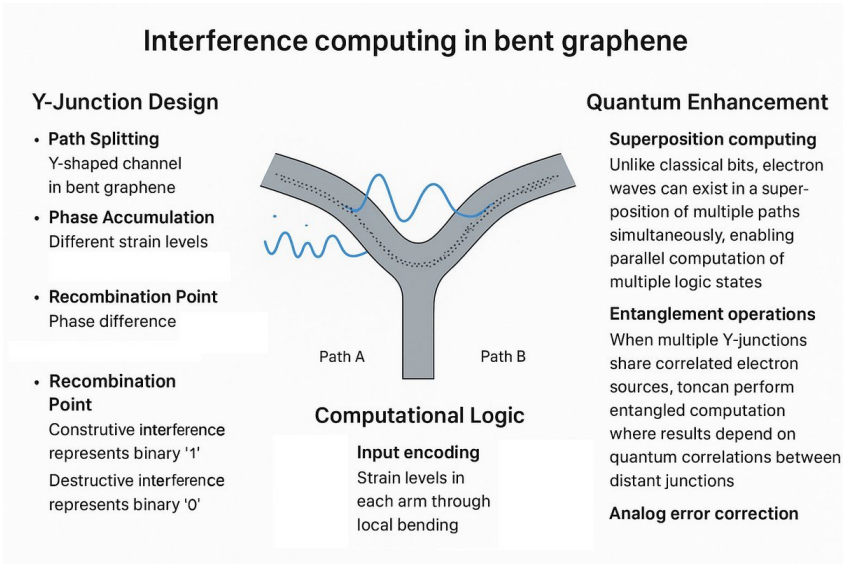
THE SCIENCE Clear casting resins and silicones are optically transparent (refractive index about 1.4–1.55) and can be moulded, so they are convenient hosts for embedded sensors, optical fibres or nanoparticles. Optical-grade materials are checked for bubbles and internal stress.

REALITY CHECK No physics claim is made by the photo itself.

Fig. 84 Bent graphene interference computing and a Y-junction test rig

MIXED





Files: 1970844883931074689-G1nY9SbbAAA8AnN.jpg, 1970844883931074689-G1nY9VNbEAALEcF.jpg

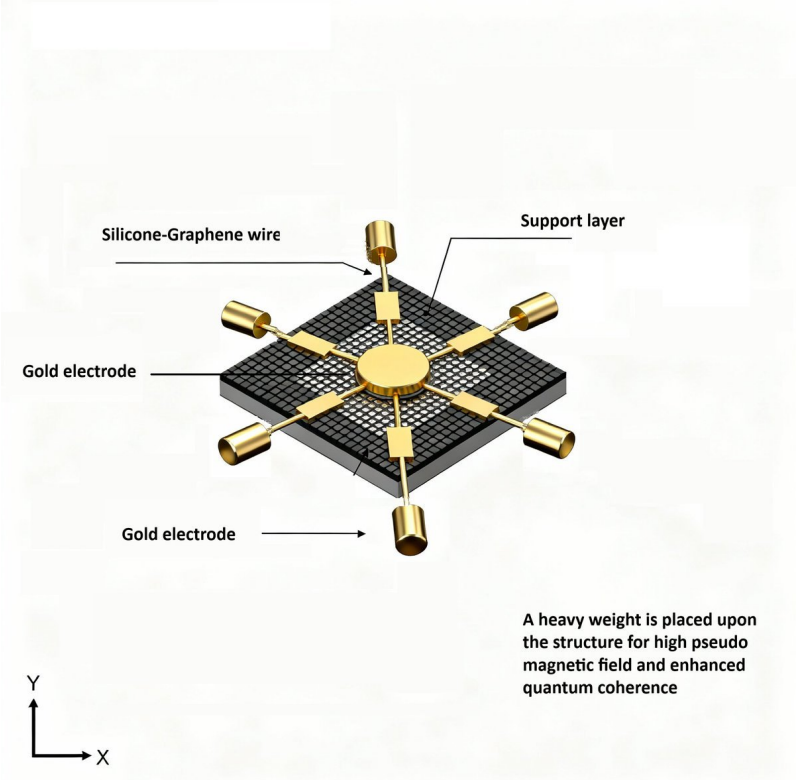
WHAT IT SHOWS Left: a bench photo labelled with a Y-shaped bent graphene strip on clear silicone, a syringe, an Arduino and a paperweight, with path A, phase φ_1 and a recombination point ($\Delta\varphi = \varphi_2$). Right: an infographic “Interference computing in bent graphene”: a Y-junction splits electron waves, strain sets the phase in each arm, and constructive interference means 1, destructive means 0.

THE SCIENCE Electron interferometers in graphene exist (Rickhaus et al., Nature Communications 2013). Strain changes the electron phase, and strained graphene generates pseudo-magnetic gauge fields (Guinea, Katsnelson and Geim, Nature Physics 2010). A Y-junction is the electron equivalent of a beam splitter, and interference-based logic (quantum-interference effect transistors) is a long-standing idea.

REALITY CHECK Coherence is the obstacle: graphene’s phase-coherence length is about a micrometre at room temperature, so a device with bending by a syringe, silicone and a paperweight is likely incoherent. Analog interference gives one bit per output, not “parallel computation of multiple states”.

Fig. 85 Silicone–graphene lattice with gold electrodes under a weight

MIXED



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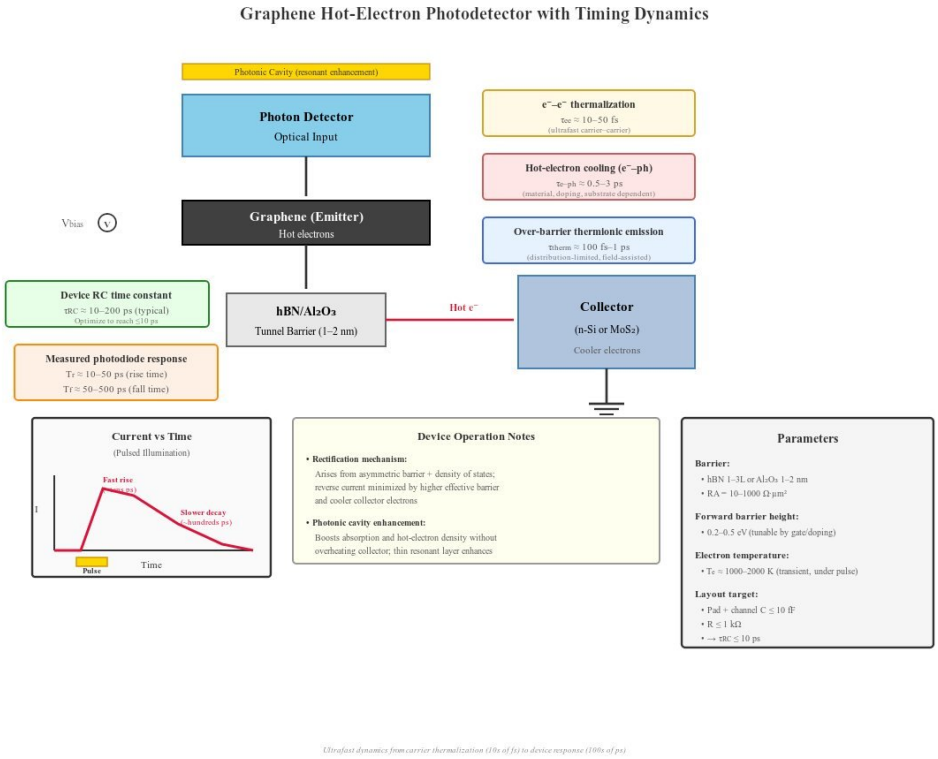
WHAT IT SHOWS An exploded isometric drawing of a grid of graphene-in-silicone with eight gold electrodes on a support layer, and the note “A heavy weight is placed upon the structure for high pseudo-magnetic field and enhanced quantum coherence”.

THE SCIENCE Strained graphene can host pseudo-magnetic fields exceeding 300 T (Levy et al., Science 2010, in nanobubbles), acting on electrons like a real field without breaking time-reversal symmetry. The pseudo-field is proportional to the strain gradient.

REALITY CHECK Uniform compression from a weight makes a uniform strain and no pseudo-field. You need a triaxial, non-uniform strain (Guinea et al., 2010), and graphene breaks at about 20–25 % strain. Nothing shows that pressing on the structure improves coherence.

Fig. 86 Graphene hot-electron photodetector with timing dynamics

MAINSTREAM



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WHAT IT SHOWS A block diagram: a photon and an optional photonic cavity feed a graphene emitter that generates hot electrons; they cross an hBN/Al_2O_3 tunnel barrier (1–2 nm) to a collector (n-Si or MoS_2). Timing boxes list e–e thermalisation of 10–50 fs, e–ph cooling of 0.5–3 ps, RC constants of 10–200 ps, and measured rise and fall times of about 10–50 ps and 50–100 ps; a current-versus-time sketch and parameter list are included.

THE SCIENCE This is a good summary of real physics. Photoexcited electrons in graphene share energy among themselves in tens of femtoseconds, then cool by phonon emission over about a picosecond, so the electron temperature can exceed the lattice temperature for a short time. Hot carriers that cross a thin barrier give a fast internal-photoemission signal, and van der Waals photodetectors with picosecond response have been demonstrated (Massicotte et al., Nature Nanotechnology 2016). Device speed is then limited by the RC time.

REALITY CHECK Nothing to correct: the timescales quoted match the literature.

Fig. 87 Black powder in a petri dish

CONTEXT

Files: 2000165115845558434-G8IDkUfakAAVFPk.jpg

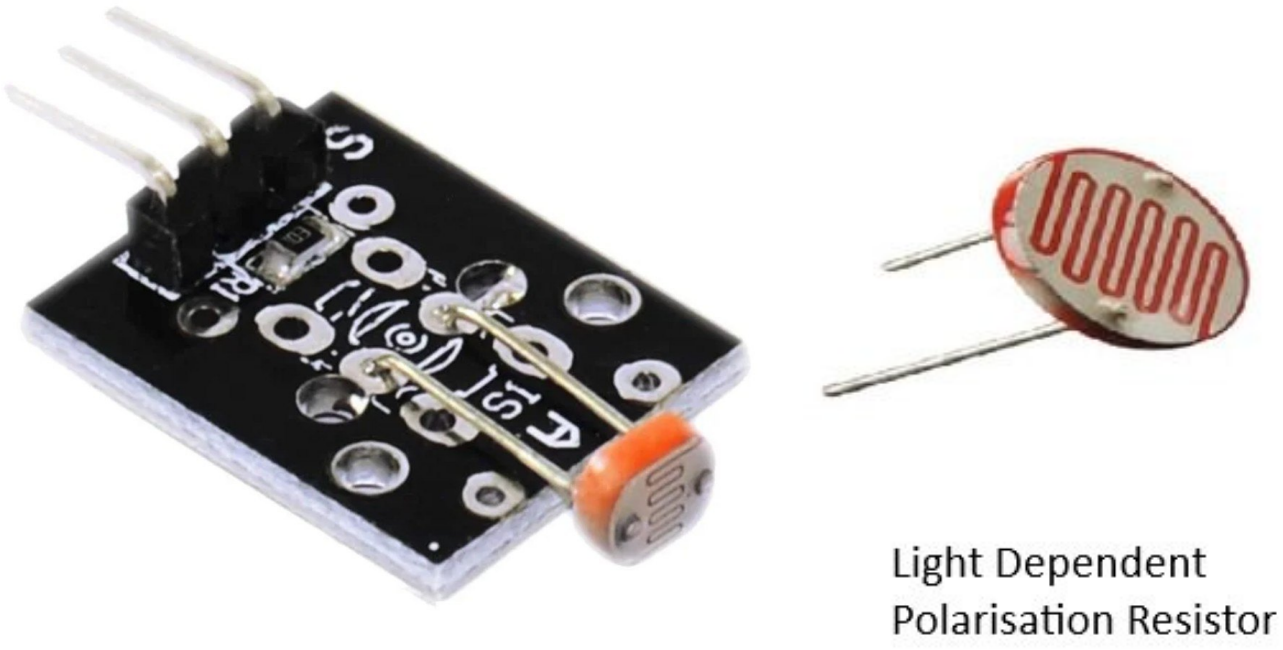
WHAT IT SHOWS A photograph of a fine black powder in a glass dish.

THE SCIENCE Carbon nanotubes, graphite and carbon black all look like this. Carbon nanotubes dispersed in epoxy form a conductive network above a percolation threshold that can be as low as a few thousandths of a weight percent in well-dispersed samples, and typical values are 0.1–1 wt %.

REALITY CHECK A photo cannot identify the powder; only its conductivity, Raman spectrum or electron microscopy can.

Fig. 88 Light-dependent resistor modules (KY-018 style)

MAINSTREAM



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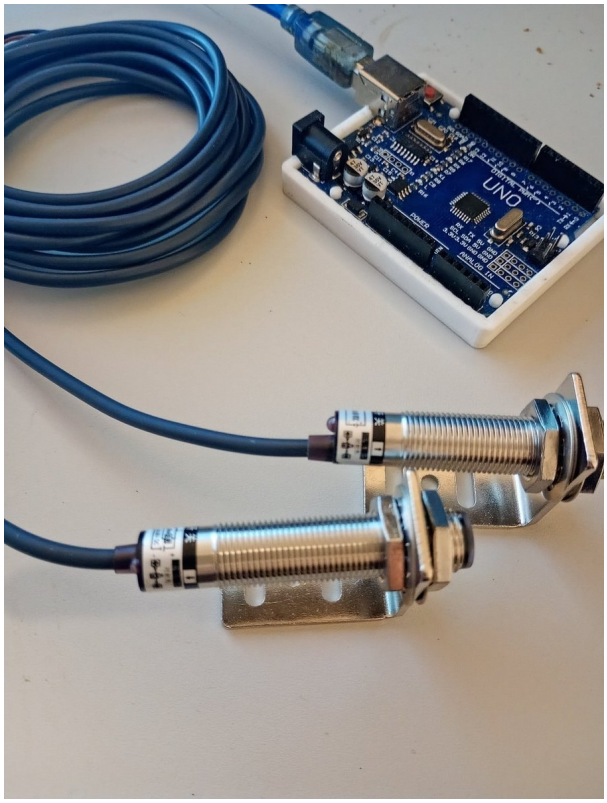
WHAT IT SHOWS Left: a small Arduino-compatible module with a photoresistor on a black board. Right: a bare photoresistor captioned “Light Dependent Polarisation Resistor”.

THE SCIENCE A photoresistor (LDR) is a cadmium sulfide film whose resistance falls from about 1 MΩ in the dark to a few kΩ in bright light. It responds in about 20–30 ms and peaks near 540 nm, and a module puts it in a voltage divider so an Arduino can read the level.

REALITY CHECK A standard LDR is not polarisation-sensitive, so the right-hand caption is a rename, not a property. Polarisation sensitivity needs an anisotropic absorber (such as black phosphorus) or a wire-grid polariser in front.

Fig. 89 Two inductive proximity sensors with an Arduino

MAINSTREAM



Files: 1987792069944021305-G5YOWmibAAAk8AX.jpg

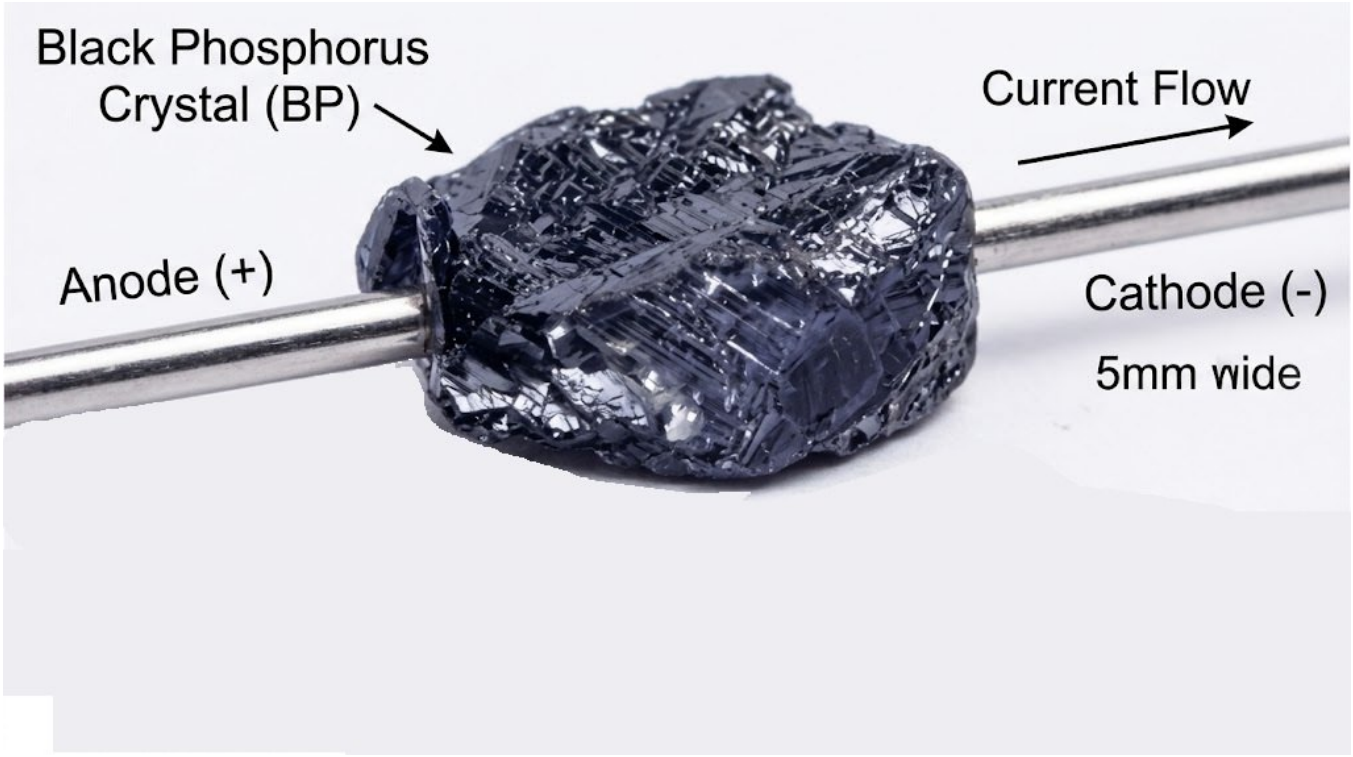
WHAT IT SHOWS A photo: an Arduino Uno clone in a white 3D-printed holder, with two M12 cylindrical inductive proximity sensors mounted on an angle bracket.

THE SCIENCE An inductive proximity sensor is an LC oscillator whose amplitude falls when a metal object enters its field (eddy-current loss); M12 types typically sense to about 4 mm. Two sensors a known distance d apart give speed $v = d/\Delta t$ from the time between triggers, like a light gate.

REALITY CHECK Most M12 sensors run at 10–30 V DC and output at supply level, so the signal must be level-shifted or divided before an Arduino’s 5 V input.

Fig. 90 Black phosphorus crystal with electrodes

MAINSTREAM



Files: 9b01abe4-60a9-4fa7-b3ec-262ea83d77e3.jpg

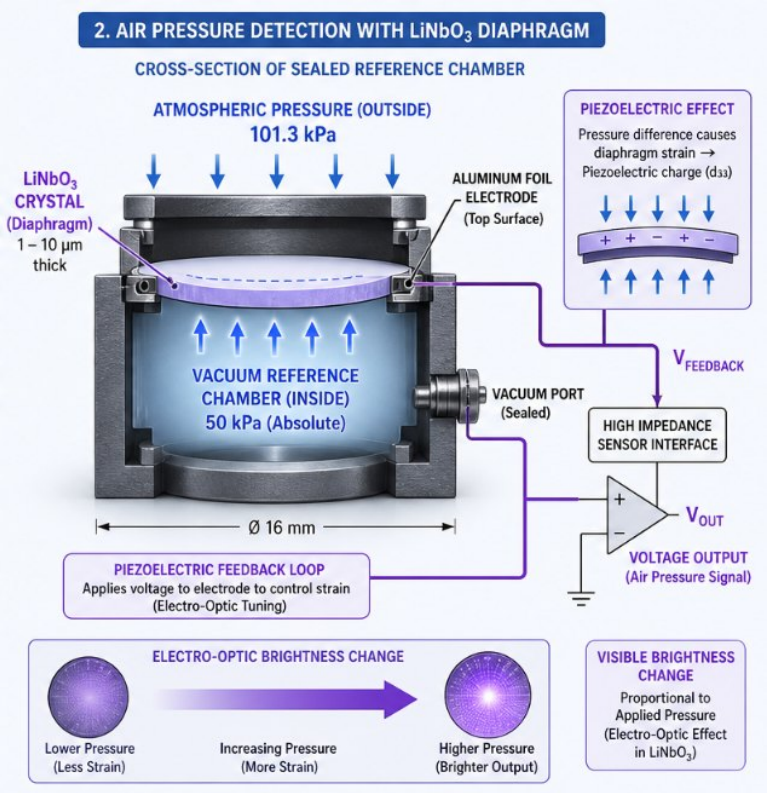
WHAT IT SHOWS A photo of a bulk black phosphorus crystal (about 5 mm wide) threaded on two metal rods labelled anode (+) and cathode (−), with “Current Flow” shown between them.

THE SCIENCE Black phosphorus is a layered semiconductor with puckered layers. Its band gap runs from about 0.3 eV in bulk to about 2 eV for one layer (Li et al., 2014), and its electrical and optical properties depend on direction (armchair versus zigzag), which makes it useful for polarisation-sensitive infrared detectors. It oxidises in air and needs encapsulation.

REALITY CHECK Two rods through a bulk crystal is a crude contact geometry; measurements need clean, well-defined contacts.

Fig. 91 Air-pressure detection with a $LiNbO_3$ diaphragm

MIXED



Files: HJ4ivxlbUAED2Zy.png

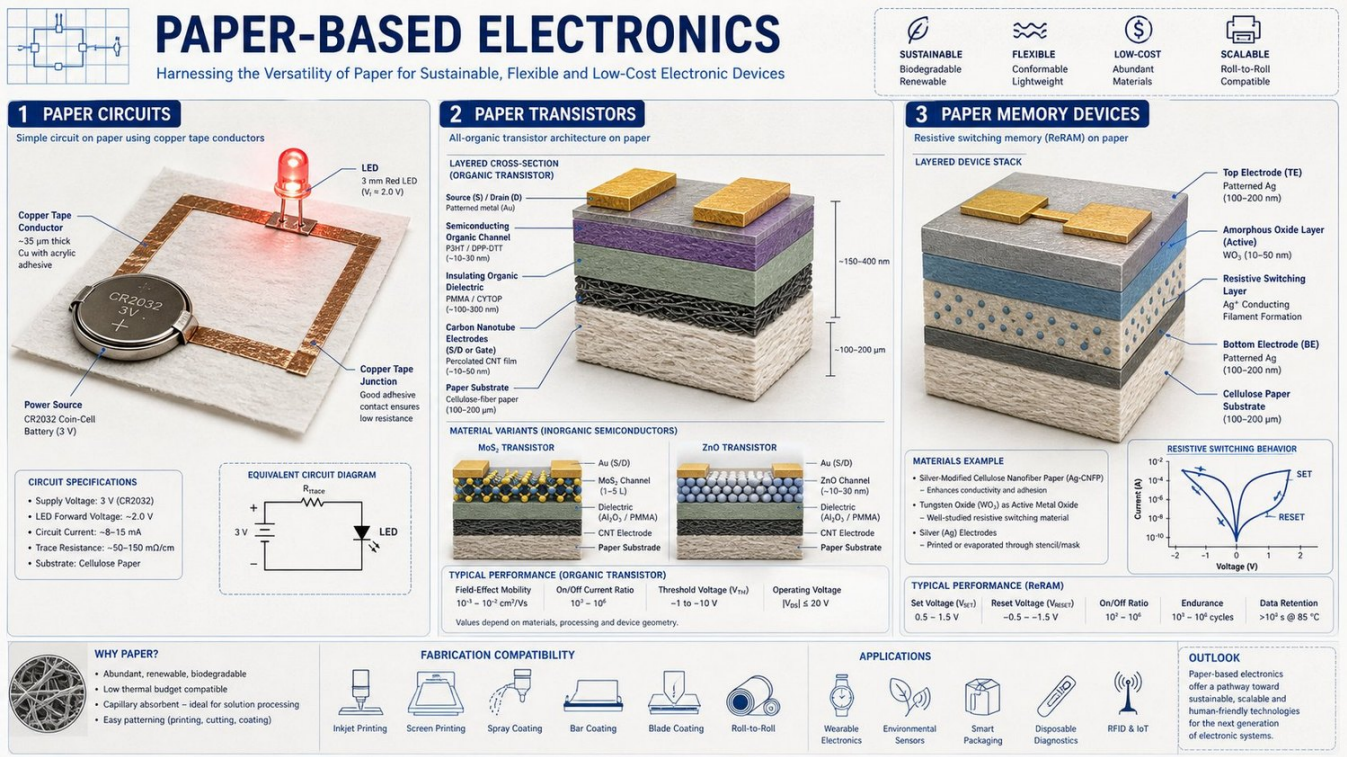
WHAT IT SHOWS A cross-section of a 16 mm sealed chamber with a 1–10 µm thick lithium niobate diaphragm as a piezoelectric and electro-optic sensor, 101.3 kPa atmosphere outside and a 50 kPa reference inside, an aluminium-foil electrode, a piezoelectric feedback loop and a high-impedance amplifier producing V_{out}.

THE SCIENCE Lithium niobate is both piezoelectric ($d_{33} \approx 6$ pC/N) and electro-optic ($r_{33} \approx 31$ pm/V), so one crystal can turn strain into charge or use voltage to change optical path. Piezoelectric sensors need a high-impedance readout because the charge leaks away.

REALITY CHECK Two flaws. A piezoelectric charge leaks away in seconds, so this arrangement measures changes in pressure, not a steady value. And the mechanics do not work: for a clamped circular plate, the edge stress is $\sigma = 3Pa^2/(4t^2)$. With $P = 50$ kPa, $a = 8$ mm and $t = 10$ µm, $\sigma \approx 24$ GPa, far above the few hundred MPa that $LiNbO_3$ can withstand, so the diaphragm would fracture.

Fig. 92 Paper-based electronics

MAINSTREAM



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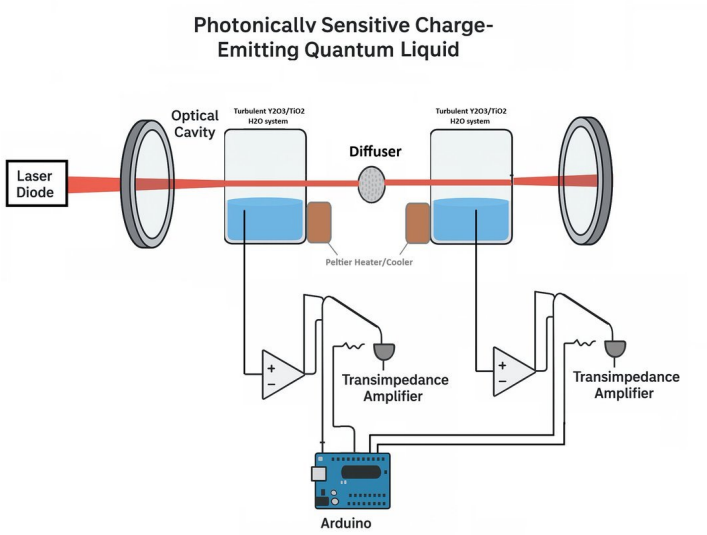
WHAT IT SHOWS An infographic: circuits made from copper tape on paper with an LED and a CR2032 cell; paper transistors using organic, MoS_2 or ZnO channels; paper-based resistive memory; fabrication by inkjet, screen printing and spray coating; and applications in wearables and disposable diagnostics.

THE SCIENCE Cellulose paper is cheap, flexible and biodegradable, and printed electronics on paper is a real field. A copper-tape circuit with an LED and a 3 V coin cell is a classroom staple, and low-voltage organic and oxide transistors have been demonstrated on paper.

REALITY CHECK Paper transistors are far slower and less uniform than silicon ones, and are aimed at disposable sensors.

Fig. 93 Photosensitive “quantum liquid”: Y_2O_3/TiO_2 water suspension in an optical cavity

MIXED



Files: 1973336917916917793-G2KxmGJaIAEEJP7.jpg

WHAT IT SHOWS A laser diode passes through a resonant cavity that contains two vessels of turbulent Y_2O_3/TiO_2 water suspension with a diffuser between them, Peltier heater/coolers, transimpedance amplifiers and an Arduino reading the photocurrents.

THE SCIENCE Titanium dioxide is the classic photocatalyst (Fujishima and Honda, 1972): UV light (band gap 3.0–3.2 eV) generates electron–hole pairs that split water or drive redox reactions. Y_2O_3 is a wide-gap oxide often used as a host. Transimpedance amplifiers turn small photocurrents into voltages, and Peltier elements hold temperature.

REALITY CHECK “Quantum liquid” normally means superfluid helium or a quantum Hall state, not an oxide slurry, and a red diode laser (about 650 nm, 1.9 eV) has too little photon energy to excite TiO_2’s band gap directly.

Fig. 94 Concept syringe-gun

CONTEXT

Files: 1974076259446763838-G2VT3slbMAA_wf6.jpg

WHAT IT SHOWS A rendered illustration of a pistol-shaped autoinjector with a transparent barrel, an orange and a blue chamber, and a needle.

THE SCIENCE Dual-chamber autoinjectors are real medical devices: they keep a dry drug and its diluent apart and mix them at the moment of injection.

REALITY CHECK It is concept art with no measurements or physics.

Fig. 95 IR polarisation switch: aluminium wire grid on COP with annealed VO_2

MIXED

Files: image.png

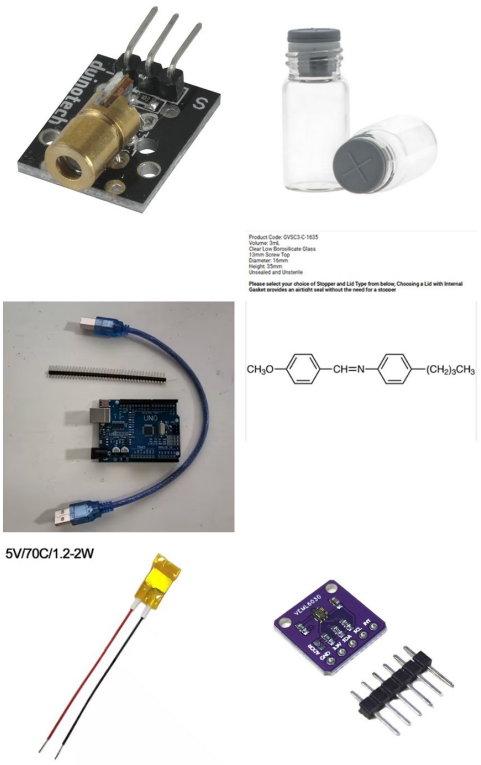
WHAT IT SHOWS A post describing an infrared polarisation switch and detector built from an electrically controlled aluminium wire grid on COP (cyclic olefin polymer) with VO_2 annealed, plus a photo of a grey sheet.

THE SCIENCE A wire-grid polariser (parallel sub-wavelength lines) transmits the polarisation across the wires and reflects the polarisation along them. Vanadium dioxide has a sharp insulator-to-metal transition at 68 °C (Morin, 1959), so its infrared transmission changes when heated electrically. COP is transparent in the infrared and terahertz.

REALITY CHECK VO_2 normally needs annealing at about 400–550 °C to crystallise, but COP softens at about 100–160 °C (grade dependent), so the film would need low-temperature deposition or a different substrate.

Fig. 96 Parts list: laser diode module, vial, Arduino, MBBA liquid crystal, Kapton heater, sensor board

MAINSTREAM



Files: parts_list.png

WHAT IT SHOWS A collage of six items: a red laser-diode module, a sealed glass sample vial, an Arduino Uno kit, the chemical structure $CH_3O–C_6H_4–CH=N–C_6H_4–(CH_2)_3CH_3$, a 5 V / 70 °C / 1.2–2 W Kapton film heater, and a purple multi-pin sensor breakout board.

THE SCIENCE The structure is MBBA (N-(4-methoxybenzylidene)-4-butylaniline), the first room-temperature nematic liquid crystal (Kelker, 1969); it is nematic between about 22 °C and 47 °C and clears to an isotropic liquid above that. Laser light through a nematic sample is scattered or rotated according to molecular alignment, and heating through the nematic–isotropic transition switches the optics.

REALITY CHECK The 70 °C heater comfortably exceeds the clearing point, so a controller is needed to avoid overheating. MBBA hydrolyses in moist air, so the vial should be sealed and dry.

Fig. 97 Tactile push-button

CONTEXT

Files: shopping.webp

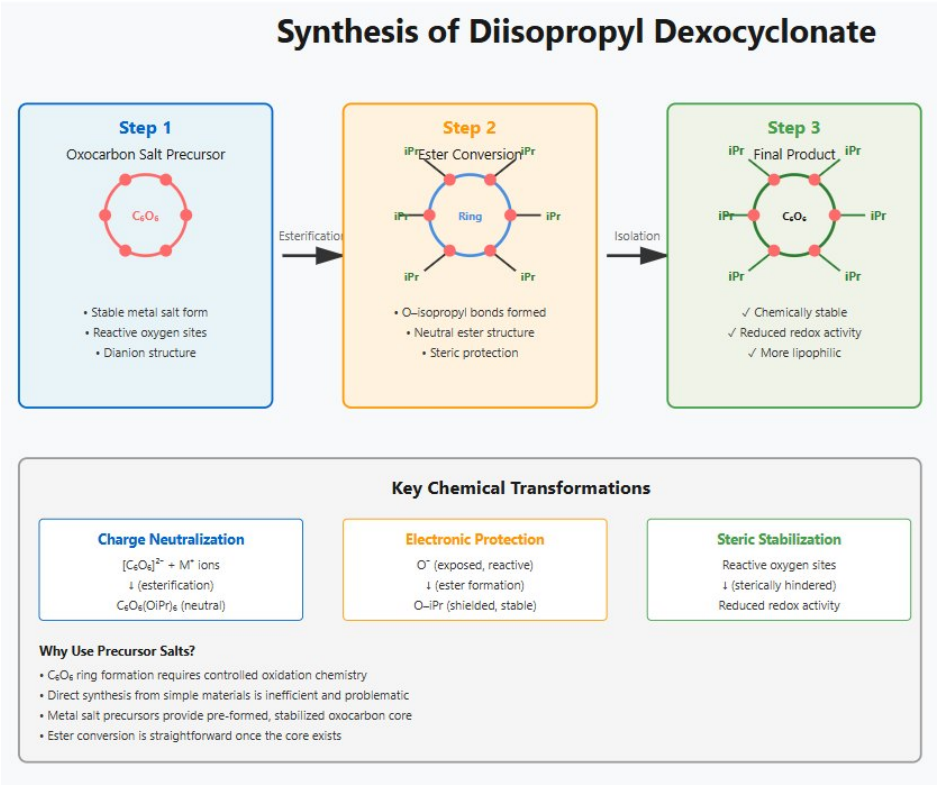
WHAT IT SHOWS A photo of a small black tactile switch with bent metal legs.

THE SCIENCE Tactile switches are momentary buttons used for user input on Arduino boards. Mechanical contacts bounce for several milliseconds on closing, so software or an RC filter should debounce them.

REALITY CHECK It contains no experiment on its own.

Fig. 98 “Synthesis of Diisopropyl Dexocyclonate”: an oxocarbon ester scheme

MIXED



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WHAT IT SHOWS A three-step flow chart: an oxocarbon salt precursor (a C_6O_6 ring dianion), an ester conversion adding isopropyl (iPr) groups, and an isolated neutral product, with boxes for charge neutralisation, electronic protection and steric

stabilisation.

THE SCIENCE Oxocarbons are cyclic carbon–oxygen anions such as croconate ($C_5O_5^{2-}$) and rhodizonate ($C_6O_6^{2-}$); lithium and sodium rhodizonate are studied as battery electrode materials. Esterification replaces an O^- charge with an O–R group, giving a neutral molecule that is more soluble in organic solvents.

REALITY CHECK "Dexocyclonate" is not a standard chemical name, and the scheme gives no reagents, conditions or characterisation, so it is a concept sketch rather than a validated route. This handbook does not add a procedure.

Fig. 99 Zinc–sodium-chlorite cell: design and chemistry

MIXED

Cell Design (Prototype)

Component	Material
Positive electrode	Vitreous carbon foam
Negative electrode	Zinc sheet
Membrane	Nafion 117 (sodiated)
Positive electrolyte	$NaClO_2$ aqueous solution
ClO_2 accumulator	Organic solvent phase (traps gaseous ClO_2)

The Core Chemistry

The battery exploits the highly reversible electrochemical reaction between **chlorite ion (ClO_2^-)** and **chlorine dioxide (ClO_2)**: pubs.acs

$$ClO_2^- \rightleftharpoons ClO_2 + e^-$$

This reaction has an equilibrium potential of **+0.954 V vs. SHE** and is notably **pH-independent**, meaning it works stably in acidic, neutral, or alkaline conditions. Paired with a zinc (Zn) negative electrode, the full cell delivers **1.72 V** — comparable to a zinc-bromine battery and far higher than vanadium redox (~1.26 V) or iron-nickel (~1.4 V) chemistries. chemrxiv +1

Key Advantages

- **No rare metals** — uses zinc and carbon electrodes, sodium chlorite electrolyte, all cheap and abundant chemrxiv
- **High Coulombic efficiency** — prototype Zn-$NaClO_2$ cells achieved **92–96% efficiency over 50 cycles** pubs.acs
- **Cold-weather performance** — ClO_2 stays stable and liquid below 11°C, making it promising for Arctic/Antarctic energy storage pubs.acs
- **Chemical cost** is extremely low compared to lithium-ion chemrxiv

Files: 2030437340607340716-HC2P9JSbIAAmu2-.png, 2030437340607340716-HC2P9MkWEAA_yac.png

WHAT IT SHOWS Left: a prototype cell table: vitreous carbon foam positive electrode, zinc sheet negative, sodiated Nafion 117 membrane, aqueous $NaClO_2$ positive electrolyte, and an organic solvent phase to trap gaseous ClO_2. Right: a chemistry summary: the reversible $ClO_2^- \rightleftharpoons ClO_2 + e^-$ reaction at +0.954 V versus SHE, pH-independent, a 1.72 V cell with zinc, no rare metals, 92–96 % Coulombic efficiency over 50 cycles, and stable liquid ClO_2 below 11 °C.

THE SCIENCE The voltage adds up: E° for ClO_2/ClO_2^- is +0.954 V and Zn^{2+}/Zn is −0.76 V, so the cell is 0.954 + 0.76 = 1.71 V, matching the 1.72 V claim. ClO_2 boils at 11 °C, consistent with the "liquid below 11 °C" statement. Zinc anodes have a theoretical capacity of 820 mAh/g.

REALITY CHECK Chlorine dioxide is a toxic, oxidising gas that can explode in air at moderate concentration, so this chemistry needs specialist handling, ventilation and containment; it is not a hobby project. The 92–96 % efficiency figure is cited to a preprint and journal, and I have not verified it.

Fig. 100 Zinc–chlorite stack scaling and charging tables

MIXED

Layers	Zinc/layer	Zn thickness	Series voltage	Parallel current	Power
5	18.6 g	0.87 mm	8.6 V	3.0 A	5.2 W
8	11.1 g	0.52 mm	13.8 V	4.8 A	8.3 W
10	8.6 g	0.40 mm	17.2 V	6.0 A	10.3 W
15	5.3 g	0.25 mm	25.8 V	9.0 A	15.5 W
20	3.6 g	0.17 mm	34.4 V	12.0 A	20.6 W
24	2.8 g	0.13 mm	41.3 V	14.4 A	24.8 W

Charging Performance (100g Battery)

Config	Total Zinc	Capacity	Charge Time @ 20mA/cm²	Energy Stored
10 layers	86.2 g	~42,400 mAh	**7.1 hours**	73 Wh
15 layers	79.3 g	~39,000 mAh	**4.3 hours**	67 Wh
20 layers	72.4 g	~35,600 mAh	**3.0 hours**	61 Wh

Files: 2030442226585944423-HC2UMilbMAAdKpx.png, 2030445289522479150-HC2XOfGaQAEdlDu.png

WHAT IT SHOWS Left: a table of 5–24 stacked layers: zinc per layer (18.6 g to 2.8 g), zinc thickness (0.87 to 0.13 mm), series voltage (8.6 to 41.3 V), parallel current (3.0 to 14.4 A) and power (5.2 to 24.8 W). Right: charging performance for a “100 g battery”: 10 layers, 86.2 g zinc, ~42,400 mAh, 7.1 h at 20 mA/cm², 73 Wh; 15 layers, 79.3 g, 39,000 mAh, 4.3 h, 67 Wh; 20 layers, 72.4 g, 35,600 mAh, 3.0 h, 61 Wh.

THE SCIENCE The tables hang together. Each layer is a cell of about 30 cm² (18.6 g / 7.14 g/cm³ / 0.087 cm), which at 20 mA/cm² passes 0.6 A. Series voltage is n × 1.72 V, parallel current is n × 0.6 A, and power is the same in either arrangement (5 layers: 1.72 × 3.0 = 5.2 W). The energy stored, 42.4 Ah × 1.72 V, is 73 Wh, and the 7.1 h charge time is 42.4 Ah / 6 A.

REALITY CHECK The mass budget does not close. Delivering 42.4 Ah takes 1.58 mol of electrons, that is 1.58 mol × 90.4 g/mol ≈ 143 g of $NaClO_2$ alone, more than the whole “100 g” battery. The theoretical limit for zinc plus chlorite is (1.72 V × 26.8 Ah) / 0.123 kg ≈ 375 Wh/kg of reactants, before water, solvent, membrane and casing, so “73 Wh in 100 g” (730 Wh/kg) counts only the zinc and overstates it by several times.

Fig. 101 Typical drone battery capacities

CONTEXT

- **Small/Consumer Drones:** Typically have batteries in the range of 15–25 Wh (0.015 – 0.025 kWh).
- **Prosumer/Industrial Drones:** Larger, heavier drones (e.g., DJI Matrice 300) might use batteries with 100–300+ Wh (0.1 – 0.3 kWh).
- **Large Specialized Drones:** High-end, custom, or fixed-wing industrial drones can carry much larger capacities, with some systems reaching up to 5 kWh. Unmanned Systems... +4

Files: 2030488255695896731-HC2-TUUagAAmCLn.jpg

WHAT IT SHOWS A chat-assistant list: small consumer drones 15–25 Wh; larger prosumer or industrial drones such as the DJI Matrice 300 about 100–300+ Wh; large specialised drones up to about 5 kWh.

THE SCIENCE Lithium-ion cells store 200–270 Wh/kg, so 25 Wh is roughly 100 g of cells, and a 5 kWh pack is about 20–25 kg. This gives a reference for what a new chemistry would have to beat at the pack level, including casing and electronics.

REALITY CHECK Numbers are typical ranges and vary by model.

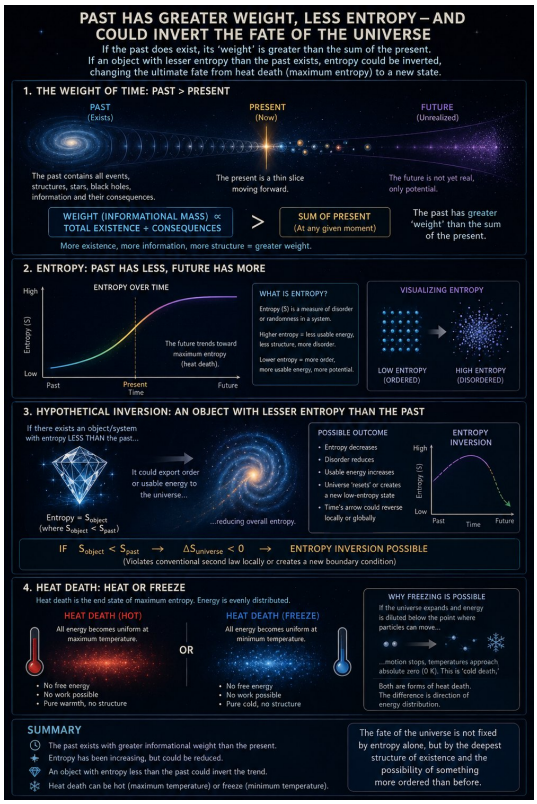
CHAPTER 12

Time, entropy and cosmology

Figures 102–108 · 7 entries

Fig. 102 “Past has greater weight, less entropy” and the fate of the universe

SPECULATIVE



Files: 224ead14-ba68-4fdf-8bf7-b945b7427478.png

WHAT IT SHOWS A four-part infographic: (1) the past “weighs” more than the present (weight = informational mass); (2) entropy rises from past to future; (3) a hypothetical object with entropy lower than the past could invert entropy; (4) heat death as either hot or freezing, ending in a claim that the universe’s fate is not fixed by entropy alone.

THE SCIENCE The second law says the entropy of an isolated system does not decrease, and the “Past Hypothesis” says the early universe had extraordinarily low entropy, which is why the past differs from the future (Boltzmann; Penrose; Albert). Heat death is the state of maximum entropy; in ΛCDM the universe approaches a de Sitter state with a horizon temperature of order 10^{-30} K, so “freeze” is the standard picture.

REALITY CHECK “Informational mass” as the weight of the past is not a defined physical quantity. Local order can appear by fluctuation or by pumping (life and refrigerators do it), but always by producing more entropy elsewhere, so the total does not fall: $\Delta S_{\text{universe}} \geq 0$.

Fig. 103 Entropy, population inversion and lasers

MIXED

Files: 6277c98f-602e-4e62-8fab-eb8d58dae8ca.png

WHAT IT SHOWS A post: “If a material (or ensemble of material states) has entropy lower than that of incoming radiation it can sustain or be driven into a population inversion or other non-thermal distributions required for stimulated, coherent emission — this is the basis for lasers and masers”, with a glowing crystal image reading “If there exists an object/system with entropy LESS THAN the past... Entropy = S_{object} (where $S_{object} < S_{past}$)”.

THE SCIENCE A population inversion, where more atoms are in an upper level than a lower one, is a non-thermal state that corresponds to a negative absolute temperature (Purcell and Pound, 1951; Braun et al., Science 2013). It needs a pump that feeds in energy from outside, and the pump’s waste heat carries the entropy away, so the second law is satisfied.

REALITY CHECK The first half is a fair statement: lasing needs a non-thermal distribution. The comparison with the “past” is a metaphor; the entropy of a crystal has no simple relation to the entropy of the early universe.

Fig. 104 Laser tripwire, a Peltier boiler and a 5th dimension

SPECULATIVE

Files: HKOV6_RbsAA30gJ.png

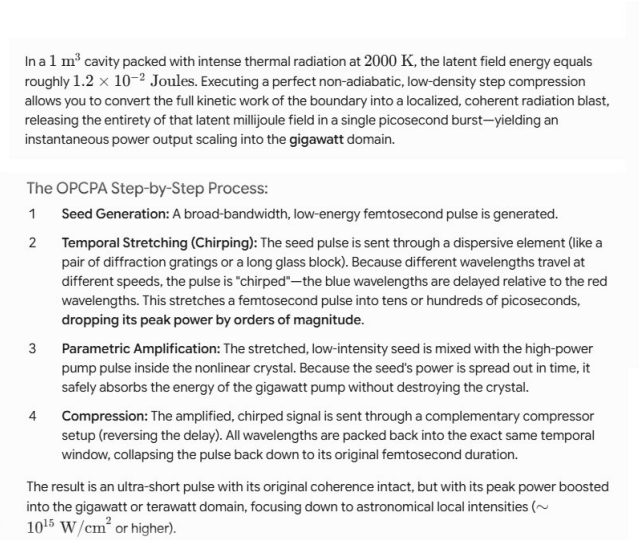
WHAT IT SHOWS A post about a portable Peltier water boiler and a laser tripwire, adding that if atoms decouple from photonic energy and T-symmetry is violated “then the carrier is in 5-D space”, and quoting an assistant’s table on how a laser partially “precluded” into a fifth dimension could show coherence distortion, modified dispersion, energy loss to the bulk and anisotropic interference.

THE SCIENCE Extra dimensions are a real research topic: Kaluza and Klein (1920s) unified gravity and electromagnetism with a fifth dimension; the ADD (1998) and Randall–Sundrum (1999) models allow large or warped extra dimensions. Experiments test Newtonian gravity down to about 50 µm and find no deviation (Kapner et al., PRL 2007; Lee et al., PRL 2020). A laser tripwire is simply a beam and a detector that registers a break.

REALITY CHECK In brane-world models, photons and other gauge fields are confined to our 3D brane; only gravity leaks into the bulk, so a laser beam cannot “partly enter” a fifth dimension. Atoms decoupling from light, or T-symmetry violation, would not indicate an extra-dimensional carrier: T violation is already seen in kaon and B-meson decays.

Fig. 105 Self-Amplified Non-Adiabatic Emission (SANAE) and how OPCPA really reaches gigawatts

MIXED



Files: HKSGjNUaAAApJpg.png, HKSnrihboAAni3c.png

WHAT IT SHOWS Left: a post proposing that a 1 m³ cavity of thermal radiation at 2000 K holds 1.2×10^{-2} J, which a perfect non-adiabatic compression could release in one picosecond, giving gigawatt power. Right: the same paragraph above a step list for optical parametric chirped-pulse amplification (OPCPA): seed a femtosecond pulse, stretch it, amplify it parametrically, recompress it to reach GW–TW peak power, focused to about 10^{15} W/cm².

THE SCIENCE The arithmetic is right: the blackbody energy density is $u = aT^4 = 7.57\times10^{-16} \times (2000)^4 \approx 1.2\times10^{-2}$ J/m³, and 12 mJ in 1 ps is 12 GW. OPCPA is a real, widely used technique (Dubietis, Jonušauskas and Piskarskas, 1992) and is a route to petawatt-class pulses, because stretching lowers the peak power so the crystal survives, and recompression restores it.

REALITY CHECK SANAE breaks down at the conversion step. Compressing radiation is work done by the piston and returns exactly that work, and thermal light has high entropy, so turning it into a coherent pulse is a heat-engine problem limited by Carnot: you must reject waste heat. Twelve millijoules is also tiny compared with the energy needed to hold a cubic metre at 2000 K. OPCPA starts from a coherent seed and a strong pump; it is not a way to amplify thermal energy.

Fig. 106 The 5D temporal ET and the 4D reality cube

CONTEXT

Files: dcc16395-1a3d-4011-b487-888593f610d9.png

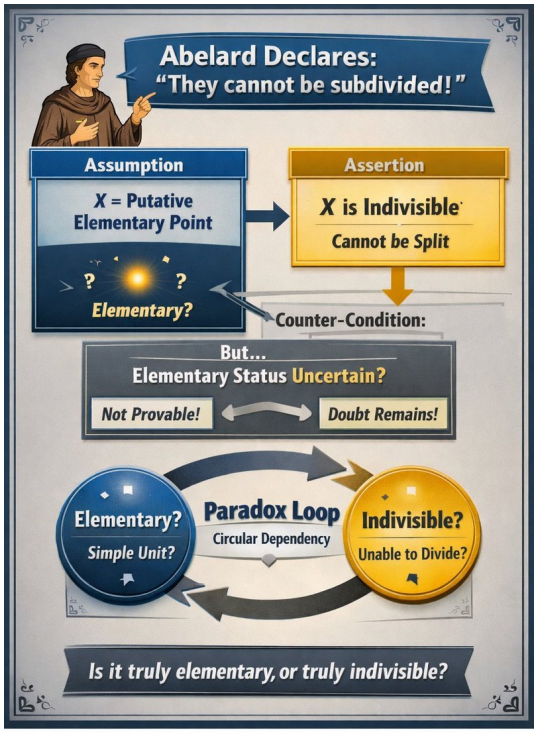
WHAT IT SHOWS A painted scene of a translucent figure touching a faceted cube, captioned: “5D temporal ET – experiences all moments simultaneously, observes change as a whole”; “4D reality cube – a fixed construct of space-time, perceived linearly by 3D beings”; “Encounter: temporal awareness meets spatial structure.”

THE SCIENCE The picture illustrates the “block universe” view (eternalism): in special relativity, spacetime is a four-dimensional manifold, and events at all times exist together in the mathematical description (Minkowski, 1908; Putnam, 1967). Our sense of a flowing present is not built into the equations.

REALITY CHECK Whether time really “flows” is a philosophical debate, and a fifth temporal dimension is not part of standard physics.

Fig. 107 “Abelard declares: they cannot be subdivided!”

CONTEXT



Files: 2026405780078104688-HB89TqNaMAgZcGS.jpg

WHAT IT SHOWS A stylised illustration with a medieval scholar declaring “They cannot be subdivided!”, a box asking whether a putative elementary point X is indivisible, and a “paradox loop” between “elementary?” and “indivisible?”.

THE SCIENCE “Atom” originally meant uncuttable, and whether a particle has parts is settled by experiment. Scattering at higher energy resolves structure: deep inelastic scattering in 1968 revealed quarks inside protons. Electrons show no sign of size or substructure down to about 10^{-18} m or less. Quarks also cannot be pulled out as free particles (confinement), a real case of “elementary yet inseparable”.

REALITY CHECK Nothing is being measured here; the picture poses a puzzle in words.

Fig. 108 Three-stage AI art: piezoelectric crystal, molten iron and spacetime-warping material

CONTEXT

Files: 2024017501324300443-HBbBK85bcAES2fp.jpg · Also in repository: 4e1fb95f-a26d-420a-bd25-f173a2d35...

WHAT IT SHOWS Three panels captioned “Stage 1: Piezoelectric crystal response” (a hammer striking a crystal with sparks), “Stage 2: Molten iron electromagnetic interaction” (glowing iron on an anvil in swirling fields), “Stage 3: Spacetime-warping advanced material” (a faceted crystal bending around a black hole). (Image 77 is a duplicate.)

THE SCIENCE Stage 1 is real: quartz and other piezoelectric crystals produce voltages of thousands of volts when struck, which is how lighter igniters work (the Curie brothers, 1880). Stage 2 has a genuine hook: iron loses ferromagnetism above its Curie temperature of 770 °C, and molten iron is a conductor, so changing fields induce eddy currents in it (induction furnaces).

REALITY CHECK Stage 3 has no counterpart: spacetime is curved by mass–energy, and no known material “warps” it more than its mass would; the crystal in the picture is decoration.

CHAPTER 13

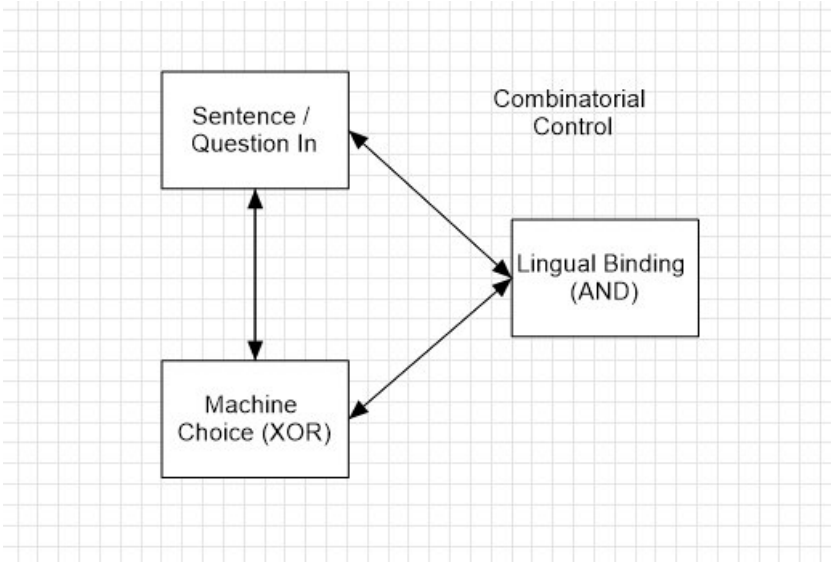
AI, cognition and diagrams that are not physics

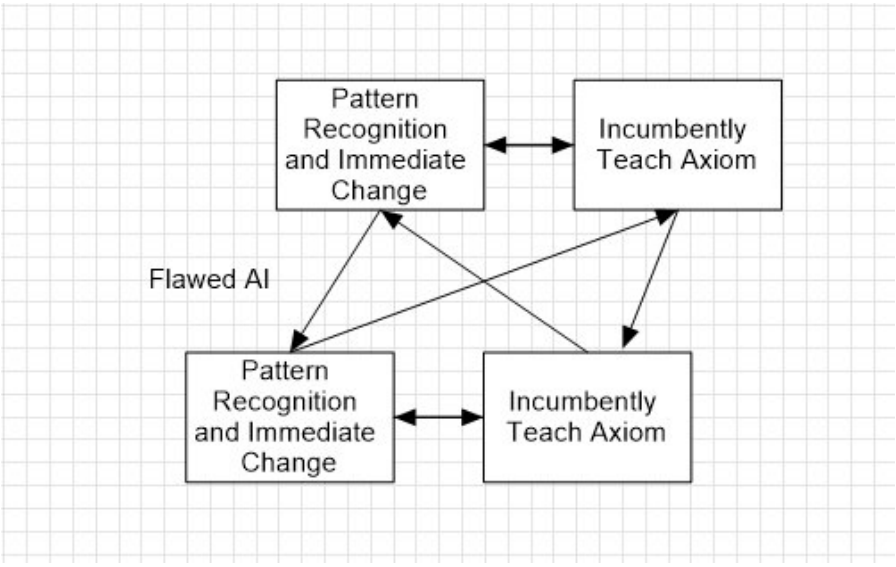
Figures 109–121 · 13 entries

This chapter groups images that are about minds, machines, language or personal systems rather than measurable physical quantities. They are described plainly and with light-touch context; where a concept has a proper scientific name, it is given.

Fig. 109 Sentence-in flowcharts: “lingual binding (AND)” and the “flawed AI”

CONTEXT





Files: 1967195736640278995-G0ziF9ZaEAEBjYR.png, 1967420400041869801-G02uaA7aYAE5lGm.png

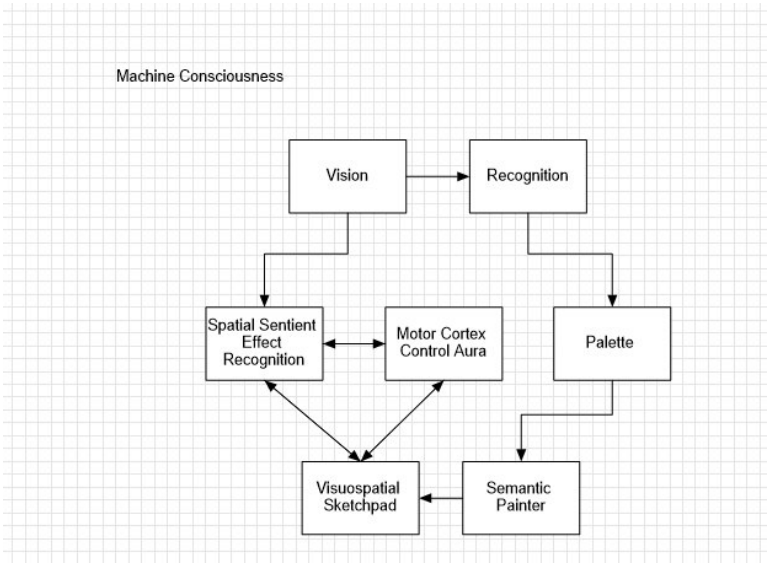
WHAT IT SHOWS Left: a flowchart with boxes “Sentence / Question In”, “Lingual Binding (AND)” and “Machine Choice (XOR)”, joined by arrows and labelled “Combinatorial Control”. Right: a “Flawed AI” diagram of two loops between “Pattern Recognition and Immediate Change” and “Incumbently Teach Axiom”.

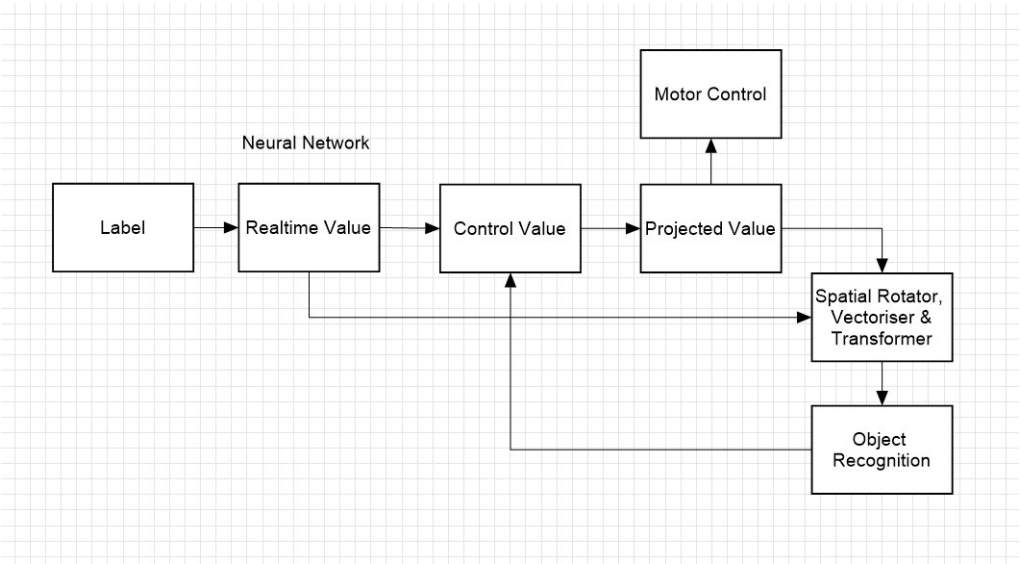
THE SCIENCE AND and XOR are Boolean gates: AND is true only if both inputs are true, XOR if exactly one is. XOR is the classic case a single-layer perceptron cannot learn (Minsky and Papert, 1969), which is why multi-layer networks matter. The right-hand loops resemble the tension between systems that learn patterns from data and systems that are given fixed rules (axioms), the core issue in neuro-symbolic AI.

REALITY CHECK Both are conceptual sketches; they define no data flows or measurements.

Fig. 110 Machine-consciousness and neural-network control diagrams

CONTEXT





Files: 1986555311017369636-G5GphkgbIAMJLv3.png, 2029309799154991207-HCmOgTjawAIW6GT.png

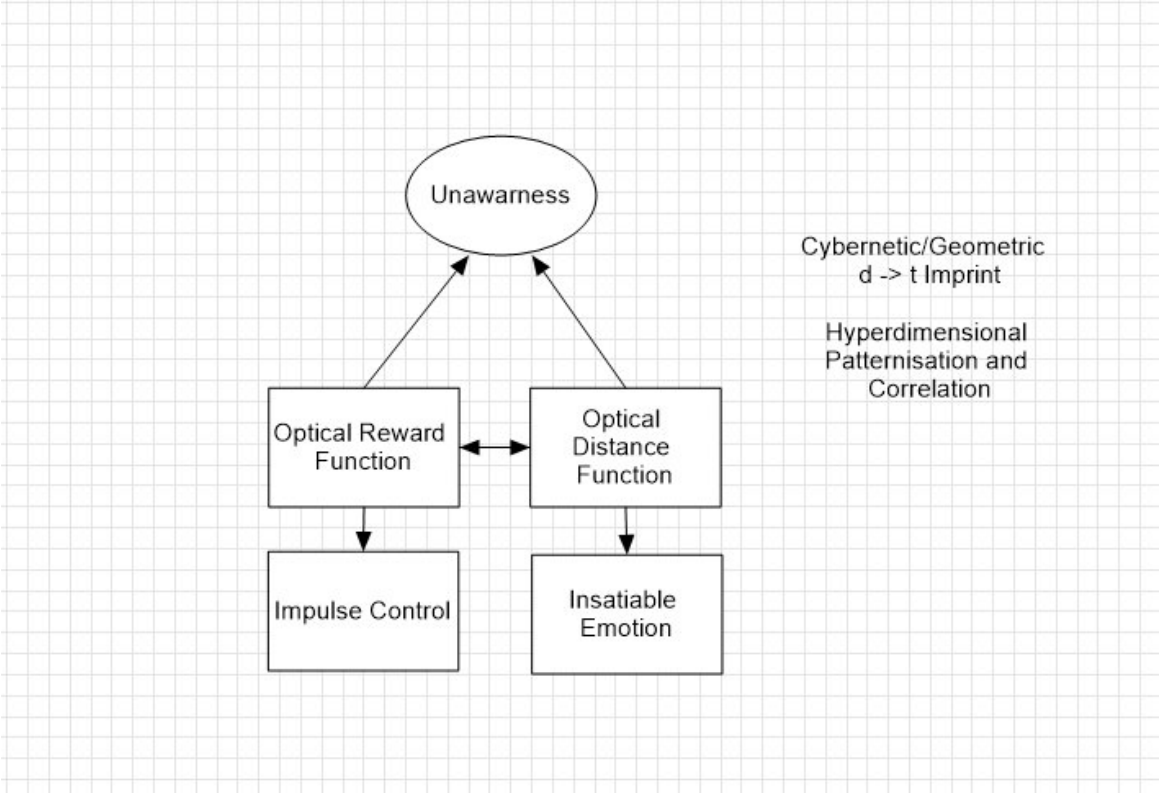
WHAT IT SHOWS Left: “Machine Consciousness”: Vision → Recognition → Spatial Sentient Effect Recognition ↔ Motor Cortex Control Aura, with Palette → Semantic Painter → Visuospatial Sketchpad. Right: “Neural Network”: Label → Realtime Value → Control Value → Projected Value → Motor Control, with a spatial rotator/vectoriser/transformer and an object-recognition feedback loop.

THE SCIENCE “Visuospatial sketchpad” is a real term from Baddeley’s model of working memory (1974): the store where visual and spatial information is held. The right-hand diagram is a standard closed perception–action loop: sense, estimate, predict, act, and feed back, as in model-predictive control and robotic manipulators.

REALITY CHECK No definition of machine consciousness is used, so the left diagram does not say what would count as success.

Fig. 111 “Unawareness”, reward and distance functions

CONTEXT



Files: 2001902694135738582-G8gv2BBakAMz7Al.png · Also in repository: 2001922025741521379-G8hBb7UbkAAei...

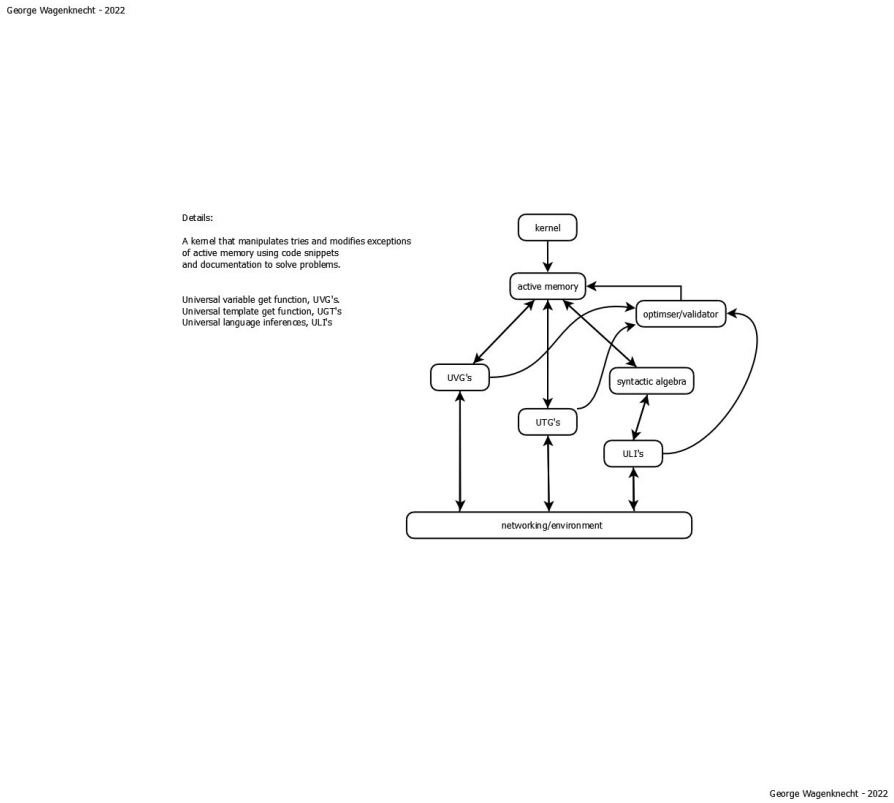
WHAT IT SHOWS A diagram in which “Unawareness” is fed by an “Optical Reward Function” and an “Optical Distance Function”, which link to “Impulse Control” and “Insatiable Emotion”, with side notes: “Cybernetic/Geometric d → t Imprint” and “Hyperdimensional Patternisation and Correlation”. (Image 47 is a copy.)

THE SCIENCE In reinforcement learning an agent maximises a reward function, and a distance-to-goal function is a common shaping term. Hyperdimensional computing represents concepts as very long (about 10,000-element) vectors and relates them by correlation (Kanerva, 2009). Cybernetics (Wiener, 1948) studies feedback and control in animals and machines.

REALITY CHECK The diagram sets out a vocabulary and does not supply equations for how the terms interact.

Fig. 112 Kernel, active memory and universal getter functions (2022 diagram)

CONTEXT



Files: Diagram1.png

WHAT IT SHOWS A signed 2022 diagram: a kernel manipulates tries and modifies exceptions in active memory using code snippets and documentation to solve problems; boxes for UVGs (universal variable get functions), UTGs (universal template get functions) and ULIs (universal language inferences), an optimiser/validator, a syntactic algebra and a networking/environment layer.

THE SCIENCE A trie is a prefix tree for storing strings. A central shared memory that several specialised modules read and write is the “blackboard architecture” of the Hearsay-II speech system (1970s), and programs that modify themselves are studied in program synthesis and genetic programming.

REALITY CHECK The diagram names components but does not say how the optimiser/validator decides whether a change is an improvement, which is the difficult part of any self-modifying system.

Fig. 113 “AI-slop war engine…”: a C++ table-automorphism snippet

CONTEXT

Files: AI_slop_war.png

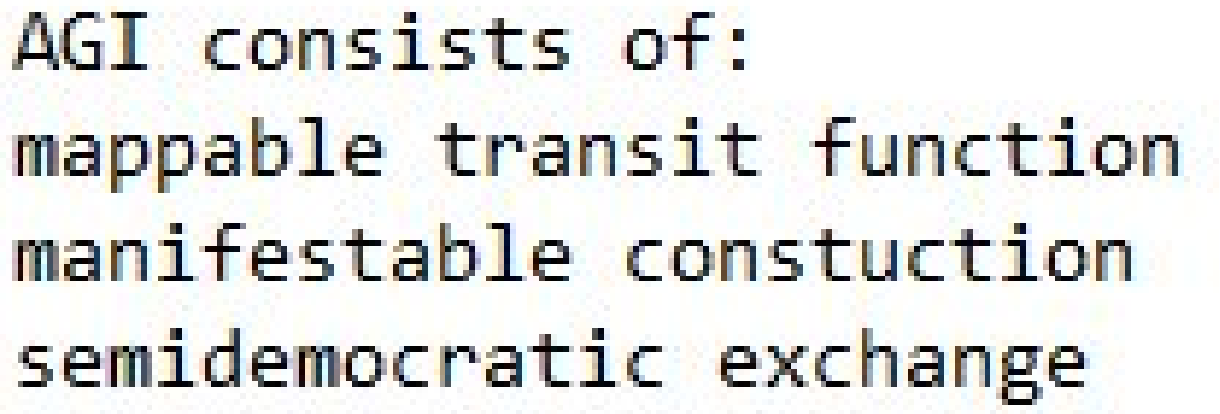
WHAT IT SHOWS A post with a short C++ condition: if a table has a non-empty name, dimension above zero, is symmetric, is not already automorphism-like, and the database is connected and the table exists, then drop the table and call makeAutomorphism(table).

THE SCIENCE In algebra, an automorphism is a structure-preserving mapping of an object onto itself; the automorphisms of a symmetric relation (for example a graph) form a group. The code treats a table as such an object.

REALITY CHECK The snippet drops the database table before the replacement is created, so a failure in makeAutomorphism would lose data. The safe order is to build the replacement first, or to run both steps inside a transaction.

Fig. 114 “AGI consists of…”

CONTEXT



Files: HLApkTQa8AAhF1T.png

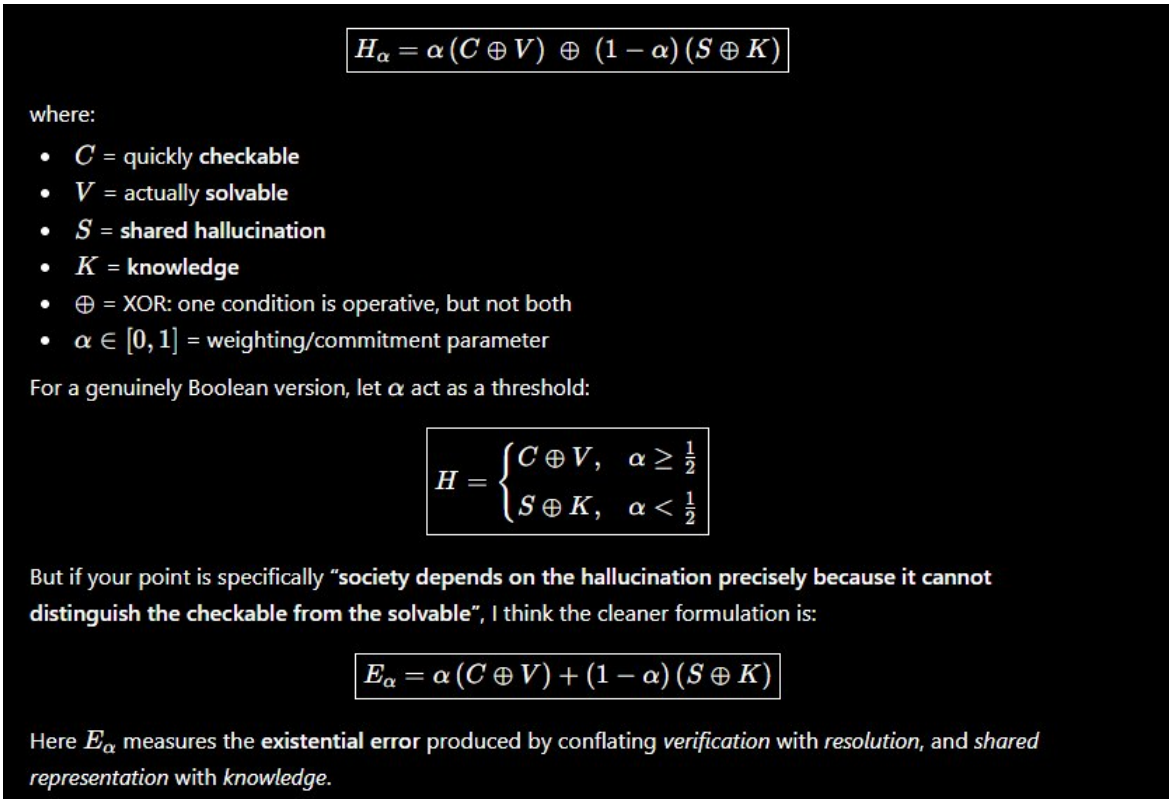
WHAT IT SHOWS Three lines of text: “AGI consists of: mappable transit function; manifestable construction; semidemocratic exchange.”

THE SCIENCE There is no agreed technical definition of artificial general intelligence. Working definitions include performing any intellectual task a human can (Goertzel and Pennachin, 2007), or Legg and Hutter’s measure of expected performance across all computable environments (2007).

REALITY CHECK The three terms are not defined here, so the statement cannot be tested.

Fig. 115 H_α: the formula for “checkable versus solvable” and shared hallucination

CONTEXT



Files: df6aec6c-0b43-4bde-bce0-db93f354f632.png

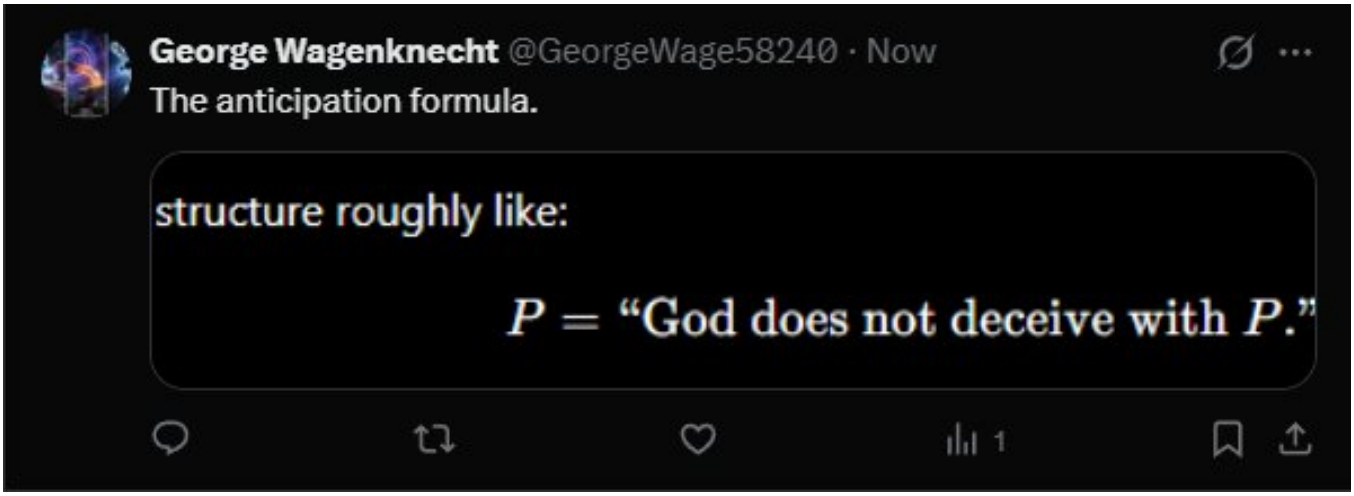
WHAT IT SHOWS A chat-assistant answer with $H_\alpha = \alpha(C \oplus V) \oplus (1-\alpha)(S \oplus K)$, where C is “quickly checkable”, V “actually solvable”, S “shared hallucination”, K “knowledge”, ⊕ is XOR and α ∈ [0, 1] weights the terms; a threshold version and an “existential error” $E_\alpha = \alpha(C \oplus V) + (1-\alpha)(S \oplus K)$.

THE SCIENCE C and V map onto the P versus NP question: a problem is in NP if a proposed answer can be checked quickly, and in P if it can be solved quickly. C ⊕ V is true for problems that are checkable but not efficiently solvable, which is exactly the gap between NP and P if P ≠ NP. The second term applies the same XOR to social beliefs, contrasting what is shared with what is known.

REALITY CHECK The formula is a compact way to state a viewpoint, not a physical law. The XOR of a weighted pair only makes sense as a real-valued measure when the terms are read as indicators, as the later E_α does.

Fig. 116 “The anticipation formula” and “Trying to noscope God”

CONTEXT



Files: 6a1db90a-5c7a-439d-be0e-d08ce1da4afd.png · Also in repository: HQ0BIsAbQAAhvK3.png, HQ0C2khawAEhFDV.png, HQ0D4PvbcAAUu11.png, HQz8qJ9aUAAAs5-.png

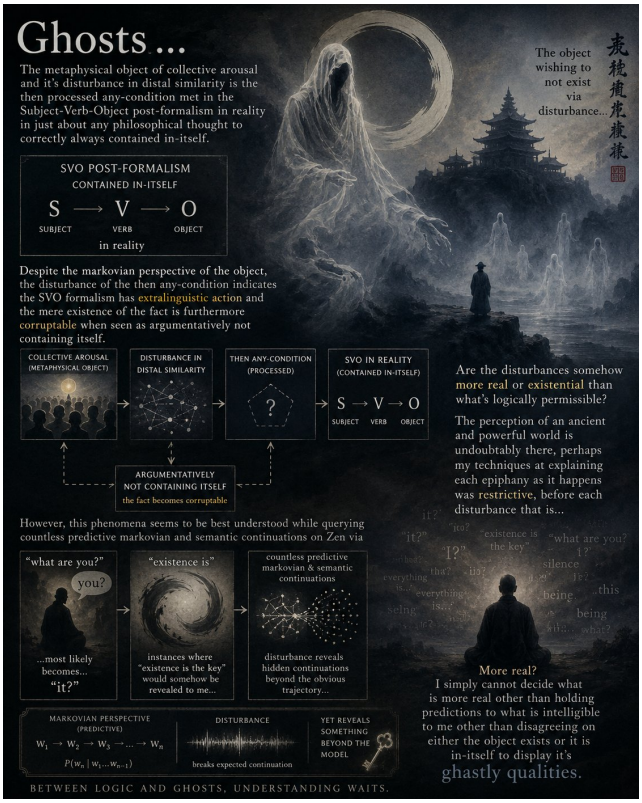
WHAT IT SHOWS Several post screenshots and plain-text versions carrying the same text: “The anticipation formula. Structure roughly like: P = ‘God does not deceive with P.’”, alongside “Trying to noscope God.”, “This was my point insertion.”, “Has modal connotations to death itself.” and “Unreal Tournament. Automatic rule bound motion, crisp.”

THE SCIENCE The wording plays on Einstein’s remark that God does not play dice. “Modal” refers to modal logic, which adds operators for necessity and possibility (Kripke, 1959). A “no-scope” is a video-game term for a sniper kill without using the scope.

REALITY CHECK There is no equation to check: P stands for an unspecified proposition.

Fig. 117 “Ghosts…”: Markov continuations and subject–verb–object post-formalism

CONTEXT



Files: de0909ae-6d7b-4a5b-81dc-2970d2e8ec22.png

WHAT IT SHOWS A dark illustrated poster: a ghostly figure over an old temple, with a text on the subject-verb-object (SVO) formalism, “collective arousal”, Markovian and semantic continuations, “what are you?”, and the closing line “Between logic and ghosts, understanding waits.”

THE SCIENCE A Markov language model predicts the next word from a limited context, $P(w_t \mid w_{t-1}, \ldots)$. A surprising word, one with low probability, carries high surprisal, $-\log P$, which is used to model reading times and the N400 brain response.

REALITY CHECK The poster is a piece of art and philosophical text and makes no measurable claim.

Fig. 118 “Fear of detachment and fake quintessentialism”

CONTEXT

Files: ecc3a97d-1348-4bc5-9904-9d46f5afa385.jpeg

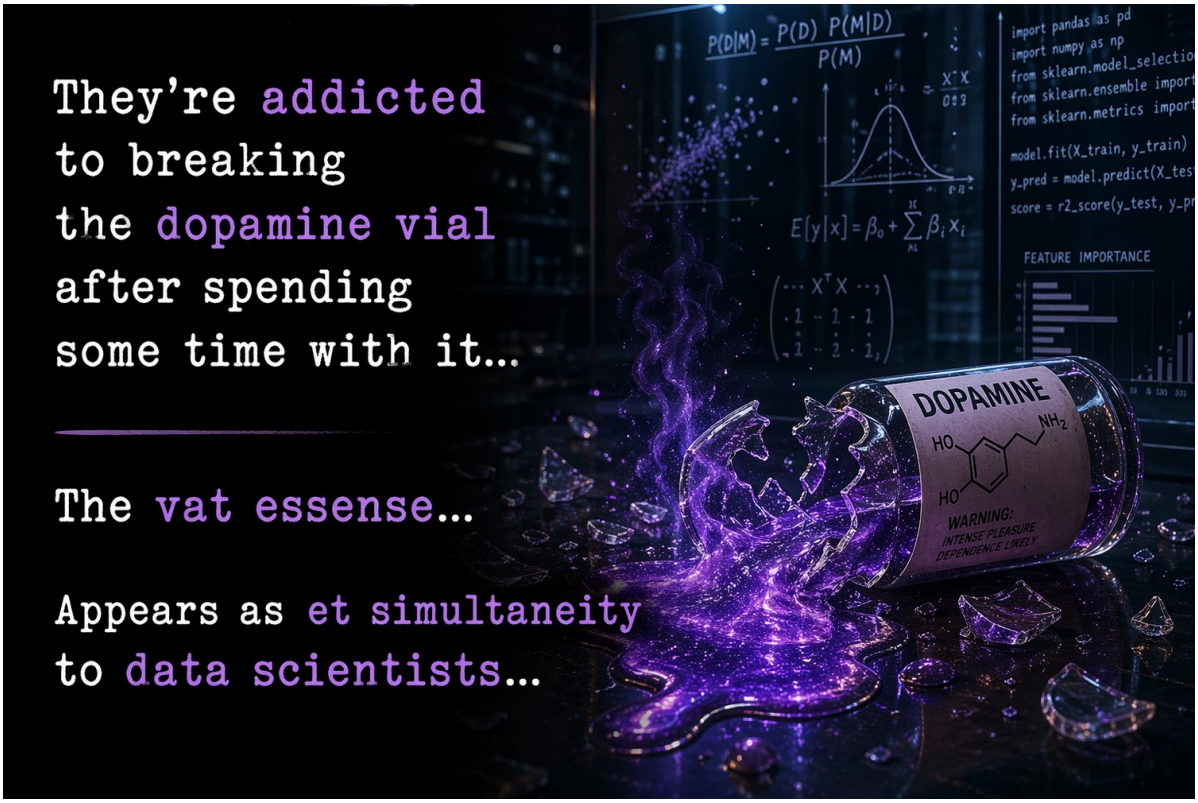
WHAT IT SHOWS A dramatic painting: a figure standing on a cliff edge above a whirlpool, holding chains, with a shattering glass-like dome and the title “Fear of detachment and fake quintessentialism”.

THE SCIENCE Quintessence in physics is a hypothesised dark-energy scalar field (Caldwell, Dave and Steinhardt, 1998), and “quintessentialism” here evidently points at the older sense of an essential nature.

REALITY CHECK It is an artwork with no experimental content.

Fig. 119 Broken dopamine vial

CONTEXT



Files: 7e99d109-c6bd-497a-9c16-320ba62b8539.png

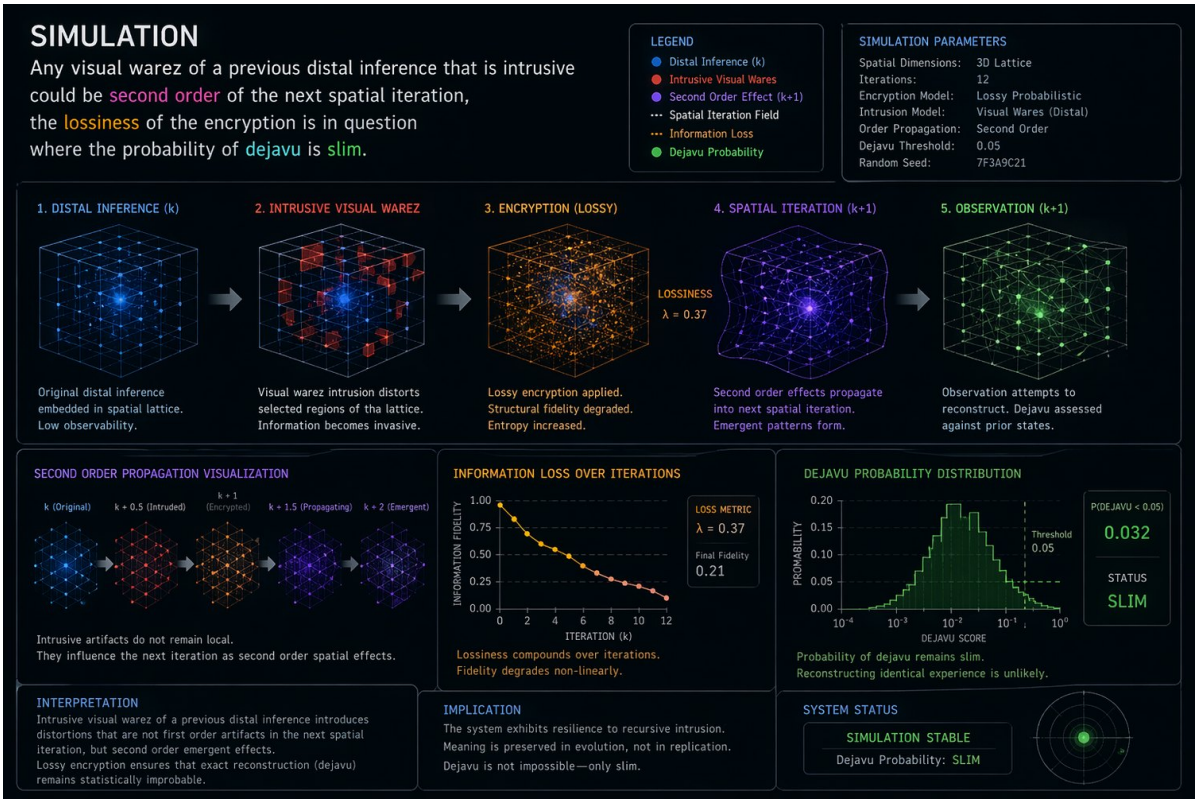
WHAT IT SHOWS A dark illustration of a smashed vial labelled “DOPAMINE” with its structure on the label, and a caption: “They’re addicted to breaking the dopamine vial after spending some time with it… The vat essense… Appears as et simultaneity to data scientists…”, over a background of scikit-learn code and Bayes’ theorem.

THE SCIENCE Dopamine is a neurotransmitter, $C_8H_{11}NO_2$. Recordings by Schultz and colleagues (1997) showed that dopamine neurons signal reward prediction error, the difference between expected and received reward, rather than pleasure directly.

REALITY CHECK The caption is poetic and its phrases (“vat essence”, “et simultaneity”) are not standard terms.

Fig. 120 Déjà-vu simulation with lossy “encryption”

CONTEXT



Files: a0401cc2-2545-40d4-9317-fadcb5271bcc.png

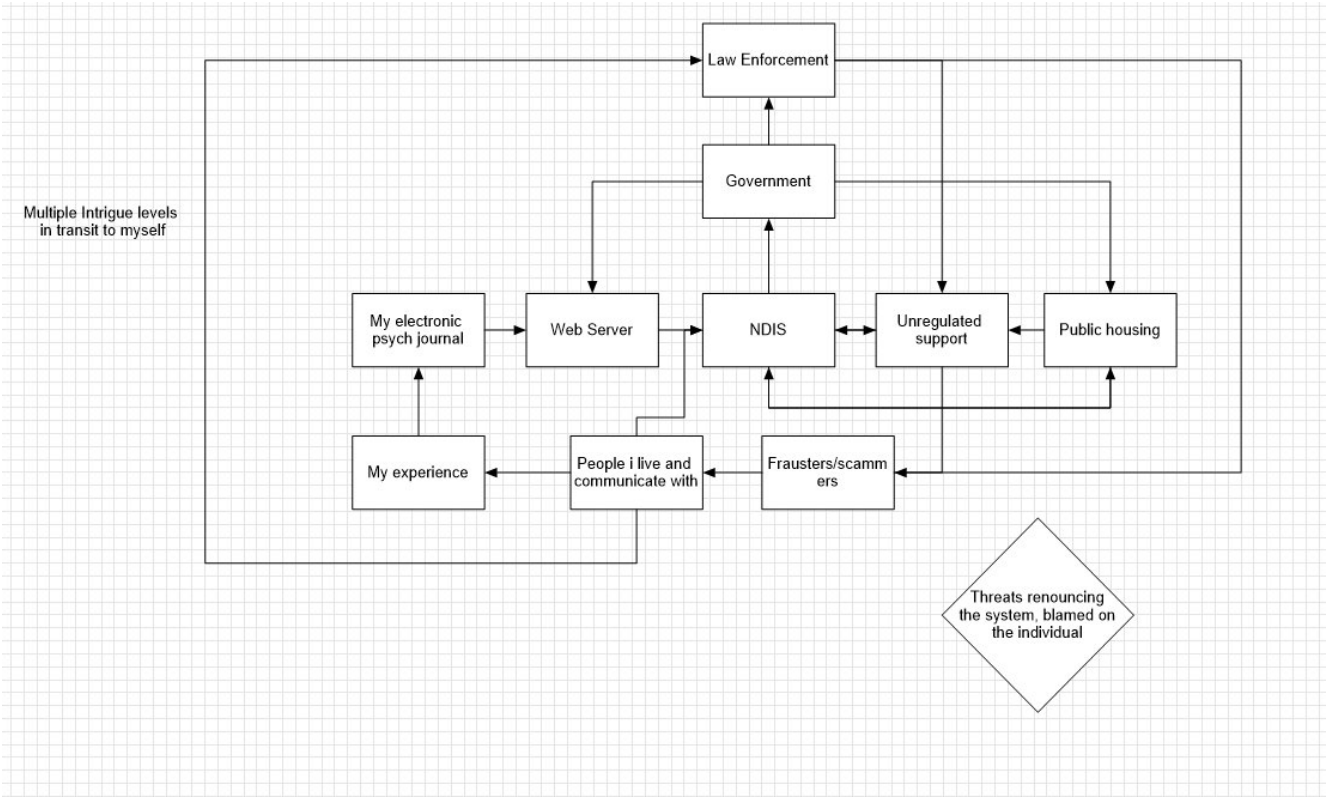
WHAT IT SHOWS A dense dashboard: a 3D lattice model with 12 iterations, distal-inference “warez” that intrude, lossy encryption with λ = 0.37, second-order spatial effects, an information-loss curve and a déjà-vu probability histogram with P(déjà vu) = 0.032, below a 0.05 threshold, marked “SLIM”.

THE SCIENCE Déjà vu is a real, common feeling of familiarity without a memory source, reported by about 60–70 % of people (Brown, 2003), and linked to temporal-lobe familiarity signals.

REALITY CHECK Encryption must be invertible, so “lossy encryption” is a contradiction; lossy processing is compression. The simulation’s probability comes from its own parameters and is not measured.

Fig. 121 A personal systems flowchart

CONTEXT



Files: 1978410394944217416-G3S5vbea0AA-SnB.png

WHAT IT SHOWS A flowchart of boxes for law enforcement, government, a disability-support scheme (NDIS), a web server, a personal electronic journal, unregulated support, public housing, people the author lives with, fraudsters and scammers, and “my experience”, with a diamond reading “Threats renouncing the system, blamed on the individual” and a side note about multiple levels of intrigue in transit.

THE SCIENCE Diagrams like this are causal-loop maps of the kind used in systems thinking: boxes are actors, arrows are influences, and feedback loops show where effects return to the source.

REALITY CHECK This is a personal map rather than a physical model, so it is not evaluated further.

Appendix A · Formula sheet

The relations behind the reality checks. Constants: $c = 2.998\times10^{8}$ m/s, $\hbar = 1.055\times10^{-34}$ J·s, $k_B = 1.381\times10^{-23}$ J/K, $\varepsilon_0 = 8.854\times10^{-12}$ F/m, $\sigma = 5.67\times10^{-8}$ W m^{-2} K^{-4}.

Topic	Relation	Example
Radiation force	F = P/c (absorbed or emitted), F = 2P/c (perfect reflection)	1 MW reflected → 6.67 mN
Blackbody	j = σT⁴; u = aT⁴ (a = 7.57×10⁻¹⁶ J m⁻³ K⁻⁴); λmax = 2898 µm·K / T	3000 K → 4.6 MW/m²; 2000 K → 12 mJ/m³
Casimir pressure	P = π²ħc / (240 d⁴)	13 Pa at 100 nm
Critical density	nc = ε₀ mₑ ω² / e² ≈ 1.1×10²¹ cm⁻³ / λ²(µm)	1 µm light: 1.1×10²¹; 1.2 nm X-ray: 7×10²⁶
Plane-wave field	I = ½ c ε₀ E²; B = E / c (so B ∝ √I)	10¹² W/m² → E = 2.7×10⁷ V/m, B = 0.09 T
Cyclotron resonance	f = eB / (2π mₑ) = 28.0 GHz per tesla	2.45 GHz ↔ 0.0875 T = 875 G
Relativistic speed	γ = 1 + KE/(mc²); β = √(1 − 1/γ²)	105.5 keV electron → 0.559 c
Tunnelling	T ≈ e^(−2κL), κ = √(2m(V−E)) / ħ	proton, 0.3 eV: κ ≈ 12 Å⁻¹; L = 0.8 Å → 5×10⁻⁹
Landauer bound	E ≥ kB T ln 2	2.87×10⁻²¹ J at 300 K
Exhaust speed	vₑ = Isp·g₀	1 MeV proton: 1.4×10⁷ m/s, Isp = 1.4×10⁶ s
Group velocity	v_g = c / n_g	n_g = 300 → 10⁶ m/s
Balmer lines	1/λ = R(1/4 − 1/n²), R = 1.097×10⁷ m⁻¹	n = 3, 4, 5 → 656, 486, 434 nm
Brillouin shift	Ω_B = 2 n vₐ / λ	silica, 1550 nm → about 11 GHz
Kittel / Larmor	f = γH, γ/2π = 2.80 MHz/Oe	1 GHz ↔ 357 Oe
Fringe visibility	V = (Imax − Imin)/(Imax + Imin)	which-path information lowers V
Shannon entropy	H = −Σ p log₂ p	fair coin: 1 bit
Enantiomeric excess	ee = (R − S)/(R + S)	75:25 mixture → 50 %
Cell voltage	E = E(cathode) − E(anode)	0.954 − (−0.76) = 1.71 V
Dice payout	multiplier = (100 − house edge) / win chance %	99 / 1.01 = 98.02

Appendix B · File-to-figure index

Every image file in the repository (in sorted filename order) and the figure that covers it. Copies of the same picture share a figure number.

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